

ID	Start time	Completion time	Email
1	3/23/26 10:26:22	3/23/26 10:30:48	10780379@polimi.it
2	3/23/26 11:10:04	3/23/26 11:18:26	10975089@polimi.it
3	3/23/26 14:11:08	3/23/26 14:36:23	11119541@polimi.it
4	3/23/26 16:39:29	3/23/26 17:15:57	11030873@polimi.it
5	3/23/26 17:30:56	3/23/26 17:36:47	10666869@polimi.it
6	3/23/26 21:32:36	3/23/26 21:42:57	11061975@polimi.it
7	3/24/26 14:39:51	3/24/26 14:42:27	10716383@polimi.it
8	3/19/26 11:40:01	3/24/26 16:32:20	10768645@polimi.it
9	3/24/26 17:12:06	3/24/26 17:13:24	10770813@polimi.it
10	3/24/26 20:10:06	3/24/26 20:12:41	10962875@polimi.it
11	3/24/26 21:41:22	3/24/26 22:41:34	11015522@polimi.it
12	3/25/26 2:51:45	3/25/26 2:56:22	10845860@polimi.it
13	3/25/26 10:08:43	3/25/26 10:09:15	11027977@polimi.it
14	3/25/26 14:28:13	3/25/26 14:45:04	10764083@polimi.it
15	3/25/26 19:17:45	3/25/26 19:27:53	11101495@polimi.it
16	3/26/26 4:33:37	3/26/26 4:52:40	10994077@polimi.it
17	3/26/26 17:35:44	3/26/26 17:40:49	10771695@polimi.it
18	3/26/26 18:25:09	3/26/26 18:32:44	10796320@polimi.it
19	3/26/26 20:51:54	3/26/26 20:56:10	10683374@polimi.it
20	3/26/26 22:40:50	3/26/26 22:47:05	11030334@polimi.it
21	3/27/26 10:06:06	3/27/26 10:09:19	10787431@polimi.it
22	3/27/26 10:36:56	3/27/26 10:43:35	10816840@polimi.it
23	3/27/26 10:48:37	3/27/26 10:52:04	10796163@polimi.it
24	3/27/26 10:47:48	3/27/26 10:54:03	10676235@polimi.it
25	3/27/26 11:05:14	3/27/26 11:11:46	10819854@polimi.it
26	3/27/26 11:16:30	3/27/26 11:31:36	10720381@polimi.it
27	3/21/26 16:58:27	3/27/26 12:55:59	10838600@polimi.it
28	3/27/26 14:35:02	3/27/26 15:01:25	10764570@polimi.it
29	3/27/26 18:28:43	3/27/26 18:34:22	10851929@polimi.it
30	3/27/26 19:11:46	3/27/26 19:16:38	10702601@polimi.it
31	3/27/26 20:30:03	3/27/26 20:33:32	10660050@polimi.it
32	3/27/26 21:18:51	3/27/26 21:29:40	10798825@polimi.it
33	3/27/26 21:36:31	3/27/26 22:08:24	11171264@polimi.it
34	3/27/26 22:15:44	3/27/26 22:20:54	11095668@polimi.it
35	3/28/26 9:24:14	3/28/26 9:27:35	10828027@polimi.it
36	3/28/26 9:48:49	3/28/26 9:52:10	10774521@polimi.it
37	3/28/26 10:11:45	3/28/26 10:27:11	10802468@polimi.it
38	3/28/26 10:19:15	3/28/26 10:30:33	10683981@polimi.it
39	3/28/26 10:37:07	3/28/26 10:38:28	10773862@polimi.it
40	3/28/26 10:54:13	3/28/26 10:58:25	10857924@polimi.it
41	3/28/26 13:42:39	3/28/26 14:08:24	10790205@polimi.it
42	3/28/26 14:30:21	3/28/26 14:39:44	10789693@polimi.it
43	3/28/26 14:34:22	3/28/26 14:59:13	11083784@polimi.it
44	3/28/26 14:42:21	3/28/26 15:10:29	10831959@polimi.it
45	3/28/26 15:07:57	3/28/26 15:27:43	10837597@polimi.it
46	3/28/26 15:22:22	3/28/26 15:30:36	11128210@polimi.it
47	3/27/26 18:04:49	3/28/26 15:56:56	11173956@polimi.it
48	3/28/26 16:19:46	3/28/26 16:28:16	10770385@polimi.it
49	3/28/26 16:15:15	3/28/26 16:29:34	10797420@polimi.it
50	3/28/26 16:34:31	3/28/26 16:44:46	10825131@polimi.it

Name	Total points	Quiz feedback	Last modified time
Andrea Rubagotti			
Alessandro Cupini			
Wang Yizhe			
Jiang Zhenglin			
Yuri Cinesi Ferrari			
Alperen Aydin			
Simone Bianchessi			
Gabriele Riboldi			
Giovanni Ferola			
Qiu Yongqi			
Yawen Yan			
Mario Monzani			
Yu Xinchun			
Marco Apollonio			
Riccardo De Giacomo			
Giovanni Orciuolo			
Edoardo Figini			
Samuele Lo Piccolo			
Zarlene Justrem De Mesa			
Bi Yiran			
Davide Vendramin			
Filippo Motti			
Stefano Castiglione			
Marco Vitali			
Giovanni Miccio			
Luca Guerra			
Daniele Crippa			
Giacomo Scampini			
Tommaso Rossetti			
Angelo Di Rosa			
Luca Donato			
Andrea Giovanni Zanelli			
Zhang Wenzhu			
Francesco Lazzaro			
Rebecca Besana			
Pietro Callandrone			
Lin Elena			
Cezar Gjinaj			
Samuele Galbiati			
Marco Coviello			
Marco Carminati			
Alessio Vittorio Rosano'			
Zheng Zihao			
Simone Azzoni			
Alessia Ragosta			
Mohannad Mohamed Fawzy Ahmed Rashwan			
Ignacia Nuyens Gotelli			
Alessia Radicula			
Pietro Pizzoccheri			
Edoardo Sartore			

Name Surname 1	Points - Name Surname	Feedback - Name Surname	Person Code 1
Andrea Rubagotti			10780379
Alessandro Cupini			10975089
Yizhe Wang			11119541
JIANG ZHENGLIN			11030873
Yuri Cinesi Ferrari			10666869
Alperen AYDIN			11061975
Simone Bianchessi			10716383
Riboldi			10768645
Giovanni Ferola			10770813
yongqi qiu			10962875
Yawen Yan			11015522
Mario Monzani			10845860
XINCHEN YU			11027977
Marco Apollonio			10764083
Riccardo De Giacomo			11101495
Giovanni Orciuolo			10994077
Edoardo Figini			10771695
Samuele Lo Piccolo			10796320
Zarlene Justrem De Mesa			10683374
Yiran Bi			11030334
Davide Vendramin			10787431
Filippo Motti			10816840
Stefano Castiglione			10796163
Marco Vitali			10676235
Giovanni Miccio			10819854
Guerra			10720381
Daniele Crippa			10838600
Giacomo Scampini			10764570
Tommaso Rossetti			10851929
Angelo Di Rosa			10702601
Luca Donato			10660050
Andrea Giovanni Zanelli			10798825
WENZHU ZHANG			11171264
Francesco Lazzaro			11095668
Rebecca Besana			10828027
Pietro Callandrone			10774521
Elena Lin			10802468
Cezar Gjinaj			10683981
Samuele Galbiati			10773862
Marco Coviello			10857924
Marco Carminati			10790205
Alessio Vittorio Rosanò			10789693
Zihao Zheng			11083784
Simone Azzoni			10831959
Alessia Ragosta			10837597
Mohannad Mohamed Fawzy Ahmed Rashwan			11128210
Ignacia Nuyens			11173956
Alessia Radicula			10770385
Pietro Pizzoccheri			10797420
Edoardo Sartore			10825131

Points - Person Code 1	Feedback - Person Code 1	Name Surname 2	Points - Name Surname 2
		Maria Concetta Santagata	
		Jiashen Zhang	
		None	
		Kayra ÜÇKİLİNÇ	
		Villa	
		Lorenzo Della Matera	
		Francesco Azzarà	
		Fabrizio Grimaldi	
		Greta Fornaciari	
		Guglielmo Caprotti	
		Shuai Zou	
		Matteo Taschini	
		Matteo Poli	
		Samuele Dario Scamozzi	
		Luca Perazzi	
		Giulia Putelli	
		Livia Polverari	
		Francesco Giuseppe Mezzera	
		XIAOWEN PAN	
		Andres Felipe Forero Salas	
		Noemi Fazzini	
		Lorenzo Bianco	
		Jiayi Zheng	
		Om Mohan Deshmukh	
		Andrea Coscia Vanoni	
		Martino Andrea Grassi	
		Fernanda Vicente	
		Lorenzo Bardelli	

Feedback - Name Surn	Person Code 2	Points - Person Code 2	Feedback - Person Cod
	10799787		
	11115748		
	None		
	11049291		
	10768851		
	10799739		
	10796209		
	11101728		
	10862436		
	10764592		
	11024119		
	10785383		
	10797758		
	10772778		
	10846682		
	10860913		
	10793007		
	10742249		
	11168874		
	11068591		
	11178565		
	11024306		
	10811585		
	11055131		
	10762257		
	10829712		
	11188762		
	10831941		

Wokwi shared public liPoints - Wokwi shared Feedback - Wokwi sharEnergy Consumption o	
null (see .zip file)	2006.879381
https://wokwi.com/projects/459278258020937729	906.6061
https://wokwi.com/projects/458677544100957185	494.72
https://wokwi.com/projects/459191430529305601	256.355
https://wokwi.com/projects/458934433326170113 (Part 1), https://wokwi.com/projects/459317167948766209	195.223 mJ
https://wokwi.com/projects/459317167948766209	289.56
	308
https://wokwi.com/projects/458934329543329793	347,70495
	664.88076
	205.76mJ in 3.183s
https://wokwi.com/projects/459291835413371905	130.22
	109.95
https://wokwi.com/projects/459376239881000961	147.02
	414.23
https://wokwi.com/projects/459490619881106433	8065.502 mJ
Unoptimized version: https://wokwi.com/projects/4590333934386984220 , 844 mJ	
	1337
Point 1: https://wokwi.com/projects/459190052135131137 Point 3: https://wokwi.com/projects/459190052135131137	241,947397 mJ
	153,557
https://wokwi.com/projects/459593469014862849	302.847890
	296.54
	455.345
	168,69
	335.864
https://wokwi.com/projects/458758202167480321	223.19
https://wokwi.com/projects/459641265618327553	307.829
https://wokwi.com/projects/459199655956686849	47.84
https://wokwi.com/projects/458681432156361729	240.526
https://wokwi.com/projects/459650053971200001	285.64
	73.69
	120.643
	266,952
https://wokwi.com/projects/459234645984063489	sensing: 22.78mJ, tx: 0.
	1364.485
https://wokwi.com/projects/458757993511312385	323,3
	263.90 mj
https://wokwi.com/projects/459282869753498625	232,586
https://wokwi.com/projects/459728207196654593	324.019
	352,70
https://wokwi.com/projects/459728920084788225	315
	95.592
	334,8
https://wokwi.com/projects/459591356890079233	238.61
https://wokwi.com/projects/459671203252974593	154.1
https://wokwi.com/projects/458738396776892417	258.052
	89.76
https://wokwi.com/projects/459738717092939777	94.75 mJ
Ho consegnato il file wokwi da eseguire sulla virtual machine	396.946 mJ
https://wokwi.com/projects/459750909568103425	264.060657
non optimized version: https://wokwi.com/projects/45867804878485344	

Points - Energy Consumed	Feedback - Energy Consumed	Total device lifetime in Points - Total device lifetime
		27712.37534
		76501
		213800
		173370
		219902
		194066
		212603
		278633
		49199.41644
		276578
		249791
		226035
		391384
		198008
		33233.36 seconds
		293400 s
		79403
		182661,804 s
		321082,3762
		197467.448
		223845.0
		174185
		198805,77
		207713
		132857
		191638
		Lifetime = 18600 J / 47.84 mJ × 565.62 ms ≈ 2195
		219699
		214 369
		161464
		91093
		226463
68mJ, deepsleep: 86.98mJ		268,229
		50827
		217588
		210200.51 s
		187909
		231039.1181
		103812
		187857
		172,263
		282771
		325162.17
		184628
		265426
		317868.9
		235606
		173.460 s
		181274
		173664

Feedback - Total device	Energy Consumption E:Points - Energy Consun	Feedback - Energy Cons
	1026.1116 mJ	
	157.5	
	1508.56	
	132.65	
	1458.754 mJ	
	262.09	
	210	
	347,70398	
	imp1: 662.4884(WiFi management); imp2: 8381.52876(extended deep sleep); imp	
	1436.96mJ in 30.183s(in the best-case scenario)	
	130	
	109.76	
	146.60	
	545.36	
	2899.733 mJ	
	150,506 mJ	
	93432	
	Depending on the sensed values: 226,072211 mJ or 345,599821mJ (higher due to	
	24,7 mJ	
	302.169737	
	352021.5	
	2980	
	Offline cycle: 124,62 ; On average: 129,03	
	14.2175 J	
	179.45	
	292.657 (TX cycle) / 180.5 (average with all improvements)	
910 s ≈ 61.08 hours	Energy consumption reduction of the 41.86%	
	114.5	
	Conditional Transmission Realistic Avg Case: 170.123 mJ, Deep Sleep (10.2s): 198.1	
	best case 45.65 mJ, worst case 73.69 mJ	
	270.383	
	8,3	
	sensing: 0.013mJ, idle: 2.71mJ, deepsleep: 86.98mJ, Total lifetime: 333,697	
	1117.4964	
	198,127	
	259.73 mJ	
	168,993	
	162mJ Please refer to the report for a detailed explanation	
	346,02	
	181	
	214.59 mJ	
	in the improved version the consumption estimation is based on the occurs of m	
	237.74	
	153.71	
	Max energy consumption 258.052 mJ, Min energy consumption 238.344 mJ. The	
	68.46	
	53.7175 mJ	
	299.95 mJ	
	152.241761	
	341	

3:no Data (Light sleep); imp4: 208.5 (sensors off)

longer sleep times)

641 mJ, Data Batching (N=3): 200.072 mJ

movements (N) over time(T): $45.215 \cdot T + 124.66 \cdot N$

average energy consumption depends on the environmental conditions and the chosen threshold

51	3/28/26 14:40:19	3/28/26 16:56:06	10853745@polimi.it
52	3/28/26 16:50:45	3/28/26 16:57:16	11120812@polimi.it
53	3/28/26 17:24:04	3/28/26 17:27:18	10619593@polimi.it
54	3/28/26 17:19:24	3/28/26 17:43:08	10828995@polimi.it
55	3/28/26 19:05:38	3/28/26 19:12:07	10730038@polimi.it
56	3/28/26 18:36:47	3/28/26 19:35:20	11166508@polimi.it
57	3/28/26 21:37:44	3/28/26 22:35:11	11022931@polimi.it
58	3/28/26 23:59:16	3/28/26 23:59:32	11167997@polimi.it
59	3/29/26 0:26:19	3/29/26 0:37:08	11111393@polimi.it
60	3/29/26 10:09:11	3/29/26 10:18:46	11097125@polimi.it
61	3/29/26 11:03:57	3/29/26 11:32:47	10781960@polimi.it
62	3/29/26 11:22:52	3/29/26 11:45:14	10767814@polimi.it
63	3/29/26 11:45:29	3/29/26 11:47:34	10808901@polimi.it
64	3/29/26 11:56:47	3/29/26 12:00:03	10798975@polimi.it
65	3/29/26 11:28:01	3/29/26 12:05:19	11031608@polimi.it
66	3/29/26 12:29:27	3/29/26 12:41:06	11051292@polimi.it
67	3/29/26 12:50:05	3/29/26 12:53:19	11048104@polimi.it
68	3/29/26 9:54:14	3/29/26 12:58:24	11090815@polimi.it
69	3/29/26 13:07:37	3/29/26 13:13:46	10698107@polimi.it
70	3/29/26 14:19:09	3/29/26 14:24:37	11053611@polimi.it
71	3/29/26 14:25:59	3/29/26 14:35:31	10900762@polimi.it
72	3/29/26 14:57:50	3/29/26 15:18:34	10850482@polimi.it
73	3/29/26 15:50:23	3/29/26 15:58:52	11019965@polimi.it
74	3/29/26 16:12:39	3/29/26 16:19:51	11033376@polimi.it
75	3/29/26 16:23:09	3/29/26 16:26:17	10922631@polimi.it
76	3/29/26 16:07:47	3/29/26 16:44:44	10797131@polimi.it
77	3/29/26 16:35:46	3/29/26 16:49:44	10783671@polimi.it
78	3/29/26 16:50:42	3/29/26 16:56:01	11006775@polimi.it
79	3/29/26 17:05:58	3/29/26 17:11:04	11044962@polimi.it
80	3/29/26 17:34:29	3/29/26 17:37:31	10813424@polimi.it
81	3/28/26 13:20:27	3/29/26 17:38:34	11034150@polimi.it
82	3/29/26 17:21:22	3/29/26 17:43:58	11044575@polimi.it
83	3/29/26 17:36:17	3/29/26 17:45:13	11157354@polimi.it
84	3/29/26 17:44:52	3/29/26 17:48:01	10732218@polimi.it
85	3/29/26 17:48:48	3/29/26 18:28:54	11154567@polimi.it
86	3/29/26 18:15:02	3/29/26 18:31:43	11101890@polimi.it
87	3/29/26 18:28:13	3/29/26 18:41:09	10806605@polimi.it
88	3/29/26 19:04:52	3/29/26 19:09:02	10817232@polimi.it
89	3/29/26 19:10:47	3/29/26 19:18:03	11021337@polimi.it
90	3/29/26 16:51:19	3/29/26 19:21:42	11109083@polimi.it
91	3/29/26 19:19:43	3/29/26 19:29:27	11143015@polimi.it
92	3/29/26 17:19:00	3/29/26 19:45:41	10830061@polimi.it
93	3/29/26 19:48:00	3/29/26 19:50:28	11110511@polimi.it
94	3/27/26 11:54:29	3/29/26 20:04:24	11152090@polimi.it
95	3/29/26 19:43:58	3/29/26 20:10:31	11068760@polimi.it
96	3/29/26 20:11:57	3/29/26 20:12:54	10793061@polimi.it
97	3/29/26 20:20:06	3/29/26 20:37:03	10797129@polimi.it
98	3/26/26 15:53:51	3/29/26 21:06:59	11019981@polimi.it
99	3/29/26 21:06:10	3/29/26 21:10:02	11189222@polimi.it
100	3/29/26 21:24:22	3/29/26 21:27:42	10711043@polimi.it
101	3/29/26 17:42:00	3/29/26 21:50:37	11120269@polimi.it

Mattia Caon	
Orhun Tunc	
Alessandro Pessina	
Jacopo Pelagalli	
Edoardo Menegotto	
Cai Chuanhao	
Peng Rao	
Han Zhaohui	
Muhammad Rian	
Yiming Hou	
Andrea Vavassori	
Alessandro Fedele	
Luca De Nicola	
Matteo Rossi	3/30/26 18:43:54
Peng Anchen	
Alireza Jahandoost	
Ting Zhang	
Zhu Wen	
Andrea Gaetano Senatore	
Ding Zhenze	
Davide Petruzzi	
Alessandro Cirillo	
Yasin Asgari	
Li Ming	
Francesco Armuzzi	
Morena Caterina Cardenas Estrada	
Raul Agolli	
Osman Bugra Dagli	
Mohammad Javad Zandiyeh	
Andrea Midulla	
Mobina Mobaraki	
Zhang Jie	
Ingrid Lyngstad	
Davide Greco	
Esteban Lopez Barreto	
Mohamed Youssef Mohamed Youssef Elgamel	
Bruno Arena	
Tommaso Neri	
Jiang Jiaxun	
Geng Zihao	
Saltuk Bugra Kirazci	
Cristian Summa	
Hong Jinxu	
Hao Johnny Vinh Nguyen	
Giulia Barresi	
Marco Zheng	
Roberta Gennari	
Jia Tongxin	
Felicia Noomi Murkes	
Marco Stefano De Bernardi	
Zhao Zhiyi	

Mattia Caon	10853745
Orhun Tunc	11120812
Alessandro Pessina	10619593
Jacopo Pelagalli	10828995
Edoardo Menegotto	10730038
CAI CHUANHAO	11166508
Peng Rao	11022931
ZHAOHUI HAN	11167997
Muhammad Rian	11111393
Dinar Gareev	11063702
Andrea Vavassori	10781960
Alessandro Fedele	10767814
Luca De Nicola	10808901
Matteo Rossi	10798975
Qianyi Wang	11031890
Alireza Jahandoost	11051292
Zhang Ting	11048104
Wen Zhu	11090815
Andrea Gaetano Senatore	10698107
Zhenze Ding	11053611
Davide Petruzzi	10900762
Alessandro Cirillo	10850482
Yasin Asgari	11019965
MING LI	11033376
Francesco Armuzzi	10922631
Morena Caterina Cardenas Estrada	10797131
Raul Agolli	10783671
OSMAN BUGRA DAGLI	11006775
Mohammad Javad Zandiyeh	11044962
Andrea Midulla	10813424
Mobina Mobaraki	11034150
JIE ZHANG	11044575
Ingrid Lyngstad	11157354
Greco	10732218
Esteban Lopez Barreto	11154567
Mohamed Youssef Mohamed Youssef Elgamal	11101890
Bruno Arena	10806605
Tommaso Neri	10817232
Jiaxun Jiang	11021337
Zihao Geng	11109083
Saltuk Bugra Kirazci	11143015
Cristian Summa	10830061
JINXU HONG	11110511
Hao Johny Vinh Nguyen	11152090
Barresi	11068760
Marco Zheng	10793061
Roberta Gennari	10797129
Tongxin Jia	11019981
Felicia Murkes	11189222
Marco Stefano De Bernardi	10711043
Zhao Zhiyi	11120269

Marco Bertazzoli Cova
ZICHEN LIU
Aakash Bhatti
Yiming Hou
Marco Gad
Dario Crosa
Edoardo Ponzelletti
Anchen Peng
Ali Nabipour Dargah
Matteo Capone
Zhaojun Liu
Matteo Gitto
Jaime Iriarte
Armin Mardoukhi
XIANGYU LIU
Shashi Bhushan
ERDEM COBANOGU
Lorenzo Coppa
Federico Costa
Bellosi
Ma Chong
Matteo Cavalleri
Anasuya SATAPATHY
Marica Balestra
-
Mathias Rodigari
TIANYU GAO
Carbajal Serrano
D'Angelo Sara Naima
Yuedan Cai
Aron Lund
Federico Mastria

10929780
10935200
11113677
11097125
10794601
10796197
10765707
11031608
11041035
10728121
11030423
11164535
11154921
11060212
11062619
11111318
11045157
10842807
10822446
10806471
11162609
10768765
11175090
10832004
-
10796466
11102304
11001933
10710507
11093193
11150698
10698676

	291.210
	480
	374.03
https://wokwi.com/projects/459754539629661185	357.7 mJ
https://wokwi.com/projects/459487673846421505	321,271
	60.736
https://wokwi.com/projects/459774034167327745	Idle: 0.34 mJ Sensor re
https://wokwi.com/projects/459772506840767489	239.69 mJ
https://wokwi.com/projects/459297610631400449	Ettotal≈0.888726 J
We did the project locally and will have the wokwi complete file in the s43,7896 mJ = 0, 043789	
https://wokwi.com/projects/459817741318748161	117.8448
https://wokwi.com/projects/459286712768378881	150.16
	80,103 mJ
	263.24
baseline: https://wokwi.com/projects/459291138541129729 ;Improme	402.491
https://wokwi.com/projects/459115285014401025	339.661
https://wokwi.com/projects/458687129966865409	205.06
	364.841
https://wokwi.com/projects/459654428915252225 , https://wokwi.com/projects/459488040849129473	106.9 mJ (over 1.3 seco
https://wokwi.com/projects/459210876803398657	(Improved 114.71
https://wokwi.com/projects/459832842940536833	0.1224J
First link: https://wokwi.com/projects/459730058841633793	Second 285.25 mJ
https://wokwi.com/projects/459730058841633793	0.13257
	282.140
https://wokwi.com/projects/459115273887955969	300.93
https://wokwi.com/projects/459110356985526273	1835.605
	248.25
https://wokwi.com/projects/459839420787885057	275.984 mJ
https://github.com/JavadZandiyeh/POLIMI-IOT	104.46
https://wokwi.com/projects/459119089941723137	290,29
	66.6315
https://wokwi.com/projects/459304901596669953	269.374
	177.4 mJ (avg. power 15
https://wokwi.com/projects/458730478243710977	157.10
https://wokwi.com/projects/458944891926996993	148.14
https://wokwi.com/projects/459845042457414657	1663 mJ
-	0,19 J
https://wokwi.com/projects/459568059656233985 ; https://wokwi.com/projects/459638342239245313	288.810
https://wokwi.com/projects/459638342239245313	4004 mJ
https://wokwi.com/projects/459833285709146113	298.84
https://wokwi.com/projects/459749463174762497 and improved vers	Between 50 and 100 mJ
	225.23
https://wokwi.com/projects/459286990417722369	117.247 mJ
https://wokwi.com/projects/459847313223563265	2704
https://wokwi.com/projects/459851801385398273	232,122
	159.5
https://wokwi.com/projects/459841626977471489 , https://wokwi.com/projects/459853200341168129	266,38
https://wokwi.com/projects/459853200341168129	294.868mJ
	249.414
	329.8
https://wokwi.com/projects/459845471133178881	385.29

	332301.669
	78162
	267892
	5,2 s
	210988,9
	353945
ading: 0.15 mJ WiFi ON in RX mode: 123.66 mJ	233280
	390,737s
	Battery lifetime≈25.1 hours (≈ 1 day) T≈90374s
5 J	308559,766 s
	227620.7088
	240485.94
	161 527 s
	230662
	198922
	234917.68
	118217.60
	87171
nds of cycle)	219536 s
	270346.22
	227975s
	211034.59 s
	311204.55
	216629.14
	224145
	338906.102
	213012
	195840 s
	1.8
	200376
	165282.6826
	233167.279
9.4 mW)	108870
	274234
	302336
	61,031 seconds
	109908 s
	232030.2
	40600 s
	255362.2
but for ESP32 it is 60.02 mJ	630525 seconds
	128182
	221894 s
	296800
	147736
	214000
	218628,30
	259358.38s
	224026
	205864.087 s
	95,563

1432.211 mJ (Note: this is based on a longer cycle of 30sec, as our solution propo:
631
310.31
27,73 J (after the improvements one cycle is longer than 10 minutes)
See the report because there are different improvements and different time peric
60.235
3.6s baseline: 293.9(mJ) 3.6s + data-driven: 229.6 3.6s + data-driven: 229.6 3.6s +
239.27mJ
0.859663 J
43,7572 mJ = 0,0437572 J
112.2209
Between 150.16 mJ and 104.32 mJ, depending on whether there is a transmissior
326,48 mJ (with WiFi On phase)
125.916 (without Deep Sleep)
Active Cycle (with TX): 0.4003 J;Standby Cycle (no TX): 0.2056 J;Weighted Average
226.208
142.61
91.52
2591.6 mJ (over 60.1 seconds of cycle)
105.70
0.049J on average (considering the conditional wifi improvement)
The energy consumed is not the same in every cycle, there are a best case scenar
0.09218
270.581mJ
165.79
1650.939
Compared to the baseline continuous-transmission cycle (0.24825Joules), the op
148.54 mJ
102.98
584,6
65.2915
138.845 mJ (for the no-significant-change case)
596.0 mJ (avg.power 58.3 mW)
132.57
107.03
250 mJ
0,117 J
288.202
2156 mJ
297.75
5 mJ
163.25 mJ (from the worst case)
76.79 mJ
2704 - Didn't calculate this
112,764
80.75
197,07 mJ on average
207.826mJ
Full cycle: 249.385 mJ, Cycle without wifi or transmission (when no change is dete
329.3
28.885 mJ

ses)

nds of one cycle. It has no sense compare energy consumptions of cycle with different period

Interrupt-Driven: 178 60s baseline: 2875.7 60s + data-driven: 2811.4

l.

: ~ 0.215 J

io in which it consumes only 229.36 mJ and a pessimistic scenario in which it consumes 343.05 mJ. The cy

timized skip cycle (0.120Joules) reduces the total energy consumption by over 51% per cycle. his effective

cted): 124.247 mJ

cle lasts between 5 and 5.19 seconds in this new estimation.

ly doubles the autonomy of the device simply by restructuring the software execution flow to utilize the I

SP32's RTC memory. Better explained in the report.

102	3/29/26 21:48:54	3/29/26 21:50:49	10773859@polimi.it
103	3/29/26 21:14:35	3/29/26 21:51:35	11112969@polimi.it
104	3/29/26 21:54:02	3/29/26 22:03:15	10911594@polimi.it
105	3/29/26 20:37:35	3/29/26 22:06:34	10813646@polimi.it
106	3/29/26 22:17:06	3/29/26 22:19:10	11119281@polimi.it
107	3/29/26 22:16:06	3/29/26 22:19:31	10677341@polimi.it
108	3/29/26 22:08:33	3/29/26 22:22:38	11046392@polimi.it
109	3/29/26 22:43:38	3/29/26 22:46:12	10766937@polimi.it
110	3/29/26 22:38:17	3/29/26 22:49:18	10708185@polimi.it
111	3/29/26 20:02:20	3/29/26 23:18:34	11058806@polimi.it
112	3/30/26 0:15:59	3/30/26 0:45:13	11115876@polimi.it
113	3/29/26 21:46:26	3/30/26 1:00:33	11032747@polimi.it
114	3/30/26 2:00:42	3/30/26 2:16:33	11111583@polimi.it
115	3/30/26 2:49:55	3/30/26 2:57:26	10948935@polimi.it
116	3/30/26 3:28:48	3/30/26 3:33:38	11147720@polimi.it
117	3/30/26 9:43:57	3/30/26 9:53:47	11152089@polimi.it
118	3/30/26 10:07:24	3/30/26 10:18:03	10777877@polimi.it
119	3/30/26 10:54:04	3/30/26 10:57:37	10811640@polimi.it
120	3/30/26 11:05:37	3/30/26 11:07:59	11151327@polimi.it
121	3/30/26 10:59:09	3/30/26 11:08:05	10766478@polimi.it
122	3/30/26 11:05:44	3/30/26 11:08:35	10764433@polimi.it
123	3/30/26 11:11:03	3/30/26 11:14:02	10839162@polimi.it
124	3/30/26 11:16:17	3/30/26 11:17:17	10839190@polimi.it
125	3/30/26 11:09:53	3/30/26 11:18:57	11187268@polimi.it
126	3/30/26 11:09:35	3/30/26 11:19:49	11063424@polimi.it
127	3/30/26 10:50:06	3/30/26 11:23:37	11118976@polimi.it
128	3/30/26 11:08:53	3/30/26 11:28:37	10844134@polimi.it
129	3/30/26 12:19:50	3/30/26 12:20:18	11137282@polimi.it
130	3/30/26 12:24:22	3/30/26 12:26:37	10697294@polimi.it
131	3/30/26 12:17:39	3/30/26 12:32:34	10744406@polimi.it
132	3/30/26 12:25:36	3/30/26 12:33:50	10849638@polimi.it
133	3/30/26 12:28:29	3/30/26 12:41:04	11009540@polimi.it
134	3/30/26 12:05:49	3/30/26 12:43:12	10847211@polimi.it
135	3/30/26 12:46:21	3/30/26 12:49:31	10805688@polimi.it
136	3/30/26 12:01:16	3/30/26 12:50:15	10769571@polimi.it
137	3/30/26 12:36:09	3/30/26 12:57:35	10705152@polimi.it
138	3/30/26 11:52:20	3/30/26 13:27:28	10829488@polimi.it
139	3/30/26 13:02:24	3/30/26 13:36:09	10502570@polimi.it
140	3/30/26 13:35:07	3/30/26 13:43:08	10711947@polimi.it
141	3/30/26 13:27:35	3/30/26 14:05:02	11121670@polimi.it
142	3/30/26 14:12:58	3/30/26 14:15:31	10777415@polimi.it
143	3/30/26 14:24:39	3/30/26 14:25:46	11078923@polimi.it
144	3/30/26 14:25:45	3/30/26 14:31:16	11109586@polimi.it
145	3/30/26 14:51:13	3/30/26 14:54:14	11062471@polimi.it
146	3/30/26 14:25:45	3/30/26 14:59:11	11048190@polimi.it
147	3/30/26 14:51:26	3/30/26 15:00:13	11033400@polimi.it
148	3/30/26 14:35:25	3/30/26 15:14:49	10910881@polimi.it
149	3/30/26 15:19:04	3/30/26 15:23:08	10831046@polimi.it
150	3/30/26 15:18:19	3/30/26 15:24:12	11026772@polimi.it
151	3/30/26 15:25:44	3/30/26 15:25:54	10972650@polimi.it
152	3/30/26 14:49:55	3/30/26 15:26:50	10815900@polimi.it

Mattia Brianti	
Li Xin	
Francesca Maragoni	
Gabriel Leonardi	
Mariyam Rashid Pitafi	
Alberto Biffi	
Amir Hossein Rashidy	
Gabriele Lorenzo Singe	
Andrea Ardu	
Sevval Karadeniz	
Zahra Sadat Parviss	
Hu Peng	
Fatemeh Kianiharchegani	
Rong Huang	
Alex Iortoman	
Christian Vaktel	
Mattia Panzeri	
Simone Giuliano	
Kelly Tran	
Dai Run Jie Simone	
Jonny Arbex Marques	
Davide Ghisolfi	
Elena Bestetti	
Nicolás Koron	
Hu Jialin	
Dosilhenbi Yumnam	
Edoardo Francesco Albini	
Yehia Mohamed Abdelmoniem Abdallah	
Salvatore Passaro	
Gabriele Bogdan Sepe	
Alessio Lika	
Emmanuel Muyololo	
Simone Teggi	
Giacomo Genco	
Francesco Street	
Alessandro Posani	
Gabriele Benedetti	
Davide Lorenzi	3/30/26 14:35:23
Eugenia Ghiglione	
Zhang Xinyue	
Andrea Nardi	
Junhua Fang	
Seyedaman Zargari	
Ahmet Eren Genis	
Qi Jinwen	
Zhang Hezhi	
Claudia Carraro	
Matteo Franken	
Minh Man Nguyen	
Guan Enci	
Marco Scianna	

Mattia Brianti	10773859
Yuanyuan Li	11110488
Maragoni	10911594
Gabriel Leonardi	10813646
Mariyam Rashid Pitafi	11119281
Alberto Biffi	10677341
Amir Hossein Rashidy	11046392
Gabriele Lorenzo Singe	10766937
Andrea Ardu	10708185
Sevval Karadeniz	11058806
Zahra Sadat Parviss	11115876
Peng Hu	11032747
Fatemeh Kianiharchegani	11111583
Rong Huang	10948935
Alex IORTOMAN	11147720
Christian Vaktel	11152089
Mattia Panzeri	10777877
Simone Giuliano	10811640
Tran	1151327
run jie simone dai	10766478
Jonny Arbex Marques	10764433
Davide Ghisolfi	10839162
Elena Bestetti	10839190
Nicolas Koron	11187268
JIALIN HU	11063424
Dosilhenbi Yumnam	11118976
Edoardo Albini	10844134
YEHIA ABDALLAH	11137282
Salvatore Passaro	10697294
Gabriele Bogdan Sepe	10744406
Alessio Lika	10849638
EMMANUEL MUYOLOLO	11009540
Simone Teggi	10847211
Giacomo Genco	10805688
Francesco Street	10769571
Alessandro Posani	10705152
Gabriele Benedetti	10829488
Davide Lorenzi	10502570
Ghiglione	10711947
Zhang Xinyue	11121670
Andrea Nardi	10777415
FANG JUNHUA	11078923
Seyedaman Zargari	11109586
Ahmet Eren Genis	11062471
Jinwen Qi	11048190
ZHANG HEZHI	11033400
Claudia Carraro	10910881
Matteo Franken	10831046
Minh Man Nguyen	11026772
Zhaochen Qiao	11021721
Ermir Balliu	11140947

	Alex Hathaway
	Xin Li
	Mettifogo
	Federico Gislon
	Erfan MohammadZaki
	Laura Maria Bosco
	Haga Zenab Mohammed Mustafa Khalil
	Ali Isapour
	Sajad Ghadiri
	Yumnam DOSILHENBI
	Magnus Lund
	Diego Butti
	Simone Luigi Gervasi
	Nguyen
	jiaxin yang
	Leonardo Bovani
	Tony David Matrundola
	Jacopo Alesina
	ZIYIN YOU
	Alex Iortoman
	Amr Abdelhalim
	Matilde Bellome
	Marco Scrivano
	Valeria Iagulli
	Andrea Venezia
	Riccardo Calzone
	Giuseppe Buccheri
	Giacomo Milianti
	Cucchetti
	Francesco Raimondi
	CHAI KE
	Amirhossein Dibaj
	Silvia Ferrara
	Chiara Cavallo
	Lorenzo Chiroli
	Enci Guan
	Marco Scianna

10829946
11112969
10901067
10768605
11049700
10793514
11091338
11107660
11062873
11118976
11152488
10806864
10790904
1157416
10719178
10809273
10723736
10859045
11022646
11147720
11116374
10709235
10768863
10765606
10797678
10740222
10795884
10860834
10813527
10767901
11110613
11109540
10795184
10856023
10797603
10972650
10815900

	180 mJ (considering als
https://wokwi.com/projects/459629260495640577	808.5 mJ
https://wokwi.com/projects/459480931426356225	350.59
https://wokwi.com/projects/459850178784044033	355.13
https://wokwi.com/projects/459566616449368065	478.2
https://wokwi.com/projects/458932344716356609	342.999 mJ
https://wokwi.com/projects/459563782386637825	525 mJ
	331.784
https://wokwi.com/projects/459861901627137025	308.311 (fixed deep slee
https://wokwi.com/projects/459860631374702593	68.769
	193.189
https://wokwi.com/projects/459860970157063169	https://wokwi.com/projects/459860970157063169
https://wokwi.com/projects/459031760349438977	https://wokwi.com/projects/459031760349438977
zip file under path /wokwi_code/test_iot_basic_calculate.zip and /wok	0,039 / 60,81 / 173,9
	342.81
	371.26
https://wokwi.com/projects/459571995313182721	416.4536
https://wokwi.com/projects/459901084950598657	(optimized https://wokwi.com/projects/459901084950598657)
https://wokwi.com/projects/458957778211643393	337,39
https://wokwi.com/projects/459827645779826689	591.59
https://wokwi.com/projects/459752428096161793	302.28mJ
	432.1837
	124,41
https://wokwi.com/projects/459679594645941249	https://wokwi.com/projects/459679594645941249
	253,4347 mJ
	162.243
https://wokwi.com/projects/459385471296442369	259.31 mJ
	371.26mJ
https://wokwi.com/projects/459908788100034561	https://wokwi.com/projects/459908788100034561
https://wokwi.com/projects/459912287770556417	296.397
https://wokwi.com/projects/459771784253359105	352.156159
https://wokwi.com/projects/459912725593064449	167 mJ
https://wokwi.com/projects/459912472010631169	716.4563
https://wokwi.com/projects/459912614315013121	277.1 mJ
https://wokwi.com/projects/459761716305884161	261.36
	1099.0728
	1157.8
https://wokwi.com/projects/459914108583290881	250.09
https://wokwi.com/projects/458732198476103681	323.402
No link, only offline VSCode	141,43966
https://wokwi.com/projects/459916039298324481	519.1172942
https://wokwi.com/projects/459852097581952001	155.796614
https://wokwi.com/projects/459843415086098433	208.947
https://wokwi.com/projects/459662494398824449	https://wokwi.com/projects/459662494398824449
	252.42
	378
	246.74
https://wokwi.com/projects/459922074756054017	99.597
https://wokwi.com/projects/459910654907682817	112.31
https://wokwi.com/projects/459922578504642561	153,187
https://wokwi.com/projects/459923650855221249	507.50
	290
https://wokwi.com/projects/459920975143578625	255.887
https://wokwi.com/projects/459924208457656321	https://wokwi.com/projects/459924208457656321
	290.103

o Boot and Idle)	175,964.4 seconds
	311508 seconds
	241,563 s
	278449.2
	168424
	243106.337 s
	162,400 s/45.1 hours/1.9 days (no improvement of 4 seconds)
	4.4585
	247 681
	319,636
	272598
	207642
	223776
	240434.72
	149935.4
	217607
	216122.9643
	231851,19
	138 708
	211230s
	189409
	273883
	354,958 s
	254960
	220536 s
	149935.4
	251650
	227813
	250622.10
	152893.7
	109673
	338,800 s
	149476
	73336.32
	45300
	176500
	271592
	316009,51
	184896
	267496.175
	182178
	224700
	232971
	2.803
	861358s
	94,613
	419967
	179170
	175238.48
	2.836996
	297240

	133.25 mJ (for the majority of the cycles)
	781.2 mJ
	350.45
	233.74
	269.6
	79.619 J
nts)	246 mJ
	193.168
	Assuming a 60 second sleep phase, reducing expensive WiFi initialization phase, €
	67.645
	279.920
	238.17
	220,08
	205.8 (equivalent 4s cycle using avg power after improvements)
	3956.9
	251.9116
	146.44 mJ
	229,12
	471.194
	580.91mJ
	188.6265
	101,54
	1390,2237 mJ
	121.414
	Energy per cycle = 3788.95 mJ (increased) Average power = 47.24 mW (reduced) Li
	305.15mJ
	218 mJ
	After the improvement, the energy of one cycle is not constant anymore, because
	10742 mJ (but the "cycle" is 60 times longer); please see report
	163.48 mJ
	255,9692 mJ
	207.93 mJ
	193.06
	2012.1895
	1136 mJ
	126.49
	197.401
	114,285793543382
	360.8724228
	132.972000
	As said in the report, in the case of event-driven transmission (impossible to test
	134.75
	138.8 mJ
	194.03
	85.577
	112.20
	~50 mJ
	308.13
	160
	196.387
	287.533

each cycle consumes: 2872.00 mJ but lifetime of the node is increased of about 35% compared to shorter

lifetime = 108.33 h (increased)

the node can operate in two different modes. If the room is dark or motion is detected, the node sends

on Wokwi due to the randomness of the process itself) with a transmission probability equal to 0.1 the ei

deep sleep

the message and the cycle energy is about 295.7 mJ, which is very close to the baseline. If the room is br

energy consumption would be 0.103670 J.

ight and no motion is detected, the node skips Wi-Fi and transmission and sleeps for a much longer time

(18.5 s), so the energy of that single cycle becomes higher, about 852.2 mJ. This does not mean the impr

ovement is worse. The quiet cycle covers a much longer time interval, so the important metric is not only

/ the energy per cycle, but the average power over time. In fact, by skipping the expensive Wi-Fi setup in |

bright/no-motion conditions, the improved node reduces the average power and increases the estimate

d battery lifetime. So the improvement is effective even if the energy of one quiet cycle is larger.

153	3/30/26 15:19:18	3/30/26 15:30:53	10846339@polimi.it
154	3/30/26 15:14:01	3/30/26 15:31:47	10773779@polimi.it
155	3/30/26 15:23:02	3/30/26 15:35:54	11135422@polimi.it
156	3/30/26 15:30:49	3/30/26 15:35:57	10832525@polimi.it
157	3/30/26 15:32:21	3/30/26 15:36:48	10764548@polimi.it
158	3/30/26 15:28:50	3/30/26 15:37:50	11140947@polimi.it
159	3/30/26 15:06:29	3/30/26 15:43:56	10903183@polimi.it
160	3/30/26 15:24:27	3/30/26 15:47:38	10806788@polimi.it
161	3/30/26 14:56:17	3/30/26 15:59:39	11102420@polimi.it
162	3/30/26 15:55:26	3/30/26 16:01:01	10782022@polimi.it
163	3/30/26 15:56:57	3/30/26 16:05:27	11161809@polimi.it
164	3/30/26 16:01:07	3/30/26 16:06:19	11164465@polimi.it
165	3/30/26 16:05:22	3/30/26 16:15:33	10889590@polimi.it
166	3/30/26 16:07:15	3/30/26 16:15:55	10773372@polimi.it
167	3/30/26 15:59:45	3/30/26 16:26:28	11039171@polimi.it
168	3/30/26 16:19:52	3/30/26 16:31:14	11154359@polimi.it
169	3/30/26 16:43:15	3/30/26 16:46:30	10861416@polimi.it
170	3/30/26 16:45:31	3/30/26 16:52:13	11032539@polimi.it
171	3/30/26 17:05:33	3/30/26 17:11:58	10727881@polimi.it
172	3/30/26 17:18:11	3/30/26 17:21:52	10808527@polimi.it
173	3/30/26 17:25:48	3/30/26 17:28:47	10724267@polimi.it
174	3/30/26 17:27:16	3/30/26 17:31:11	10888593@polimi.it
175	3/30/26 17:43:23	3/30/26 17:48:00	10810440@polimi.it
176	3/30/26 17:50:24	3/30/26 17:54:58	10704990@polimi.it
177	3/30/26 17:29:15	3/30/26 17:58:11	10917333@polimi.it
178	3/30/26 17:42:47	3/30/26 17:59:20	10841085@polimi.it
179	3/30/26 18:07:59	3/30/26 18:11:20	11175090@polimi.it
180	3/30/26 18:05:06	3/30/26 18:18:30	10778484@polimi.it
181	3/30/26 17:49:34	3/30/26 18:30:55	10821199@polimi.it
182	3/30/26 18:37:46	3/30/26 18:44:38	11117655@polimi.it
183	3/30/26 18:37:17	3/30/26 18:46:37	11046642@polimi.it
184	3/30/26 18:22:27	3/30/26 18:59:24	11026049@polimi.it
185	3/30/26 19:01:32	3/30/26 19:06:15	10710384@polimi.it
186	3/30/26 15:58:10	3/30/26 19:13:35	11057250@polimi.it
187	3/30/26 19:12:21	3/30/26 19:14:47	10861979@polimi.it
188	3/30/26 19:09:33	3/30/26 19:15:50	10772713@polimi.it
189	3/30/26 19:00:39	3/30/26 19:15:58	10778074@polimi.it
190	3/30/26 19:21:09	3/30/26 19:56:48	10938858@polimi.it
191	3/30/26 20:05:27	3/30/26 20:10:53	10768882@polimi.it
192	3/30/26 15:46:57	3/30/26 20:14:59	10812603@polimi.it
193	3/30/26 20:14:55	3/30/26 20:18:15	10816921@polimi.it
194	3/30/26 20:02:44	3/30/26 20:18:42	11123116@polimi.it
195	3/30/26 19:43:38	3/30/26 20:22:53	10712273@polimi.it
196	3/30/26 20:11:24	3/30/26 20:25:29	10774792@polimi.it
197	3/30/26 20:07:17	3/30/26 20:33:58	10823873@polimi.it
198	3/30/26 20:25:39	3/30/26 20:38:27	10772022@polimi.it
199	3/30/26 20:23:09	3/30/26 20:45:13	10709837@polimi.it
200	3/30/26 20:42:37	3/30/26 20:55:09	11063739@polimi.it
201	3/30/26 20:42:14	3/30/26 21:07:51	10712008@polimi.it
202	3/30/26 16:59:49	3/30/26 21:10:57	10855966@polimi.it
203	3/30/26 21:04:33	3/30/26 21:11:24	11162444@polimi.it

Andrea Gianfelice
Francesco Marchi
Luca Pezzoli
Michele Tavecchio
Andrea Callari
Ermir Balliu
Matteo Ferrari
Gabriele Maria Michieletto
Matteo Ghiaroni
Stefano Longhi
Eric Lombardo Facciale
Tommaso Ernst
Alice Berta
Simone Perugini
Kristi Duro
Wu Tianqi
Susanna Lo Cascio
Pan Xiaohan
Andrea Luca Miele
Manoueil Michael Halim Riad Hanna
Alessandra Romano
Leonardo Di Giovanni
Samuele Grisoni
Riccardo Baiardi
Edoardo Bidini
Federico Fasola
Anasuya Satapathy
Eliseo Paoletti
Golsa Shokri
Fang Yuhang
Shaghayegh Vesali
Fernando Javier Barcelata Fuentes
Lorenzo Francesco Luigi Minorini
Arzu Yildiz
Alessandro Nipoti
Duilio Diego Giorgetti
Enrico Grillo
Alessandro Ferrando
Ronald Cammaroto
Andrea Sau
Federico Trubian
Lucía Fernández Giner
Matteo Vicentini
Alessandro Andrea Maria Costa
Alberto Pini
Davide Rutigliano
Carmen Giaccotto
Sajjad Shaffaf
Alessia Maria Di Capua
Marco Puddu
Yihang Ji

Andrea Gianfelice	10846339
Francesco Marchi	10773779
Luca Pezzoli	11135422
Michele Tavecchio	10832525
Andrea Callari	10764548
Ermir Balliu	11140947
Matteo Ferrari	10903183
Gabriele Michieletto	10806788
Matteo Ghiaroni	11102420
Stefano Longhi	10782022
Eric Lombardo Facciale	11161809
Tommaso Ernst	11164465
Alice Berta	10889590
Simone Perugini	10773372
Kristi Duro	11039171
Tianqi Wu	11154359
Susanna Lo Cascio	10861416
Xiaohan Pan	11032539
Andrea Luca Miele	10727881
Manoueil Michael Halim Riad Hanna	10808527
Alessandra Romano	10724267
leonardo di giovanni	10888593
Samuele Grisoni	10810440
Riccardo Baiardi	10704990
Edoardo Bidini	10917333
Federico Fasola	10841085
Anasuya Satapathy	11175090
Eliseo Paoletti	10778484
Golsa Shokri	10821199
yuhang fang	11117655
Shaghayegh Vesali	11046642
Fernando Javier Barcelata Fuentes	11026049
Lorenzo Francesco Luigi Minorini	10710384
Arzu YILDIZ	11057250
Alessandro Nipoti	10861979
Grillo	10778074
Enrico Grillo	10778074
Alessandro Ferrando	10938858
Ronald Cammaroto	10768882
Andrea Sau	10812603
Federico Trubian	10816921
Fernandez Giner	11123116
Matteo Vicentini	10712273
Alessandro Andrea Maria Costa	10774792
Alberto Pini	10823873
Davide Rutigliano	10772022
Carmen Giaccotto	10709837
sajjad shaffaf	11063739
Alessia Di Capua	10712008
Marco Puddu	10855966
Yihang Ji	11162444

Juan Marco Patagoc
Mattia Rainieri
Alessandro Sola
Simone Checchia
Marco Scianna

Chiara Fantini
Beatrice Spazzadeschi
Filippo Palazzetti
Alessandro Paolo Gianni Callegari
Marco Abbadessa

Arturo Amoretti
Qinglong Huang
Mengshuang Tang
Federico Previtali

Alessio Guarisco

Mattia Oglier Badessi
Marco Fontana
Jiaxun Jiang
Gabriele Nicolò

Aref Akbari

Andrea Venturini
Yaser Enes AKAN
Xiaoyang Guo
Giorgetti
Diego Giorgetti

Daniele Toniolo

Alessandro Modica
Emiliano Finetti
Alessia Franchetti-Rosada
seyed taha mousavi
Francesco Fioretti

10767244

11011342

10777259

10764758

10815900

10846947

10766466

11084799

10809515

10854816

10799588

11151255

11035477

10796043

10795876

10724359

10767413

11021337

10848911

11130192

10721613

11057249

11108477

10772713

10772713

10859228

10822654

11095908

10773289

11060363

10766440

https://wokwi.com/projects/459916833903067137	https://wokwi.com/projects/459924669562749953	360.93
https://wokwi.com/projects/459924669562749953		1224.002
https://github.com/PezzoliLuca/InternetOfThings_challenges		189,24
		314,92
https://wokwi.com/projects/459925205089479681	https://wokwi.com/projects/459924208457656321	371,35
https://wokwi.com/projects/459924208457656321	after improvement	290.103
		305
https://wokwi.com/projects/459298609504022529		316.85
https://wokwi.com/projects/459815152887635969		533.325
		244,182
		92.86 mJ
https://wokwi.com/projects/459927018220370945		150.10
https://wokwi.com/projects/459926821148858369	(original code) - https://wokwi.com/projects/459926155961468929	650.4 mJ
https://wokwi.com/projects/459926155961468929		Etot = 0,249 J, Ein = 0,12
https://wokwi.com/projects/459763176147178497		317.9 mJ
FOR Question1 : https://wokwi.com/projects/459870922215764993		107.073
		144,074
https://wokwi.com/projects/459546363416424449		244.3
https://wokwi.com/projects/459911929475212289		284.5
		313.347
		148.05
https://wokwi.com/projects/459720542901937153		165.19
		421.56mJ
		326,58
https://wokwi.com/projects/459922728525497345	https://wokwi.com/projects/459922728525497345	301.0
		315,009
Baseline: https://wokwi.com/projects/459638342239245313	Improvement: https://wokwi.com/projects/459934987017053185	4004.27 mJ
https://wokwi.com/projects/459934987017053185	https://wokwi.com/projects/459359124285689857	308.413 mJ
https://wokwi.com/projects/459359124285689857		438.38
https://wokwi.com/projects/459867140493641729	https://wokwi.com/projects/459867140493641729	191.17
https://github.com/sh-ves/IoT-challenge-1.git		340.8552 mJ
		367.8
none		313.23
https://wokwi.com/projects/459938365829479425		0.71 mJ
		688.582
https://wokwi.com/projects/459286612774048769		289.74 (0.161, 0.102, 13
https://wokwi.com/projects/459286612774048769		289.74 (0.161, 0.102, 13
https://wokwi.com/projects/458944400667995137		104.92 mJ
https://wokwi.com/projects/459941529415286785	https://wokwi.com/projects/459941529415286785	373,57
		245
https://wokwi.com/projects/459512468051506177		243.25
https://wokwi.com/projects/458732718711492609		198.7235
improved: https://wokwi.com/projects/459506032228854785	baseline: https://wokwi.com/projects/459506032228854785	238.8
		420.37
https://wokwi.com/projects/459940680630614017		291.69
		220.97
		315,06
https://wokwi.com/projects/459945245321779201		408 (including boot time)
		198.15
		283.24
		350.630

	213408
	57010.872
	230546
	187000
	290779,67
	297240
	238831.59
	238870
	101419,255
	201461
	267942
	272880
	174537 s
24 J, E sens = 0,000160 J, E tx = 0,000955 J, E deep	214046,378 s
	181493.5 seconds
	264792.1
	250 500
	317168
	248263.6
	209965
	296298.198
	591039
	183129
	4,7
	230799.55
	214868,1031
	40699.6 seconds
	246240 seconds
	221600
	113625
	239201 s
	244410.17
	202105
	12172000
	120280.05
2)	198000
2)	198000
	247857.323 s
	206640
	57360
	194600
	191477.8095246027
	215302
	240242
	205005.3
	227186.49
	276905
e)	218408
	135877.03
	139461
	253617.142

229.53 (on average)
205.928
252,31 (One of the improvements is the Deep sleep augmented, so one cycle con
68681,74 (25 min cycle)
326,56
287.533
187
207.35
360,465
122,851 + 118,562*p (p = probability of sensors state change). With p=0,05, E=128
294.2 mJ increased with respect to baseline estimation having a longer deep sleep
90.41
333 mJ
$E_{new} \approx 0.247875 - 0.00467 = 0.2432 \text{ J}$
190.3 mJ
93.360
112,396 (second optimization in Report)
241.2
446.5
The estimated consumption is approximately 193.2 mJ (in case of unchanged stai
115.34 mJ
40
253.098mJ
217,35
237.9
184,429 mJ in case of avoidable message (the message is not sent). 310,768 mJ in
2156.70 mJ
128.17 mJ (TX Cycle) / 0.816 mJ (Skip TX Cycle) (No Deep Sleep counted)
348.318
46.42
257.16013 mJ
1493.95
Between 179.03mJ and 312.98mJ based on motion detection frequency
0.11 mJ
656.935
302
302
In a scenario with 1 minute of continuous motion the improved version consume
373,57 (mixed interrupt-driven approach)
244,6
119.49
149.6856 mJ
207.34
320.43
184
160.06
198,2
344 mJ
102.48
55.31 J
224.947 mJ ~ 350.238 mJ

sumes more than before but it is longer)

3,779.

o time when no motion is detected

es) or approximately 313.3 mJ (in case of different states). Thanks to Optimization 2 (Conditional Transm

case of message transmission

is 2.74 J vs 4.51 J of the original version

ission) , if the environment is unchanged, the node skips the Wi-Fi initialization: the active cycle consump

tion drops from approximately 166 mJ to approximately 46 mJ. Added to the 147.2 mJ of the standard Di

Deep Sleep, this brings the optimized cycle to 193.2 mJ. If a real change is detected, the cycle performs the

full transmission, consuming 313.3 mJ. This estimation is strictly limited to the "single cycle" metric. The c

Other architectural optimizations discussed in the report (Event-Driven Wake-up and Dynamic Deep Sleep

) cannot be quantified within a single approximately 3.5s cycle, as their massive energy impact derives from

om the total elimination of redundant wake-ups over prolonged use and the temporal extension of the di

deep sleep phases.

204	3/30/26 21:16:06	3/30/26 21:19:14	11156535@polimi.it
205	3/30/26 21:12:42	3/30/26 21:21:00	10777399@polimi.it
206	3/30/26 21:17:15	3/30/26 21:21:20	10862592@polimi.it
207	3/30/26 18:21:47	3/30/26 21:25:12	10863400@polimi.it
208	3/30/26 21:19:42	3/30/26 21:25:25	10742323@polimi.it
209	3/30/26 20:33:51	3/30/26 21:36:51	10793255@polimi.it
210	3/30/26 21:34:04	3/30/26 21:39:49	10861756@polimi.it
211	3/30/26 21:35:45	3/30/26 21:40:37	10832147@polimi.it
212	3/30/26 21:43:23	3/30/26 21:48:58	10717254@polimi.it
213	3/30/26 21:46:06	3/30/26 21:52:10	11191225@polimi.it
214	3/30/26 21:47:48	3/30/26 21:57:47	11087290@polimi.it
215	3/30/26 21:45:42	3/30/26 21:59:58	10914222@polimi.it
216	3/30/26 21:56:09	3/30/26 22:00:22	10738139@polimi.it
217	3/30/26 22:05:10	3/30/26 22:09:29	11059833@polimi.it
218	3/30/26 21:30:52	3/30/26 22:28:20	11119582@polimi.it
219	3/30/26 22:18:45	3/30/26 22:33:55	11107611@polimi.it
220	3/30/26 22:27:40	3/30/26 22:34:49	11035971@polimi.it
221	3/30/26 22:24:56	3/30/26 22:38:39	10852883@polimi.it
222	3/30/26 22:37:34	3/30/26 22:41:08	10828129@polimi.it
223	3/30/26 21:57:52	3/30/26 22:52:35	10863960@polimi.it
224	3/30/26 22:31:49	3/30/26 22:57:59	11036293@polimi.it
225	3/30/26 22:57:51	3/30/26 23:00:57	11140550@polimi.it
226	3/30/26 22:21:53	3/30/26 23:02:15	10799170@polimi.it
227	3/30/26 22:49:36	3/30/26 23:10:32	10778819@polimi.it
228	3/30/26 16:06:28	3/30/26 23:11:32	11119918@polimi.it
229	3/30/26 23:01:13	3/30/26 23:13:59	11010656@polimi.it
230	3/30/26 23:13:23	3/30/26 23:15:57	10607397@polimi.it
231	3/30/26 23:19:17	3/30/26 23:23:57	11145475@polimi.it
232	3/30/26 23:13:11	3/30/26 23:27:45	11154362@polimi.it
233	3/30/26 23:27:21	3/30/26 23:31:50	10497245@polimi.it
234	3/30/26 23:29:39	3/30/26 23:36:06	10771308@polimi.it
235	3/30/26 23:37:19	3/30/26 23:44:41	10830365@polimi.it
236	3/30/26 20:26:53	3/30/26 23:46:05	11061927@polimi.it
237	3/30/26 23:44:35	3/30/26 23:47:52	10892569@polimi.it
238	3/30/26 16:16:55	3/30/26 23:49:27	10810258@polimi.it
239	3/30/26 21:56:32	3/30/26 23:49:46	11114435@polimi.it
240	3/30/26 22:49:24	3/30/26 23:51:12	11036887@polimi.it
241	3/30/26 23:33:48	3/30/26 23:53:02	11121758@polimi.it
242	3/30/26 23:51:36	3/30/26 23:53:15	10847803@polimi.it
243	3/30/26 23:48:36	3/30/26 23:55:41	11115207@polimi.it
244	3/30/26 20:12:27	3/30/26 23:55:58	11052807@polimi.it
245	3/30/26 23:51:25	3/30/26 23:56:31	11118429@polimi.it
246	3/30/26 23:34:23	3/30/26 23:58:33	11191868@polimi.it
247	3/31/26 0:00:54	3/31/26 0:02:14	11159736@polimi.it
248	3/30/26 23:59:12	3/31/26 0:02:34	10495016@polimi.it
249	3/31/26 0:00:46	3/31/26 0:05:16	10835836@polimi.it

Torodd Urdshals	
Mario Pesca	
Enrico Mantegazza	
Wang Yuan	
Giacomo Pasturensi	
Flavio Riccobono	
Federico Ciarniello	
Simone Amighini	
Gabriele Clara Di Gioacchino	
Angelica Ortiz Ballesteros	
Han Mengzhen	
Martín Corchuelo Verjano	
Davide Lupo	
Zhanat Argimbayeva	
Amin Sakhaei	
Ning Zhang	
Özgür Efehan Kati	
Francesco Pio Lo Monaco	
Lorenzo Foini	
Simone Rodari	
Amr Ahmed Yehia Elsaid Elmasry	
Hadi El Hajj	
Andrea Rossi	
Lorenzo Pedranzini	
Wang Yunxuan	3/30/26 23:39:04
Ghazal Sepehrirad	
Ivana Lin	
Lorena Lam Diez	
Cheng Keyan	
Mattia Angelo Brunetti	
Matteo Bergagna	3/30/26 23:36:28
Andrea Santaniello	
Haoyu Zhang	
Giacomo Aragnetti	
Jennifer Valeria Feijoo Sandoya	
Mobin Khatib	
Hu Kaipeng	
Razia Jafari	
Maciej Krzysztof Rzewuski	
Juan David Vargas Galindo	
Erdal Yalçın	
Wang Guangyi	
Oscar Per Lindqvist	
Muhammad Hassaan Javed	
Dalila Samr	
Andrea Ziccardi	

Torodd Urdshals	11156535
Mario Pesca	10777399
Mantegazza Enrico	10862592
Yuan Wang	10863400
Giacomo Pasturensi	10742323
Flavio Riccobono	10793255
Federico Ciarniello	10861756
Simone Amighini	10832147
Gabriele Clara Di Gioacchino	10717254
Angelica Ortiz	11191225
Mengzhen Han	11087290
Martin Corchuelo Verjano	10914222
Davide Lupo	10738139
Zhanat Argimbayeva	11059833
Amin Sakhaei	11119582
Ning Zhang	11107611
Özgür Efehan KATI	11035971
Francesco Pio Lo Monaco	10852883
Foini	10828129
Simone Rodari	10863960
Elmasry	11036293
Hadi EL Hajj	11140550
Andrea Rossi	10799170
Lorenzo Pedranzini	10778819
WANG YUNXUAN	11119918
Ghazal Sepehrirad	11010656
Lin	10607397
Lorena Lam	11145475
Keyan Cheng	11154362
Mattia Angelo Brunetti	10497245
Matteo Bergagna	10771308
Andrea Santaniello	10830365
Haoyu Zhang	11061927
Giacomo Aragnetti	10892569
jennifer valeria feijoo sandoya	10810258
Mobin Khatib	11114435
Kaipeng Hu	11036887
RAZIA	11121758
Rzewuski	10847803
Juan David Vargas Galindo	11115207
ERDAL	YALCIN
Guangyi Wang	11118429
Lindqvist	11191868
Muhammad Hassaan Javed	11159736
Dalila Samr	10495016
Andrea Ziccardi	10835836

Osama Atiqi

Andrea Mori

Andrea Pazienza

Daniel Nägele

Shiming Gao

Hamidreza Khodadadi

Fabio Agrati

Amirhossein Sharifisadr

Firat Adem BİLGE

Gotta

Gabriele Pignataro

Elsadig

Francesco Sarra

Hedieh Raeisi

Sofia Merino

Haonan Pang

NA

Ali Edareh Heidarabadi

Cristian David Sandoval Diaz

10772098
11212726
10716103
11180610
11111794
11057471
10769569
11121331
11035164
10774461
10766370
11061031
10797304
11044024
11156291
11028952
NA
11036028
11097500

https://wokwi.com/projects/459656291463453697	305
https://wokwi.com/projects/459666214085621761	378.33
all in the zip file	ETot= 0.206 (idle) + 0.36
	64.85 mJ
https://wokwi.com/projects/459946459333869569	0.0627 J
https://wokwi.com/projects/459947939445227521	95.47
https://wokwi.com/projects/459129070602724353	https://wokwi.com/projects/459129070602724353
	92,432
	364.549
baseline: https://wokwi.com/projects/459833599042593793 ; improve	82.302
https://github.com/lotChallenges/Challenge-1.git	257.1917mJ
baseline version: https://wokwi.com/projects/459841203205905409 ; c	263.37
https://wokwi.com/projects/459795258959690753 ; https://wokwi.com/projects/459795258959690753	271.0577
	422.297
	312.9
https://wokwi.com/projects/459854066148355073	309.69
	94.9731 mJ
	333.77
https://wokwi.com/projects/459939128706410497	https://wokwi.com/projects/459939128706410497
	297,810
	277.715
	134.774
https://wokwi.com/projects/459849251343575041	354.25
	207,06
	246.78
https://wokwi.com/projects/459952703197837313	283.52
https://wokwi.com/projects/459932716055713793	25.11
https://wokwi.com/projects/459952910802288641	96
https://wokwi.com/projects/459846663925265409	368,65
	254.19
https://wokwi.com/projects/459954773720430593 (original) https://wokwi.com/projects/459954773720430593	94.321
https://wokwi.com/projects/459949437908754433	955.63
	100,025
	135.2131
	0.272, 131.363, 146.486
https://wokwi.com/projects/459949751488587777	259.877
	89,644
	342.20
https://wokwi.com/projects/459116569983647745	254
	249.136
https://wokwi.com/projects/459950520192393217	0
https://wokwi.com/projects/459946928957574145	57.82
	55.66 mJ
https://wokwi.com/projects/459921370275394561	156.526
	197.7
https://polimi365-my.sharepoint.com/:u:/g/personal/11159736_polin@polimi.it	695.86
	183201
	0,238J

227736
257980
0.1 (sensing) + 125.9 (wifi on) + 0.171 (tx)+ 2.32 (wi
250943
160767 s
829.440
1.079
1,167
254468.548
202347
56.161 h
308983
209345.22
214563
252970
248824
397,828
149292
240,013 s
235284
229805
232325.8207
68900
210569
181803864
1995000
178200
255445
194,650
362623
92,038
211961
235764
206528.59
183210
232360
246600
300767
87451
0
316124
302,400
117737.89
218880
117,408
34172
3 days 7 hours

302
273.99
$E_{Tot} = 0.168 + 0.163 + 54.424 + 0.059 + 1.301 + 215.166 = 271.281 \text{ [mJ]}$
25.13 mJ
0.48 J/hour
The proposed optimization does not reduce the energy of an individual cycle, but
60,510
Cannot answer, it depends on parameters: see relation.
41.8 (under scenario assumed in the report)
1452.0088 mJ
261.30
133.4739
213.16 mJ
306.6
307.21
670mj(but the we use a method which makes the cycle longer)
258.60
175,615
155.2405
Version 1 (RTC_MEM, best case): 88.029mj - Version 2 (interrupt): see report
The energy consumption per cycle does not change significantly after applying th
229,56
154.47
236.16
Extended Sleep (10 s):28.96mJ, Event-Driven (10% TX):4.84mJ
51 (explanation in pdf)
not done yet
391.51
77.551
923.88
Not calculable deterministically, see PDF
91.4791
0.269013 J
141.4
-
341.46
248
21
0
94.33
55.66 mJ
155.95
-
1.497
not computed
0,0594

t rather the long-term average consumption. Its effectiveness depends on the event frequency. Over a 10

e optimizations, as the transmission phase represents a very small portion of the total energy. However,

) s interval, the periodic approach consumes approximately 955 mJ, while the event-driven approach con

the system lifetime is expected to improve considerably. This is because the enhanced implementation ii

sumes about 456 mJ in the absence of events.

Introduces conditional transmission, resulting in many cycles where no data is sent when no significant ch

anges in motion or illumination are detected. In such cases, the transmission energy becomes effectively

zero, reducing the average energy consumption over time. Consequently, although the per-cycle energy

τ remains nearly unchanged, the reduction in transmission frequency leads to a noticeable extension in tl

re overall battery lifetime.

student_id	student_name	section_1	section_1	section_1	section_1	section_2
11137282-11116374	Yehia Abdallah - Amr Abdelhalir	IoT Motio	1.5	1.5	Baseline fu	Energy Co
10783671	Raul Agolli	IoT Motio	1.1	1.5	Baseline fu	Energy Co
10844134	Edoardo Albini	IoT Motio	1.5	1.5	Baseline n	Energy Co
11140550	Hadi El Hajj	IoT Motio	1.0	1.5	Baseline fu	Energy Co
10832147	Simone Amighini	IoT Motio	1.2	1.5	Most base	Energy Co
10764083	Marco Apollonio - Francesco Az	IoT Motio	1.2	1.5	The baseli	Energy Co
10892569	Giacomo Aragnetti	IoT Motio	1.2	1.5	Baseline n	Energy Co
10922631	Francesco Armuzzi	IoT Motio	1.5	1.5	Baseline n	Energy Co
10708185	Andrea Ardu - Laura Maria Bos	IoT Motio	1.25	1.5	Baseline n	Energy Co
10806605	Bruno Arena - Matteo Cavalleri	IoT Motio	1.5	1.5	Baseline n	Energy Co
11059833	Zhanat Argimbayeva	IoT Motio	1.3	1.5	Core node	Energy Co
11019965	Yasin Asgari - Armin Mardoukh	IoT Motio	1.50	1.5	Baseline n	Energy Co
11061975	Alperen Aydin - Kayra Uckilinc	IoT Motio	1.5	1.5	Baseline n	Energy Co
10704990	Riccardo Baiardi	IoT Motio	1.30	1.5	Core node	Energy Co
11140947	Ermir Balliu - Marco Scianna	IoT Motio	1.50	1.5	Baseline n	Energy Co
11026049	Fernando Javier Barcelata Fuer	IoT Motio	1.4	1.5	Baseline n	Energy Co
11068760	Giulia Barresi + Cesar Adrian Ca	IoT Motio	1.50	1.5	Baseline n	Energy Co
10829488	Gabriele Benedetti + Giuseppe	IoT Motio	1.5	1.5	Baseline n	Energy Co
10771308	Matteo Bergagna	IoT Motio	1.50	1.5	Baseline n	Energy Co
10828027	Rebecca Besana + Noemi Fazzir	IoT Motio	1.2	1.5	Baseline fu	Energy Co
10716383	Simone Bianchessi	IoT Motio	1.0	1.5	Core basel	Energy Co
10917333	Bidini Edoardo - Oglier Badessi	IoT Motio	1.50	1.5	Baseline n	Energy Co
10677341	Alberto Biffi	IoT Motio	1.50	1.5	Baseline n	Energy Co
11030334	Yiran BI + Shuai ZOU	IoT Motio	1.50	1.5	Baseline n	Energy Co
10773859	Mattia Brianti - Alex Hathaway	IoT Motio	1.50	1.5	Baseline n	Energy Co
10497245	Mattia Angelo Brunetti	IoT Motio	1.50	1.5	Baseline n	Energy Co
11166508	CAI CHUANHAO	IoT Motio	1.10	1.5	Baseline fu	Energy Co
10774521	Pietro Callandrone - Lorenzo Bi	IoT Motio	1.50	1.5	Baseline n	Energy Co
10764548	Andrea Callari - Simone Checch	IoT Motio	1.5	1.5	Baseline n	Energy Co
10768882	Ronald Cammaroto	IoT Motio	1.35	1.5	Baseline n	Energy Co
10853745	Caon Mattia - Bertazzoli Cova Mo	T Motio	1.00	1.5	The baseli	Energy Co
10797131	Morena Caterina Cardenas Est	IoT Motio	1.5	1.5	Baseline n	Energy Co
10790205	Marco Carminati - Om Mohan	IoT Motio	1.50	1.5	Baseline n	Energy Co
10910881	Claudia Carraro - Chiara Cavallc	IoT Motio	1.10	1.5	Baseline fu	Energy Co
10796163	Stefano Castiglione - Samuele C	IoT Motio	1.50	1.5	Baseline n	Energy Co
11154362	Keyan Cheng	IoT Motio	1.50	1.5	Baseline n	Energy Co
10861756	Federico Ciarniello - Andrea Mc	IoT Motio	1.50	1.5	Baseline n	Energy Co
10666869	Yuri Cinesi Ferrari	IoT Motio	1.5	1.5	Baseline n	Energy Co
10857924	Marco Coviello	IoT Motio	1.1	1.5	Baseline n	Energy Co
10850482	Alessandro Cirillo - Jaime Iriarte	IoT Motio	1.50	1.5	Baseline n	Energy Co
10717254	Gabriele Clara Di Gioacchino - A	IoT Motio	1.5	1.5	Baseline n	Energy Co
10914222	Martin Corchuelo Verjano - Hai	IoT Motio	1.5	1.5	Baseline n	Energy Co
10774792	Alessandro Andrea Maria Costa	IoT Motio	1.10	1.5	Baseline n	Energy Co
10838600	Daniele Crippa	IoT Motio	1.10	1.5	Baseline fu	Energy Co
10975089	Alessandro Cupini	IoT Motio	1.50	1.5	Baseline n	Energy Co
11006775	Osman Bugra DAGLI - Erdem Cc	IoT Motio	1.50	1.5	Baseline n	Energy Co
10766478	Dai Run Jie - Simone	IoT Motio	1.10	1.5	Baseline n	Energy Co
10711043	Marco Stefano De Bernardi - Fe	IoT Motio	0.90	1.5	The baseli	Energy Co
11101495	Riccardo De Giacomo - Fabrizio	IoT Motio	1.50	1.5	Delivered s	Energy Co
10683374	Zarlene Justrem De Mesa - Gugl	IoT Motio	1.20	1.5	Baseline fu	Energy Co

section_2	section_2	section_2	section_3	section_3	section_3	section_3	section_4	section_4	section_4
0.75	1.0	Cycle ener	Comment	2.0	2.5	TX power	Deliverable	0.75	1.0
0.9	1.0	Q2 is most	Comment	1.1	2.5	`better.ino	Deliverable	0.8	1.0
0.85	1.0	The section	Comment	2.0	2.5	The impro	Deliverable	0.9	1.0
0.55	1.0	Cycle ener	Comment	0.75	2.5	The impro	Deliverable	0.6	1.0
0.85	1.0	Q2 is num	Comment	1.1	2.5	The report	Deliverable	0.8	1.0
0.8	1.0	Q2 include	Comment	1.4	2.5	The impro	Deliverable	0.75	1.0
0.8	1.0	Q2 is most	Comment	1.5	2.5	The impro	Deliverable	0.9	1.0
0.8	1.0	Cycle ener	Comment	1.1	2.5	The report	Deliverable	0.8	1.0
0.75	1.0	Q2 is num	Comment	1.7	2.5	The submi	Deliverable	0.6	1.0
0.85	1.0	Q2 values	Comment	1.7	2.5	The impro	Deliverable	0.8	1.0
0.6	1.0	Q2 gives p	Comment	1.4	2.5	The deliver	Deliverable	0.8	1.0
0.75	1.0	Power stat	Comment	1.40	2.5	The impro	Deliverable	0.80	1.0
0.8	1.0	Q2 is num	Comment	1.2	2.5	The report	Deliverable	0.7	1.0
0.70	1.0	Cycle ener	Comment	1.30	2.5	The sketch	Deliverable	0.70	1.0
0.65	1.0	Cycle ener	Comment	1.50	2.5	The impro	Deliverable	0.75	1.0
0.8	1.0	Q2 is num	Comment	1.4	2.5	The impro	Deliverable	0.8	1.0
0.80	1.0	Per-state p	Comment	1.50	2.5	The impro	Deliverable	0.80	1.0
0.8	1.0	Q2 uses pl	Comment	1.6	2.5	The impro	Deliverable	0.8	1.0
0.90	1.0	Q2 is num	Comment	2.00	2.5	The impro	Deliverable	0.90	1.0
0.8	1.0	The Q2 for	Comment	1.2	2.5	The report	Deliverable	0.8	1.0
0.5	1.0	Power, cyc	Comment	1.8	2.5	The impro	Deliverable	0.7	1.0
0.85	1.0	Q2 is num	Comment	2.00	2.5	The impro	Deliverable	0.90	1.0
0.65	1.0	Q2 uses C	Comment	1.70	2.5	The optim	Deliverable	0.90	1.0
0.70	1.0	The Q2 nu	Comment	0.70	2.5	The report	Deliverable	0.70	1.0
0.75	1.0	State pow	Comment	1.40	2.5	The impro	Deliverable	0.70	1.0
0.70	1.0	Cycle ener	Comment	1.00	2.5	The report	Deliverable	0.80	1.0
0.60	1.0	The submi	Comment	1.50	2.5	The event-	Deliverable	0.80	1.0
0.90	1.0	The submi	Comment	1.60	2.5	The impro	Deliverable	0.90	1.0
0.85	1.0	Q2 is tech	Comment	2.0	2.5	The optim	Deliverable	0.85	1.0
0.60	1.0	Q2 reports	Comment	1.20	2.5	The impro	Deliverable	0.80	1.0
0.85	1.0	The Q2 sta	Comment	1.50	2.5	The report	Deliverable	0.75	1.0
0.55	1.0	The report	Comment	1.70	2.5	The optim	Deliverable	0.80	1.0
0.60	1.0	Q2 uses p	Comment	1.25	2.5	The impro	Deliverable	0.70	1.0
0.65	1.0	Q2 reports	Comment	1.20	2.5	The report	Deliverable	0.65	1.0
0.85	1.0	The per-st	Comment	1.50	2.5	The impro	Deliverable	0.90	1.0
0.55	1.0	The report	Comment	1.45	2.5	The impro	Deliverable	0.75	1.0
0.90	1.0	Per-state p	Comment	1.80	2.5	The optim	Deliverable	0.90	1.0
0.85	1.0	Q2 is num	Comment	2.1	2.5	The impro	Deliverable	0.9	1.0
0.65	1.0	Cycle ener	Comment	1.6	2.5	The report	Deliverable	0.7	1.0
0.70	1.0	The report	Comment	1.80	2.5	The impro	Deliverable	0.75	1.0
0.85	1.0	Q2 is well	Comment	1.8	2.5	The impro	Deliverable	0.9	1.0
0.85	1.0	Q2 is num	Comment	1.5	2.5	The impro	Deliverable	0.9	1.0
0.85	1.0	Q2 is deta	Comment	2.10	2.5	Two real in	Deliverable	0.85	1.0
0.60	1.0	Q2 reports	Comment	1.00	2.5	The upgra	Deliverable	0.70	1.0
0.45	1.0	Cycle ener	Comment	0.80	2.5	The report	Deliverable	0.60	1.0
0.65	1.0	Q2 reports	Comment	1.10	2.5	The submi	Deliverable	0.80	1.0
0.65	1.0	Q2 reports	Comment	0.90	2.5	The report	Deliverable	0.75	1.0
0.60	1.0	Q2 reports	Comment	0.80	2.5	The report	Deliverable	0.70	1.0
0.55	1.0	Question 2	Comment	1.25	2.5	The deliver	Deliverable	0.70	1.0
0.75	1.0	Q2 include	Comment	1.50	2.5	The submi	Deliverable	0.75	1.0

section_4_total_score total_max deduction final_comment

Report and	5.0	6.0	-0.25	Section 1	Correct baseline submission with a meaningful but only partial
Report and	3.9	6.0	-0.4	Section 1	Solid baseline submission, but the updated Section 1 rule and 1
Files and c	5.25	6.0	-0.15	Section 1	Strong baseline implementation and a meaningful optimized v
The baseli	2.9	6.0	-0.50	Section 1	The submission shows the basic baseline node and correct X/Y
Files and r	3.95	6.0	-0.30	Section 1	Mostly correct baseline submission with a credible Q2 analysis,
The PDF ar	4.15	6.0	-0.30	Section 1	Good baseline implementation with a real improved prototype
Report, co	4.4	6.0	-0.30	Section 1	Good baseline submission with a real improved implementation
The baseli	4.2	6.0	-0.10	Section 1	Correct and well-documented baseline submission, with most l
Report qu	4.3	6.0	-0.25	Section 1	Strong report and a meaningful interrupt-driven idea, but the c
PDF, code,	4.85	6.0	-0.15	Section 1	Strong baseline submission with two important implemented c
Report, co	4.1	6.0	-0.20	Section 1	Mostly correct core implementation, but the grade is reduced l
The PDF ar	4.45	6.0	-0.25	Section 1	Compliant baseline submission with a meaningful but only par
The baseli	4.2	6.0	-0.20	Section 1	Strong baseline submission with a coherent energy analysis, bu
Code and i	4.00	6.0	-0.20	Section 1	Correct core sensing and ESP-NOW implementation with the u
Code and i	4.40	6.0	-0.35	Section 1	Technically solid baseline submission with a real but only partia
The PDF pl	4.4	6.0	-0.10	Section 1	Correct baseline submission with a meaningful but only partial
The PDF, b	4.60	6.0	-0.20	Section 1	Compliant baseline submission with a meaningful but only par
Report and	4.7	6.0	-0.2	Section 1	Correct and well-documented baseline submission with a mean
The PDF, b	5.30	6.0	-0.10	Section 1	Strong baseline submission with a meaningful but only partially
The PDF ar	4.0	6.0	-0.3	Section 1	Functional baseline submission with a structured report, but it
The report	4.0	6.0	-0.30	Section 1	Technically substantial baseline submission with a meaningful i
The PDF, b	5.25	6.0	-0.15	Section 1	Strong submission with a fully compliant baseline and a meani
Baseline ar	4.75	6.0	-0.35	Section 1	Strong baseline submission with a meaningful optimized variar
The baseli	3.60	6.0	-0.30	Section 1	Correct baseline submission with the main losses coming from
The report	4.35	6.0	-0.25	Section 1	Strong baseline submission with a meaningful but only partially
The baseli	4.00	6.0	-0.30	Section 1	Clean and compliant baseline submission, but the final grade is
The PDF, d	4.00	6.0	-0.40	Section 1	The submission delivers a working baseline and a meaningful e
The report	4.90	6.0	-0.10	Section 1	Strong and reproducible baseline submission with a measured
Report and	5.2	6.0	-0.15	Section 1	Strong baseline implementation with a measured and meaning
Report, co	3.95	6.0	-0.15	Section 1	Functional baseline submission with a real interrupt-driven imp
The report	4.10	6.0	-0.20	Section 1	Readable submission with a plausible Q2 analysis and a well-m
Baseline/ir	4.55	6.0	-0.45	Section 1	Functional baseline submission with a real improved variant, b
Files are re	4.05	6.0	-0.40	Section 1	Functional baseline submission with correct sensor-before-Wi-l
The PDF is	3.60	6.0	-0.40	Section 1	Readable baseline report with a mostly correct design, but the
The report	4.75	6.0	-0.15	Section 1	Compliant baseline submission with a meaningful conditional-t
The report	4.25	6.0	-0.30	Section 1	Functionally strong baseline submission with correct sensor-be
Report and	5.10	6.0	-0.10	Section 1	Strong and fully compliant baseline submission with sensor-be
The report	5.35	6.0	-0.15	Section 1	Strong and fully compliant baseline submission with a real imp
Baseline fil	4.05	6.0	-0.40	Section 1	Baseline submission is mostly complete, but the updated sens
The PDF ar	4.75	6.0	-0.30	Section 1	Compliant baseline submission with correct sensor-before-Wi-l
The report	5.05	6.0	-0.15	Section 1	Strong and fully compliant baseline submission with a meaning
The report	4.75	6.0	-0.15	Section 1	Strong and compliant baseline submission with correct sensor-
The PDF ar	4.90	6.0	-0.40	Section 1	Strong report and solid baseline / improved code. The main iss
The PDF ar	3.40	6.0	-0.40	Section 1	Functional baseline with a real but limited improved variant. Th
The PDF, s	3.35	6.0	-0.55	Section 1	Functionally correct baseline node, but the Q2 energy model is
The PDF ar	4.05	6.0	-0.20	Section 1	Complete and compliant baseline functionality, but Q2 is only j
The baseli	3.40	6.0	-0.40	Section 1	The baseline submission is functional and documented, but th
The PDF, b	3.00	6.0	-0.60	Section 1	The baseline node is close to complete, but the updated senso
PDF, code,	4.00	6.0	-0.45	Section 1	Functionally strong submission with correct sensor-before-Wi-l
Report and	4.20	6.0	-0.30	Section 1	The submission shows a solid sensor-before-Wi-Fi implementa

rating the strongest expected low-energy design choices.

evidence packaging in Q2.

remains a report-only proposal instead of an implemented improved design.

n instead of a measured improved implementation.

and a Q3 design that is still timer-driven and not lower-power on TX than the baseline.

nd a Q3 design that reduces transmission frequency without applying the expected TX-power/payload/ir
g a measured improved implementation.

/reproducibility artifacts needed for a cleaner grade.

energy estimate, and the unused `motionChanged` branch logic.

ayload-plus-interrupt low-power design.

id the absence of the expected TX-power, payload-size, and interrupt-driven low-power improvements.

ved implementation artifacts.

he incomplete Q2 state methodology, and a Q3 improvement that is only partially aligned with the expected

uestion 3 missing payload reduction and PIR-interrupt wake-up.

analysis that implements PIR-interrupt wake-up but not TX-power/payload reduction or a fully measured
oes not deliver the strongest expected low-power techniques.

nd from Q3 not delivering payload reduction or PIR-interrupt wake-up.

ack of raw Q2 timing traceability.

being delivered as a reproducible improved implementation.

plementation and by code/report mismatches.

tion 3 and from limited raw timing traceability in Question 2.

ot implementing the expected TX-power and payload reductions or a measured post-improvement energy
oved implementation.

plementing the expected payload-reduction and interrupt-driven wake-up strategy.

ad of delivering a measured improved implementation.

s only partially reproducible, and Question 3 does not package the full expected low-power optimization s

t yet include payload reduction or interrupt-driven wake-up.

f the expected interrupt-driven wake-up design.

mismatch, and the lack of measured post-improvement energy evidence.

id the improved version is not packaged as a reproducible standalone deliverable.

uction plus interrupt-driven wake-up.

at is only partially implemented and only partially remeasured.

s keep the score in the mid range.

plementation that keeps full TX power, full payload size, and timer-based wake-up.

ode mismatch on the actual cycle timing and by Section 3 stopping short of the strongest expected low-p
proxy and from Q3 stopping short of payload reduction and PIR-interrupt wake-up.

om Q3 not also reducing the payload size.

mposition, missing raw artifacts, and the lack of a separate runnable improved implementation.

ate model and a Q3 optimization that reduces transmission frequency but does not also shrink the paylo
cing payload size or using PIR-interrupt wake-up.

roducibility artifacts and from Q3 not implementing the expected payload reduction, relative TX-power r
d raw artifacts / boot estimation, and Q3 missing payload reduction in the delivered optimized sketches.

proved design that does not implement the strongest expected low-power techniques.

evel discussion instead of a measured improved implementation.

stops short of a clearly separated improved implementation with payload reduction and interrupt-drive

only proposal rather than a correct delivered low-power reimplementation.

: because Question 2 is only partially reproducible and Question 3 remains a report-level proposal instead
ates a different baseline than the one submitted. Question 3 shows some real improvements, yet it stops
baseline grading path is not packaged cleanly. Q2 values are plausible yet not fully reproducible from the

interrupt optimizations.

ted low-power design.

ed improved energy workflow.

y workflow.

set as a clean measured improved configuration.

ower implementation choices.

oad or improve TX power relative to the already-low-power baseline.

reduction, and PIR-interrupt wake-up.

n wake-up.

l of a measured improved implementation.

short of the expected TX-power, payload, and interrupt-wake low-power implementation.

default submitted build, and Q3 implements only part of the expected optimization strategy.

10808901	Luca De Nicola	IoT Motio	1.50	1.5 The baseli	Energy Co
10712008	Alessia Di Capua - Francesco Fic	IoT Motio	1.50	1.5 Baseline n	Energy Co
10888593	Leonardo Di Giovanni	IoT Motio	1.10	1.5 The baseli	Energy Co
11053611	Zhenze Ding - Zhaojun Liu	IoT Motio	1.50	1.5 Baseline n	Energy Co
10702601	Angelo Di Rosa - Livia Polverari	IoT Motio	1.50	1.5 Baseline n	Energy Co
10660050	Luca Donato	IoT Motio	1.50	1.5 Baseline n	Energy Co
10799588	Arturo Amoretti - Kristi Duro	IoT Motio	1.50	1.5 Baseline n	Energy Co
11101890	Mohamed Youssef Mohamed `	IoT Motio	0.80	1.5 Baseline n	Energy Co
11036293	Amr Elmasry - Amjed Elsadig	IoT Motio	1.25	1.5 The node i	Energy Co
11164465	Tommaso Ernst - Alessandro Pi	IoT Motio	1.25	1.5 The submi	Energy Co
11078923	Fang Junhua - Chai Ke	IoT Motio	1.50	1.5 Baseline n	Energy Co
11117655	Yuhang Fang	IoT Motio	1.50	1.5 Baseline n	Energy Co
10841085	Federico Fasola - Marco Fontar	IoT Motio	1.50	1.5 Baseline n	Energy Co
10767814	Alessandro Fedele - Dario Cros	IoT Motio	1.50	1.5 Baseline n	Energy Co
10810258	Feijoo Sandoya Jennifer Valeria	IoT Motio	1.10	1.5 Baseline fu	Energy Co
11123116	Lucia Fernandez Giner	IoT Motio	1.10	1.5 Baseline fu	Energy Co
10770813	Giovanni Ferola - Lorenzo Della	IoT Motio	1.10	1.5 Baseline n	Energy Co
10938858	Ferrando Alessandro	IoT Motio	1.50	1.5 Baseline n	Energy Co
10903183	Ferrari Matteo	IoT Motio	1.50	1.5 Baseline re	Energy Co
10771695	Edoardo Figini - Greta Fornacia	IoT Motio	1.10	1.5 Baseline n	Energy Co
10828129	Foini Lorenzo - Gotta Filippo	IoT Motio	1.50	1.5 Baseline n	Energy Co
10831046	Matteo Franken - Lorenzo Chir	IoT Motio	1.50	1.5 Baseline n	Energy Co
10773862	Galbiati Samuele	IoT Motio	0.40	1.5 Only the V	Energy Co
11063702	Dinar Gareev - Yiming Hou	IoT Motio	1.50	1.5 Baseline n	Energy Co
10805688	Giacomo Genco - Valeria Iagulli	IoT Motio	1.15	1.5 Baseline n	Energy Co
11109083	Zihao Geng - Marica Balestra	IoT Motio	1.10	1.5 Baseline n	Energy Co
10797129	Roberta Gennari		1.5	1.5 Full credit: PIR+LDR V	
11102420	Ghiaroni Matteo		1.5	1.5 Full credit: baseline V	
10711947	Eugenia Ghiglione - Andrea Cucchetti		1.5	1.5 Full credit: Wokwi ske	
2662391	Davide Ghisolfi		1.5	1.5 Satisfied: baseline Wc	
10709837	Carmen Giaccotto		1.5	1.5 Full credit: PIR+LDR V	
10846339	Andrea Gianfelice		1.4	1.5 Mostly satisfied: PIR/	
10811640	Simone Giuliano		1.4	1.5 Nearly complete; the	
10732218	Davide Greco		1.5	1.5 Full credit: Wokwi PIF	
10778074	Enrico Grillo		1.5	1.5 Full credit: Wokwi PIF	
10810440	Samuele Grisoni		1.0	1.5 Baseline behavior is n	
10720381	Luca Guerra		1.5	1.5 No deduction: Wokw	
11087290	Mengzhen Han & Shiming Gao		1.3	1.5 Baseline Wokwi node	
11167997	HAN ZHAOHUI		1.3	1.5 Mostly complete, bu	
11110511	Jinxu Hong		1.3	1.5 Points removed: the	
10948935	Rong Huang		1.3	1.5 Baseline Wokwi moti	
11063424	Jialin Hu - Ziyin You		1.3	1.5 Mostly satisfied: the :	
11036887	Hu Kaipeng		1.3	1.5 Baseline Wokwi moti	
11032747	Peng Hu		1.0	1.5 Partial credit: the rep	
10808527	Manoueil Michael Halim Riad Hanna - Fed		1.5	1.5 Full credit: baseline V	
11147720	Alex IORTOMAN		1.3	1.5 Partial credit: the bas	
11051292	Alireza Jahandoost		1.5	1.5 Full credit: baseline V	
11030873	JIANG ZHENGLIN		1.3	1.5 Mostly satisfied: the :	
11019981	JIA_TONGXIN		1.3	1.5 Baseline Wokwi node	
11162444	Yihang Ji		1.3	1.5 Points removed: the	
11121758	Razia Jafari		1.3	1.5 Points removed: the	

0.80	1.0 Q2 reports	Comment	0.90	2.5 The report	Deliverable	0.75	1.0
0.70	1.0 Repeated	Comment	1.50	2.5 A real impr	Deliverable	0.85	1.0
0.60	1.0 The PDF re	Comment	1.00	2.5 The report	Deliverable	0.40	1.0
0.85	1.0 Q2 is num	Comment	1.30	2.5 An improv	Deliverable	0.85	1.0
0.80	1.0 Q2 is num	Comment	1.50	2.5 A real impr	Deliverable	0.75	1.0
0.90	1.0 Q2 is stron	Comment	2.00	2.5 A real impr	Deliverable	1.00	1.0
0.90	1.0 Q2 is stron	Comment	1.70	2.5 A real impr	Deliverable	0.90	1.0
0.30	1.0 A cycle-en	Comment	0.50	2.5 The report	Deliverable	0.60	1.0
0.60	1.0 The report	Comment	1.30	2.5 There are r	Deliverable	0.75	1.0
0.55	1.0 The PDF gi	Comment	1.30	2.5 There is re	Deliverable	0.65	1.0
0.85	1.0 Q2 is techr	Comment	1.60	2.5 A real impr	Deliverable	0.90	1.0
0.75	1.0 The Q2 en	Comment	1.60	2.5 A real impr	Deliverable	0.90	1.0
0.80	1.0 The Q2 mc	Comment	2.00	2.5 A real impr	Deliverable	0.90	1.0
0.85	1.0 The Q2 mc	Comment	1.45	2.5 A real impr	Deliverable	0.90	1.0
0.75	1.0 The Q2 an	Comment	0.90	2.5 The report	Deliverable	0.70	1.0
0.45	1.0 The report	Comment	0.90	2.5 The report	Deliverable	0.70	1.0
0.65	1.0 Q2 is num	Comment	1.40	2.5 The submi	Deliverable	0.80	1.0
0.75	1.0 The Q2 mc	Comment	1.45	2.5 The impro	Deliverable	0.80	1.0
0.75	1.0 The Q2 mc	Comment	1.70	2.5 A real impr	Deliverable	0.70	1.0
0.55	1.0 Q2 uses th	Comment	1.30	2.5 The optim	Deliverable	0.80	1.0
0.85	1.0 Average pr	Comment	1.85	2.5 The impro	Deliverable	0.85	1.0
0.85	1.0 Q2 is well c	Comment	1.35	2.5 The optim	Deliverable	0.80	1.0
0.35	1.0 The PDF re	Comment	1.00	2.5 The report	Deliverable	0.25	1.0
0.75	1.0 The report	Comment	1.70	2.5 The impro	Deliverable	0.85	1.0
0.60	1.0 Q2 include	Comment	1.25	2.5 The submi	Deliverable	0.70	1.0
0.70	1.0 Q2 reports	Comment	1.00	2.5 The report	Deliverable	0.75	1.0
0.9	1.0 Mostly complete: per		2.3	2.5 Good optii	4.7	5.0	Section 2 -
0.9	1.0 Minus 0.1: the energy		2.5	2.5 Full credit:	4.9	5.0	Section 2 -
0.9	1.0 Minor deduction: the		2.3	2.5 Good impr	4.7	5.0	Section 2 -
0.8	1.0 Mostly satisfied: stati		2.0	2.5 Partially sa	4.3	5.0	Section 2 -
0.9	1.0 Mostly complete: per		2.3	2.5 Good optii	4.7	5.0	Section 2 -
0.8	1.0 Good state-by-state e		2.3	2.5 Solid optin	4.5	5.0	-0.1 section
0.9	1.0 Traceable formulas ar		2.0	2.5 Improved	4.3	5.0	Section 1 -
1.0	1.0 Full credit: per-state p		2.3	2.5 0.2 removi	4.8	5.0	Section 3 -
0.9	1.0 All required power, ti		2.3	2.5 Improved	4.7	5.0	Section 2 -
0.8	1.0 Energy math and Y ar		2.0	2.5 Good discu	3.8	5.0	Section 1 -
0.9	1.0 Minus 0.1: the energy		2.3	2.5 Minus 0.2:	4.7	5.0	-0.1 section
0.9	1.0 Energy math, repeate		2.3	2.5 Optimized	4.5	5.0	Section 1 -
1.0	1.0 Full credit: CSV-derive		2.5	2.5 Full credit:	4.8	5.0	Section 1 -
1.0	1.0 Full credit: the report		2.3	2.5 Good optii	4.6	5.0	Section 1 -
0.9	1.0 Energy estimation is v		2.0	2.5 The report	4.2	5.0	Removed
0.9	1.0 Mostly satisfied: the i		2.0	2.5 Partial crec	4.2	5.0	Section 1 -
0.9	1.0 Energy estimation is s		2.0	2.5 The report	4.2	5.0	Removed
0.9	1.0 Mostly complete: the		2.0	2.5 Good repo	3.9	5.0	Section 1 -
0.9	1.0 Mostly complete: the		2.0	2.5 Improvem	4.4	5.0	Section 2 -
0.9	1.0 Mostly complete: the		2.5	2.5 Full credit:	4.7	5.0	Section 1 -
1.0	1.0 Full credit: per-state p		2.3	2.5 Strong imx	4.8	5.0	Section 3 -
0.9	1.0 Mostly satisfied: the i		2.0	2.5 Partial crec	4.2	5.0	Section 1 -
1.0	1.0 Full credit: the report		2.3	2.5 Strong imx	4.6	5.0	Section 1 -
0.9	1.0 Mostly satisfied: the i		2.3	2.5 Good optii	4.5	5.0	Section 1 -
0.5	1.0 Points removed: batt		1.8	2.5 Improvem	3.6	5.0	Section 1 -

The PDF, b	3.95	6.0 -0.20 Secti	The baseline submission is correct and respects the updated se
The report	4.55	6.0 -0.30 Secti	Strong baseline submission with correct sensor-before-Wi-Fi or
The PDF is	3.10	6.0 -0.40 Secti	The submission documents a plausible baseline design and incl
Report, ba	4.50	6.0 -0.15 Secti	The baseline submission is correct and respects the updated se
The report	4.55	6.0 -0.20 Secti	Strong and compliant baseline submission with a real but only
Report, RE	5.40	6.0 -0.10 Secti	Strong submission with a fully compliant baseline and a meani
Report, RE	5.00	6.0 -0.10 Secti	Strong and compliant baseline submission with a meaningful o
The baseli	2.20	6.0 -0.40 Secti	Baseline functionality is present, but the score is heavily reduce
The PDF ar	3.90	6.0 -0.25 Secti	Technically solid sensor-before-Wi-Fi baseline core with meanir
The code a	3.75	6.0 -0.25 Secti	Technically solid sensor-before-Wi-Fi core with a meaningful lov
Report, co	4.85	6.0 -0.15 Secti	Strong and fully compliant baseline submission with a real RTC.
Report, co	4.75	6.0 -0.25 Secti	Strong and fully compliant baseline submission with a real RTC.
Report anc	5.20	6.0 -0.20 Secti	Strong and fully compliant baseline submission with a meaning
The report	4.70	6.0 -0.15 Secti	Strong and fully compliant baseline submission with a meaning
The baseli	3.45	6.0 -0.40 Secti	Correct baseline structure with a plausible Q2 analysis, but the
The baseli	3.15	6.0 -0.40 Secti	The submission shows a workable baseline node and a comple
Report anc	3.95	6.0 -0.40 Secti	Complete baseline submission with several real improvement a
The PDF ar	4.50	6.0 -0.25 Secti	Strong baseline submission with a meaningful hybrid wake-up
The report	4.65	6.0 -0.15 Secti	Solid submission with correct sensor-before-Wi-Fi ordering and
Report anc	3.75	6.0 -0.40 Secti	The submission has a correct overall baseline structure and a r
The PDF, V	5.05	6.0 -0.15 Secti	Strong baseline submission with a meaningful improved imple
The report	4.50	6.0 -0.10 Secti	Strong baseline submission with a clear report and plausible Q:
Deliverabl	2.00	6.0 -1.10 Secti	Partial submission with a plausible but weakly documented Q2
The PDF, b	4.80	6.0 -0.25 Secti	Strong baseline submission with a genuine improved impleme
The PDF, V	3.70	6.0 -0.35 Secti	Technically solid core submission, but the delivered sketch cor
The baseli	3.55	6.0 -0.40 Secti	Correct baseline submission with a readable report, but the up
No major report/code mismatch; the main gap is that the optimized sketch remains timer-based, while int			
Section 4 is present but ignored for scoring. The main inconsistency is that section 2 describes CSV-based c			
Overall consistent submission; the main gap is that the reported improvement is not event-driven and the			
Main inconsistency: the report claims improvement evaluation, but the delivered code remains the baseli			
No major code/report mismatch; the main gap is the missing CSV artifact and the optimized sketch remain			
Main inconsistency: the report says the Wokwi screenshots use a 1 s deep sleep for simulation while the e			
Main inconsistency: Point 3 is only a timer-based duty-cycle optimization, not the requested interrupt-bas			
Major inconsistency: the PDF references a CSV-based power workflow, but the submission folder only incl			
Major inconsistency: the notebook comment labels the sleep-time formula with `72`, but the first student			
Major inconsistency: the PDF claims a Wokwi/ESP-NOW implementation and optimized code, but no .ino,			
Baseline work is strong, but the most capable optimizations are confined to sketch_improved.ino and the			
Main gap is reproducibility: the PDF and code are consistent, but the submission does not include a stand			
Main mismatch: the baseline sketch still uses `LIGHT` and turns Wi-Fi on before sensing, while those fixes a			
Main anomaly: Question 2 does not meet the required sensor-before-Wi-Fi ordering, but Question 3 corre			
Strong report quality overall, with correct baseline sensing/ESP-NOW timing analysis and a coherent recor			
Main mismatch: the report mistypes the full person code as 11053424 in the battery-energy calculation, ar			
The baseline implementation is mostly correct, but the submission is not fully reproducible for grading bec			
Major inconsistency: the submission is PDF-only, so the baseline and improved implementations cannot b			
Main anomaly: the submission folder contains only the PDF, sketch, and Wokwi diagram, so the CSV-base			
Main anomaly: the top-level README is an unrelated NTP clock template, so the PDF and Wokwi motion-s			
Graded using the first person code only (11051292); the second code in the submission was ignored as inst			
Main anomaly: the report describes an improved event-driven strategy, but the delivered code does not i			
Theoretical section 4 ignored for scoring. Main anomaly: both baseline and improved sketches enable Wi-			
Calibrated with eval_example.csv: baseline is solid overall, but the original sketch reads sensors after Wi-Fi			
Rubric-based review from the delivered report and the first code archive only; theoretical WSN section was			

sensor-before-Wi-Fi rule, but Question 2 is only partially reproducible and Question 3 remains mostly analytical and a real improved sketch. Most lost points come from Question 2 traceability/reporting gaps and includes a public Wokwi link, but grading confidence is limited because the exported source artifacts are misordered. Most lost points come from Question 3, where the delivered improvement is real and partially optimized improved version. Most lost points come from Question 2 reproducibility gaps and from a meaningful measured optimized version. Most lost points come only from the small Q2 reproducibility gap and from an optimized implementation. Most lost points come from Question 3, where the delivered improvement is real and partially optimized by the wrong duty-cycle value, the sensor-after-Wi-Fi ordering, weak and partly incorrect Q2 analysis, and a meaningful low-power ideas, but the package does not separate baseline from improved code and the Q2/Q3 evaluation variant, but the submission does not separate baseline from improved work. Most lost points come from a baseline-based conditional-send improvement. Most lost points come from Question 3, where the optimized design is baseline-based conditional-send improvement. Most lost points come from Question 3, where the optimized design is a meaningful measured improved version. Most lost points come from Q2 reproducibility and from Q3 not reducing a meaningful conditional-send improvement. Most lost points come from Question 3, where the optimized version is an updated Wi-Fi ordering rule is violated and Question 3 remains mostly analytical instead of delivering a more complete report structure, but the updated sensor-before-Wi-Fi rule is violated, the Q2 timing results are not complete attempts. Main lost points come from the updated sensor-before-Wi-Fi rule in the baseline, limited Q2 reproducibility improvement. Most lost points come from limited Q2 reproducibility and from Question 3 not delivering a meaningful implemented optimization. Most lost points come from Q2 reproducibility/provenance and from a real improved variant, but the updated sensor-before-Wi-Fi rule is violated in the baseline and Question 2 is not a meaningful implementation. Most lost points come from the weak Q2 boot-evidence packaging and from Question 3 not providing Q2 numbers. Most lost points come from Question 3 being delivered as a report-only optimization diff instead of a table and a discussion-only Q3. Most lost points come from the missing Arduino source, which makes the submission incomplete. Most lost points come from Q2 reproducibility and boot-timing gaps and from Question 3 keeping the baseline responds to the optimized variant rather than the baseline used in Question 2. Most lost points come from the updated sensor-before-Wi-Fi rule is violated and Question 3 stays report-only. Most lost points come from the interrupt-driven wake-up is only discussed as a future improvement.

rtical because the expected low-power reimplementation is not delivered.
 nd from Question 3 stopping short of the expected payload reduction and interrupt-driven wake-up design. The main technical issue is the updated sensor-before-Wi-Fi rule violation in the shown code, while i
 but only partial: it skips redundant transmissions, yet it does not implement reduced TX power, payload
 om Question 3 not implementing the expected reduced-TX-power and reduced-payload design.
 from Question 3 not implementing compact payload encoding plus PIR-interrupt deep-sleep wake-up.
 duces transmissions and TX power but still misses compact payload encoding and PIR-interrupt deep-slee
 and the absence of a delivered improved implementation.
 /vidence is only partially reproducible. Most lost points come from the non-clean baseline deliverable and
 ome from Question 2 traceability and from Question 3 missing payload reduction, PIR-interrupt wake-up
 ign does not implement the expected TX-power reduction, payload reduction, and PIR-interrupt wake-up
 ign does not implement the expected TX-power reduction, payload reduction, and PIR-interrupt wake-up
 g payload size or switching to PIR-interrupt wake-up.
 does not deliver the expected interrupt-driven deep-sleep redesign or payload reduction, and from Que
 reasured improved implementation.
 edible for the delivered code, and Question 3 remains hypothetical instead of being implemented and ren
 producibility, and Question 3 not delivering the expected payload-plus-interrupt low-power redesign.
 the expected payload-plus-TX-power reduction with a fully remeasured interrupt-driven design.
 id from Question 3 stopping short of payload reduction plus interrupt-driven deep-sleep wake-up.
 is weakened by an overlapping always-on sensor model. Question 3 adds useful reordering and TX-powe
 delivering a real TX-power reduction relative to the baseline while also showing a report/code mismatch ir
 ead of a runnable improved implementation, and from limited Q2 reproducibility artifacts.
 ie core baseline behavior and the improved implementation impossible to verify.
 ng the original TX power and payload size while only partially quantifying the interrupt-driven case.
 m that baseline/improved mismatch and from Question 3 not delivering payload reduction plus interrup
 the missing improved implementation and from Q2 traceability/internal-consistency gaps.

ifetime is plausible.

code, section 3 should be revisited.

erables are otherwise coherent and the Section 2 energy math is traceable.
 y GitHub link file, so the source code, Wokwi export artifacts, and improved implementation are not dire
 etch also stays timer-based, so it does not reach the higher-end improvements requested for section 3.
 distinct improved sketch. Major anomaly: the report compares a 19.5 dBm baseline against a 2 dBm opti
 ility is limited because the Wokwi export/code bundle is missing.

/i-Fi ordering, but the improved sketch fixes that with ext1 PIR wake-up. Theoretical section 4 was ignorec
 n Wokwi rather than a photoresistor, and the improvement is event-driven rather than PIR-interrupt wa
 g, and the submission does not include the requested CSV artifact for direct energy-side comparison.
 mprovement is implemented as send-every-3-cycles rather than a true motion-interrupt-driven wake-up
 ck: compared with the current shared rows, the absolute magnitudes are broadly plausible, but the repo
 omitted baseline model uses $P_{\text{sleep}} = 178.96 \text{ mW}$, far above the shared deep-sleep band of about 45 m\

gn.

most remaining lost points come from Question 2 reproducibility gaps and a Question 3 section that stay
l reduction, or PIR-interrupt deep-sleep wake-up. Question 2 is numerically plausible but would be strong

ap wake-up.

from Question 3 missing payload reduction, PIR-interrupt wake-up, and a measured improved energy w
, and a remeasured improved energy workflow.

o, while Question 2 remains only partially reproducible from the submitted artifacts.

o, and from Question 2 traceability because the repeated raw timing/CSV evidence is not packaged.

Question 2 traceability because the raw timing artifacts are not included.

measured.

r reduction, yet it still misses the expected payload reduction and interrupt-driven wake-up.

l the optimized first-boot logic.

it-driven wake-up.

ctly verifiable.

mized transmission, yet the only delivered code already sets `WIFI\_POWER\_2dBm`.

d for scoring.

ke-up, so section 3 is capped below full credit.

rt defines `Sensing = 689.85 mW` as `Idle (Wi-Fi on)` plus incremental sensor power and keeps `Waiting` a:
N, while Wi-Fi/setup is omitted and `Idle = 80 mW` has no clear CSV provenance.

s analytical instead of delivering a measured improved implementation.
ger with the raw traces and an explicit outlier-handling rule.

orkflow.

s a separate Wi-Fi-on block, so its state split is not directly comparable to the shared CSV schema. Theore

tical WSN section was ignored for scoring.

11058806 Seval Karadeniz - Haga Zenab Mohammed	1.3	1.5 Points removed: the
11035971 Ozgur Efehan KATI - Firat Adem BILGE	1.5	1.5 Full credit: baseline V
11114435 Mobin Khatib - Ali Edareh Heidarabadi	1.5	1.5 Full credit: baseline V
11111583 Fatemeh Kianiharchegani	1.3	1.5 Points removed: sens
11143015 Saltuk Bugra Kirazci	1.5	1.5 Full credit: baseline V
11187268 Nicolas Koron	1.5	1.5 Full credit: baseline V
11145475 Lorena Lam Díez - Sofía Merino López-Bra	1.5	1.5 Full credit: baseline V
11095668 Francesco Lazzaro - Andrés Felipe Forero	1.5	1.5 Full credit: baseline V
11191868 Oscar Lindqvist	1.5	1.5 Full credit: baseline V
11033376 LI MING - LIU XIANGYU	1.3	1.5 Points removed: sens
11110488 Yuanyuan Li	1.3	1.5 Baseline Wokwi imple
11161809 Eric Lombardo Facciale - Filippo Palazzetti	1.3	1.5 Points removed: the
11154567 Esteban Lopez Barreto - Ma Chong	1.5	1.5 Full credit: baseline V
11157354 Ingrid Lyngstad	1.3	1.5 Points removed: sens
10813646 Gabriel Leonardi	1.5	1.5 Full credit: baseline V
10849638 Alessio Lika	1.3	1.5 Points removed: the
10802468 Elena Lin - Jiayi Zheng	1.5	1.5 Full credit: baseline V
10607397 Ivana Lin	1.3	1.5 Points removed: no \
10861416 Susanna Lo Cascio	1.5	1.5 Full credit: delivered \
10852883 Lo Monaco Francesco Pio	1.5	1.5 Full credit: baseline V
10796320 Samuele Lo Piccolo	1.2	1.5 Points removed: exac
10782022 Stefano Longhi - Beatrice Spazzadeschi	1.4	1.5 Points removed: the
10502570 Lorenzi Davide	1.3	1.5 Mostly satisfied: Wol
10738139 Davide Lupo	1.5	1.5 Full credit: baseline V
10730038 Edoardo Menegotto	1.5	##### Full credit: baseline V
11034150 Mobina Mobaraki	1.5	1.5 Full credit: Wokwi no
11189222 Felicia Murkes	1.5	1.5 Full credit: baseline V
10862592 Enrico Mantegazza	1.3	1.5 Baseline node and br
10911594 Francesca Maragoni	1.3	1.5 Baseline hardware, E
10773779 Francesco Marchi	1.5	1.5 Full credit: baseline a
10819854 Giovanni Miccio	1.5	1.5 Full credit: the Wokw
10806788 Gabriele Michieletto	1.5	1.5 Full credit: the report
10813424 Andrea Midulla	1.5	1.5 Full credit: baseline a
10727881 Andrea Luca Miele	1.5	1.5 Full credit: the sketch
10710384 Lorenzo Francesco Luigi Minorini	1.3	1.5 The report describes
10845860 Monzani Mario	1.5	1.5 Full credit: baseline V
10816840 Motti Filippo	1.5	1.5 Full credit: the node r
11009540 Muyololo Emmanuel	1.0	1.5 Points removed: Wi-f
11152090 NGUYEN HAO JOHNY VINH	1.0	1.5 Points removed: the
11026772 NGUYEN MINH MAN	1.5	1.5 Full credit: the sketch
11173956 NUYENS GOTELLI IGNACIA	1.3	1.5 Points removed: the
10777415 Nardi Andrea	1.5	1.5 Full credit: the sketch
10817232 Neri Tommaso	1.3	1.5 Points removed: Wi-f
10861979 Nipoti Alessandro	1.5	1.5 Full credit: the local s
11191225 ORTIZ BALLESTEROS ANGELICA	1.3	1.5 Points removed: the
10994077 Giovanni Orciuolo	1.5	1.5 Full credit: the baseli
11032539 Xiaohan Pan - Mengshuang Tang	1.3	1.5 Points removed: the
11115876 Zahra Sadat Parviss	1.5	1.5 Full credit: the submi
11031890 Anchen Peng - Qianyi Wang	1.5	1.5 Full credit: the firmw
11135422 Luca Pezzoli - Mattia Rainieri	1.5	1.5 Full credit: baseline c
11119281 Mariyam Rashid Pitafi	1.5	1.5 Full credit: the baseli

0.9	1.0 Mostly complete: the	2.5	2.5 Full credit:	4.7	5.0 Section 1 -
0.9	1.0 Points removed: repe	2.0	2.5 Improvem	4.4	5.0 Section 2 -
1.0	1.0 Full credit: per-state p	2.5	2.5 Full credit:	5.0	5.0 No deduct
0.8	1.0 Points removed: repe	2.0	2.5 Improvem	4.1	5.0 Section 1 -
0.3	1.0 Points removed: per-	1.5	2.5 Improvem	3.3	5.0 Section 2 -
1.0	1.0 Full credit: per-state p	2.5	2.5 Full credit:	5.0	5.0 No deduct
1.0	1.0 Full credit: per-state p	2.0	2.5 Improvem	4.5	5.0 Section 3 -
0.8	1.0 Points removed: repe	2.3	2.5 Good impr	4.6	5.0 Section 2 -
0.8	1.0 Points removed: the	1.5	2.5 Improvem	3.8	5.0 Section 2 -
0.9	1.0 Points removed: the	2.0	2.5 Improvem	4.2	5.0 Section 1 -
0.8	1.0 Cycle energy, average	1.5	2.5 A realistic i	3.6	5.0 Section 1 -
1.0	1.0 Full credit: per-state p	2.3	2.5 Good opti	4.6	5.0 Section 1 -
1.0	1.0 Full credit: per-state p	2.3	2.5 Good impr	4.8	5.0 Section 3 -
0.9	1.0 Points removed: repe	1.5	2.5 Improvem	3.7	5.0 Section 1 -
0.9	1.0 Points removed: the	2.5	2.5 Full credit:	4.9	5.0 Section 2 -
0.8	1.0 Points removed: the	2.0	2.5 Good impr	4.1	5.0 Section 1 -
0.9	1.0 Energy estimation is v	2.0	2.5 Improvem	4.4	5.0 Section 2 -
0.8	1.0 Points removed: repe	1.5	2.5 Improvem	3.6	5.0 Section 1 -
1.0	1.0 Full credit: per-state p	2.5	2.5 Full credit:	5.0	5.0 No deduct
1.0	1.0 Full credit: per-state p	2.3	2.5 Good impr	4.8	5.0 Section 3 -
0.9	1.0 Points removed: repe	2.3	2.5 Good impr	4.4	5.0 Section 1 -
1.0	1.0 Full credit: per-state p	2.3	2.5 Strong im	4.7	5.0 Section 1 -
0.7	1.0 Partially satisfied: the	2.3	2.5 Strong par	4.3	5.0 Section 1 -
0.9	1.0 Points removed: the	2.5	2.5 Full credit:	4.9	5.0 Section 2 -
0.8	1.0 Points removed: the	2.4	2.5 Strong im	4.7	5.0 Section 2 -
1.0	1.0 Full credit: cycle timin	2.3	2.5 Strong im	4.8	5.0 Section 3 -
0.8	1.0 Good cycle model wit	2.3	2.5 Improvem	4.6	5.0 Section 2 -
1.0	1.0 Energy estimation is c	2.3	2.5 Improved	4.6	5.0 Section 1 -
1.0	1.0 Cycle energy, battery	2.0	2.5 The optimi	4.3	5.0 Section 1 -
0.9	1.0 Baseline cycle accoun	2.5	2.5 Full improv	4.9	5.0 Section 2 -
1.0	1.0 Full credit: repeated t	2.3	2.5 Improved	4.8	5.0 Section 3 -
1.0	1.0 Full credit: average p	2.0	2.5 The report	4.5	5.0 Section 3 -
1.0	1.0 Full credit: state pow	2.5	2.5 Full improv	5.0	5.0 No deduct
1.0	1.0 Full credit: power esti	2.0	2.5 The report	4.5	5.0 Section 3 -
1.0	1.0 Full credit: the energy	2.0	2.5 The optimi	4.3	5.0 Section 1 -
0.8	1.0 Points removed: the	2.0	2.5 Strong im	4.3	5.0 Section 2 -
1.0	1.0 Full credit: the report	1.5	2.5 Improvem	4.0	5.0 Section 3 -
1.0	1.0 Full credit: the energy	2.0	2.5 Good impr	4.0	5.0 Section 1 -
1.0	1.0 Full credit: the report	2.0	2.5 Improvem	4.0	5.0 Section 1 -
1.0	1.0 Full credit: the report	2.3	2.5 Strong im	4.8	5.0 Section 3 -
1.0	1.0 Full credit: the report	2.0	2.5 The impro	4.3	5.0 Section 1 -
1.0	1.0 Full credit: the report	2.3	2.5 Very good	4.8	5.0 Section 3 -
1.0	1.0 Full credit: the report	2.3	2.5 Implemen	4.6	5.0 Section 1 -
1.0	1.0 Full credit: the report	2.0	2.5 The impro	4.5	5.0 Section 3 -
1.0	1.0 Full credit: the report	2.5	2.5 Full credit:	4.8	5.0 Section 1 -
1.0	1.0 Full credit: per-state p	2.3	2.5 Strong im	4.8	5.0 Section 3 -
1.0	1.0 Full credit: the energy	2.3	2.5 Good opti	4.6	5.0 Section 1 -
1.0	1.0 Full credit: the energy	2.0	2.5 Points rerr	4.5	5.0 Section 3 -
1.0	1.0 Full credit: the report	2.3	2.5 Strong opt	4.8	5.0 Section 3 -
0.2	1.0 Points removed: the	2.3	2.5 Good impr	4.0	5.0 Section 2 -
1.0	1.0 Full credit: the report	2.3	2.5 Strong im	4.8	5.0 Section 3 -

Calibrated with eval_example.csv: baseline functionality is strong and the improvement package is above :
 Calibrated with eval_example.csv: baseline functionality is solid, but the improvement work misses payload
 Calibrated with eval_example.csv: strong submission with implemented TX-power reduction, payload corr
 Calibrated with eval_example.csv: baseline functionality and improved code are both present, but the bas
 Calibrated with eval_example.csv: baseline implementation is solid, but Q2 remains mostly formula-level a
 Rubric-based review from the delivered report/code artifacts; theoretical WSN section was ignored for sco
 Calibrated with eval_example.csv: baseline implementation and Q2 are complete; Section 3 remains discus
 Calibrated with eval_example.csv: strong baseline and solid optimized code with RTC-based conditional se
 Rubric-based review from the delivered report/code artifacts; theoretical WSN section was ignored for sco
 Calibrated with eval_example.csv: baseline is solid, but the sketches read sensors after enabling Wi-Fi and
 Calibrated with eval_example.csv: baseline implementation is mostly complete, but the sensor-before-Wi-
 Calibrated with eval_example.csv: baseline/report quality is strong overall, but the original sketch is not fu
 Calibrated with eval_example.csv: strong baseline and a solid RTC-backed delta-send improvement, but it
 Calibrated with eval_example.csv: baseline PIR plus LDR ESP-NOW functionality is delivered and the energy
 Calibrated with eval_example.csv: baseline section is complete and the improvement section is strong bec
 Rubric-based review from the delivered PDF plus public Wokwi link; theoretical WSN section was ignored fi
Calibrated with eval_example.csv: strong baseline implementation and solid Q2/Q3 reasoning, but the im
 Rubric-based review from the delivered report and analysis notebooks only; no .ino/.json/Wokwi link is in
 Rubric-based review from the delivered report/code artifacts; theoretical WSN section was ignored for sco
 Calibrated with eval_example.csv: strong baseline and a valid delta-send optimization with recomputed er
 Major inconsistency: the report says the baseline/improved implementations are in folders `Point 3 v1` an
 Calibrated with eval_example.csv: strong report and code overall, but the delivered Wokwi artifacts have t
 Major inconsistency: the delivered baseline and optimized sketches both call `setup()` from `loop()`, so the
 Q2 cross-check: Compared with the current shared rows, the reported idle, sensor-read, Wi-Fi/setup, TX, ;
 Rubric-based review from the delivered code/report artifacts; theoretical WSN section was ignored for sco
 Baseline and improved artifacts are consistent; the main missing optimization is payload reduction / inter
 The report is usable, but the battery/lifetime comparison should be rechecked against the omitted boot t
 The improved build is substantial, but the baseline payload formatting and the absence of payload-size re
 The analysis is good, but the optimization claims are not backed by a separate implemented code path.
 Strong submission overall; the only noticeable traceability issue is the timing split around transmission.
 The submission is solid and internally consistent, but it stops short of the most effective low-power optimi
 The energy discussion is clear, but the missing optimized implementation limits the score.
 Baseline and optimized versions are both delivered; the implementation covers trigger-based wake-up and
 Usable report, but the optimization is not backed by a delivered code variant.
 This is usable as a report, but the missing implementation artifacts make grading less reliable.
 Q2 needs a check against the person-code formula, and the improvement section should include the rewo
 Baseline is fine, but the improvement section needs an actual optimized implementation and recomputed
 The main mismatch is in section 1: sensing order and lux conversion are not compliant, even though the e
 The energy math is mostly traceable, but the baseline sketch does not satisfy the sensor-before-Wi-Fi and
 The improvement is credible and quantified; the remaining gap is that the optimization is still based on co
 The baseline implementation is acceptable, but the grading package would be easier to verify if the improv
 This is a strong submission overall; the main improvement path is to push the optimization from event-dr
 The report is well structured and the improved variant is included, but the baseline sensing order should b
 The main scoring risk is reproducibility: the folder points to the improved implementation, but the graded
 The only material issue is the payload formatting; the improvement package itself is strong and well supp
 Baseline and improved implementations are both present. No major anomaly beyond the missing payload
 The baseline report and code are usable for grading. Main limitation is that light is treated as a raw ADC va
 Baseline and improved sender code are present. The optimization is acceptable but remains timer-based a
 Good baseline plus improved implementation. The main remaining gap is that the optimization stays time
 The submission has a workable baseline and a real optimization, but the Q2 battery-energy calculation is n
 Baseline and improved versions are both delivered. The only real gap is that the optimization remains mo

average because both conditional transmission and interrupt-based wake-up code are delivered. Q2 cross-check: Compared with the current shared rows, the baseline still enables Wi-Fi before sensor sampling and the Q2 power constants are internally inconsistent. Q2 cross-check: Compared with the current shared rows, the reported 0.162243 J / 2.395493 s baseline-only and misses improved code plus PIR-interrupt wake-up consideration. Q2 cross-check: Comparing, but not full mark because Q2 lacks repeated timing evidence and Q3 omits payload reduction and PIR-interrupt wake-up. Major inconsistencies: the report text says `23 000 000 microseconds` for X although the submitted improvement section does not reach the expected payload-reduction / interrupt-wake-up level. Q2 cross-check: compared with the current low-power compliant because sensing happens after Wi-Fi/ESP-NOW initialization. Q2 cross-check: comparison stops short of the expected TX-power / payload / interrupt-wake optimizations. Major inconsistency: the formulas are mostly correct, but `sketch.ino` enables Wi-Fi before reading sensors. Q2 cross-check: comparison shows both delta-send and interrupt-based wake-up are delivered. Q2 cross-check: Compared with the current scoring. Major issues to recheck: the baseline code computes `X` with integer division, so `(38 % 50 + 5)`. **Improved implementation is only linked in the PDF and not delivered in the submission.** Q2 cross-check: Concluded, so the baseline and improvement implementation cannot be verified directly. Theoretical section missing. Q2 cross-check: the reported powers, timings, and lifetime are broadly consistent with the shared data, but not full mark because payload reduction is not implemented and trigger-based wake-up is not implemented. Q2 cross-check: the submission only contains a PDF, so the Wokwi project, source files, and CSV artifacts are missing. The LDR power-gating wiring reversed between baseline and improved projects, so the implementation is incorrect. Q2 cross-check: the node can execute the sensing/transmission logic twice per wake cycle while the report models only one. Q2 cross-check: deep-sleep magnitudes are broadly plausible. The main anomaly is that the cycle-energy model starts at a non-zero value. Q2 cross-check: Compared with the current shared rows, the reported idle, sensor-read, Wi-Fi-on, Triggered wake-up, and the lack of a concrete trigger-wake implementation. Q2 cross-check: The payload reduction keep it short of full marks.

ization.

- d low-power operation cleanly.

arked sketch or a clearly packaged improved project.
energy results to reach full credit.
nergy analysis is otherwise solid.
lux-output requirements.
nditional transmission rather than a wake-up trigger.
red sketch were included alongside the report.
iven transmission into a more explicit wake-up strategy.
ie corrected to match the stated specification.
package itself does not contain the full optimized source tree.
orted by code, scripts, and comparisons.
d-size reduction in the optimization.
lue rather than a clearly justified illuminance measurement.
nd does not show the stronger event-driven behavior expected for full credit.
r/state-change based rather than fully event-driven.
niscalculated and should be corrected before use.
stly state-based rather than fully event-driven.

s-check: Compared with the current shared rows, the power levels are plausible, but the major anomaly is energy gain relies on an assumed boot reduction. Q2 cross-check: Compared with the current shared row line power and timing values are broadly plausible; the main anomaly is that Boot and Post-TX Idle timing. Q2 cross-check: Compared with the current shared rows, the deep-sleep and Wi-Fi values are broadly plausible. `ch.ino` is labeled `FINAL OPTIMIZED VERSION` even though the report presents a separate improved system. The operating cycle is somewhat lower than most 2.3 s submissions but still broadly plausible; the main anomaly is related with the current shared rows, the reported power and timing magnitudes are broadly plausible and not PIR-interrupt wake-up. Major report/code misalignments: the report states `LUX = 1000 * analogRead(3)` but the code uses the correct `2300000ULL`, and the report itself acknowledges that the measured transmission time. Q2 cross-check: Compared with the current shared rows, `Transmission = 667.411 mW` and `Deep Sleep = 45.178.96 mW` for both `Idle / Boot / Wi-Fi initialization` and `Deep Sleep` is compared with the current shared rows, the baseline powers are broadly plausible, though the boot/setup time the report derives lux with a 10 kOhm divider, while the baseline `sketch.ino` uses `2000 * voltage / (1 - voltage)`. Compared with the current shared rows, the reported 630 mW / 215.876 ms setup block and 607 mW TX value. Compared with the current shared rows, the reported idle, sensing, Wi-Fi ON, TX, and deep-sleep magnitudes are broadly plausible. `/ 10` becomes `4` instead of the required `4.3 s`, and the Q2 baseline row is not directly comparable to the current baseline row. **Compared with the current shared rows, the power magnitudes are broadly plausible, but the active timing magnitudes are not directly comparable to the current baseline row.** Q2 cross-check: there is no existing shared power-time row for student 10607. The dataset, with no major anomaly beyond limited raw timing / CSV traceability. The analysis is not discussed. Major report/code note: Section 3.1 claims an implemented optimization by skipping explicit TX timing. The TX timing cannot be verified directly. Q2 cross-check: compared with the shared rows, the reported phase magnitudes are not fully reproducible as submitted. Q2 cross-check: Compared with the current shared rows, the report presents a separate improved system. The operating cycle. The Q2 numeric magnitudes are broadly in family with the shared `power-time-estimation` after wake/boot and the TX timing only wraps `esp_now_send()`, so the stored baseline energy and lifetime. The TX, and deep-sleep magnitudes are broadly plausible, but the shipped analysis scripts reference a non-existent row.

completion, so the reported `WiFi-on = 2710 us` and `TX = 282 us` should be treated as optimistic and not as traceability: the reported `Boot + Init = ~300 ms` is not isolated with repeated `micros()` measurements/sketch, so those phase boundaries should be treated as approximate.

Baseline sketch/report use `607.41 mW` for TX and `294.47 mW` for sensing; this state mapping should be consistent with the current shared dataset.

as and the repeated raw timing logs are not included.

4 W` for the same sensing state. Q2 cross-check: Compared with the current shared rows, the cycle-energy for the current shared rows, the power magnitudes are broadly plausible, but the Wi-Fi-enabled duration is likely merged Wi-Fi/setup power rather than pure boot.

/, so the CSV/state mapping should be rechecked.

it / WiFi-On` at `342.98 mW`. Major Section 3 misalignment: `opt3.ino` senses before enabling Wi-Fi and is likely plausible, but `Wi-Fi + ESP-NOW Init = 64.54 ms` is shorter than the common shared range and the report is merged into one state and no raw repeated timing list is provided. Major inconsistency: Section 4 says the report is staged, but the submission does not package the CSV append/person-code row or the raw repeated timing list instead of isolating those states.

in the baseline cycle-energy model.

likely in family with the dataset, but Boot Average Power = 389.53 mW is on the high side and the notebook

sleep_start()`), so that item is not a real change relative to the submitted baseline. Q2 cross-check: Compare overhead.

likely plausible; the strange Q2 item is that `WiFi.mode(WIFI\_STA)` is outside the measured Wi-Fi-on window reported, so the energy estimate is only partially reproducible.

theoretical WSN section was ignored for scoring.

time), so the energy workflow is only partly reproducible. Improvement code is present in repository branch

directly comparable to the usual shared setup windows.

its and appears inferred from total active time with Serial overhead, so the cycle-energy model is only partially checked before comparing the row with the shared dataset.

My result is much higher because the whole 2 s polling window is treated as sensing, so the state split is on a lower power than the typical setup band and the TX timing is not directly comparable because it captures only the

time it goes straight back to deep sleep when no change is detected, but the improved no-TX energy model still includes start folds message formatting, Wi-Fi shutdown, and sleep preparation into one 'Idle Wait' block, so that the sleep interval was increased to 5 s, while the submitted 'sketch.ino' still defines 'SLEEP_DURATION_SECONDS' as 3 s. The log logs, and the printed 'TCycle = 5303308 us' does not exactly equal the listed state durations. Minor noise

is due to the fact that the model relies on thresholded CSV bins plus hard-coded timing values, so the cycle-energy model is only partially reconciled with the current shared rows, the reported powers and timings are broadly plausible, with the main caveat being that, so part of radio startup is charged to idle instead of Wi-Fi setup.

which occurs ('light', 'interrupt') rather than only in the main sketch.

tially reproducible.

ly partially comparable.

ie API call rather than confirmed transmission.

ncludes the full boot term, so the reported optimized savings are conservative and not fully code-aligned
ate mapping should be treated as approximate. Theoretical WSN section was ignored for scoring.

),9`.

te: the interrupt sketch uses `NO\_MOTION-LUMINOSITY:` instead of the original `MOTION_NOT_DETECTI

roducible.

:aution being the indirectly derived boot state.

l.

ED-LUMINOSITY: ` format.

10777877 Mattia Panzeri - Diego Butti	1.5	1.5 Full credit: the baselir
10778484 Eliseo Paoletti - Gabriele Nicolo	1.5	1.5 Full credit: the baselir
10697294 Salvatore Passaro	1.5	1.5 Full credit: baseline c
10742323 Giacomo Pasturensi	1.5	1.5 Full credit: the baselir
10778819 Lorenzo Pedranzini	1.5	1.5 Full credit: the baselir
10828995 Jacopo Pelagalli	1.3	1.5 Points removed beca
10773372 Simone Perugini	1.3	1.5 Points removed beca
10777399 Mario Pesca	1.5	1.5 Full credit: the baselir
10619593 Alessandro Pessina	1.5	1.5 Full credit: the node t
10900762 Davide Petruzzi	1.5	1.5 Full credit: the baselir
10823873 Alberto Pini	1.5	1.5 Full credit: the baselir
10797420 Pietro Pizzoccheri	1.5	1.5 Full credit: the baselir
10705152 Alessandro Posani	1.5	1.5 Full credit: the baselir
10855966 Marco Puddu	1.5	1.5 Full credit: the Wokw
11048190 JinWen Qi	1.5	1.5 Full credit: baseline V
10962875 Yongqi Qiu	1.5	1.5 Full credit: the baselir
11022931 Peng Rao	1.5	1.5 Full credit: the submi
11046392 Amir Hossein Rashidy	1.5	1.5 Full credit: baseline V
11128210 Rashwan Mohannad Mohamed Fawzy Al	1.5	1.5 Full credit: the baselir
11111393 Muhammad Rian	1.5	1.5 Full credit: the baselir
10770385 Alessia Radicula	1.5	1.5 Full credit: the baselir
10837597 Alessia Ragosta	1.5	1.5 Full credit: the code a
10768645 Gabriele Riboldi - Luca Villa	1.5	1.5 Full credit: the deliver
10793255 Flavio Riccobono	1.5	1.5 Full credit: the submi
10863960 Rodari Simone	1.3	1.5 Good baseline impler
10724267 Alessandra Romano	1.3	1.5 Core functionality is c
10789693 Alessio Vittorio Rosano	1.3	1.5 Most baseline require
10851929 Tommaso Rossetti	1.5	1.5 All baseline requirem
10799170 Andrea Rossi	1.0	1.5 Baseline code shows
10798975 Matteo Rossi	1.3	1.5 The baseline specifica
10780379 Andrea Rubagotti	1.5	1.5 All baseline requirem
10772022 Davide Rutigliano	1.3	1.5 Baseline functionality
10847803 Maciej Krzysztof Rzewuski	1.5	1.5 All baseline requirem
11119582 Amin Sakhaei	1.3	1.5 Baseline behavior is v
11175090 Anasuya Satapathy	1.5	1.5 Full credit: improved
11010656 Ghazal Sepehrirad	1.5	1.5 Full credit: the evalua
11063739 Sajjad Shaffaf	1.1	1.5 Points removed: the
10495016 Dalila Samr	1.3	1.5 Points removed: the
10830365 Andrea Santaniello	1.5	1.5 Full credit: the evalua
10825131 Edoardo Sartore	1.5	1.5 Full credit: the evalua
10812603 Andrea Sau	1.5	1.5 Full credit: the baselir
10764570 Giacomo Scampini	1.5	1.5 Full credit: the baselir
10698107 Andrea Gaetano Senatore	1.5	1.5 Full credit: the baselir
10744406 Gabriele Bogdan Sepe	1.3	1.5 Points removed: the
10821199 Golsa Shokri	1.3	1.5 Baseline Wokwi setup
10766937 Gabriele Lorenzo Singe	1.5	1.5 Full credit: the baselir
10769571 Francesco Street	0.8	1.5 Partial credit only: th
10830061 Cristian Summa	1.5	1.5 Full credit: the baselir
11151327 Kelly Tran	1.5	1.5 Full credit: the submi
11120812 Orhun Tunc	1.3	1.5 Good baseline functio
10832525 Michele Tavecchio	1.5	1.5 Full credit: the baselir

1.0	1.0 Full credit: the report	2.3	2.5 Good opti	4.8	5.0 Section 3 -
1.0	1.0 Full credit: the energy	2.5	2.5 Full credit:	5.0	5.0 No deduct
1.0	1.0 Full credit: timing is r	2.5	2.5 Full credit:	5.0	5.0 No deduct
0.2	1.0 Points removed: the	2.0	2.5 Points rerr	3.7	5.0 Section 2 -
1.0	1.0 Full credit: per-state p	2.5	2.5 Full credit:	5.0	5.0 No deduct
1.0	1.0 Full credit: the report	2.5	2.5 Full credit:	4.8	5.0 Section 1 -
1.0	1.0 Full credit: the energy	2.0	2.5 Improvem	4.3	5.0 Section 1 -
1.0	1.0 Full credit: the report	2.3	2.5 Strong im	4.8	5.0 Section 3 -
1.0	1.0 Full credit: the report	2.5	2.5 Full credit:	5.0	5.0 No deduct
1.0	1.0 Full credit: the report	2.0	2.5 The report	4.5	5.0 Section 3 -
1.0	1.0 Full credit: the power	2.5	2.5 Full credit:	5.0	5.0 No deduct
1.0	1.0 Full credit: the report	2.5	2.5 Full credit:	5.0	5.0 No deduct
1.0	1.0 Full credit: the report	2.3	2.5 The impro	4.8	5.0 Section 3 -
1.0	1.0 Full credit: the report	2.5	2.5 Full credit:	5.0	5.0 No deduct
1.0	1.0 Full credit: per-state p	2.0	2.5 Points rerr	4.5	5.0 Section 3 -
1.0	1.0 Full credit: the report	1.8	2.5 Points rerr	4.3	5.0 Section 3 -
0.9	1.0 Mostly complete: the	2.3	2.5 Strong im	4.7	5.0 Section 2 -
1.0	1.0 Full credit: the report	2.3	2.5 Good impr	4.8	5.0 Section 3 -
1.0	1.0 Full credit: the report	2.3	2.5 Improvem	4.8	5.0 Section 3 -
0.8	1.0 Points removed: the	1.5	2.5 Points rerr	3.8	5.0 Section 2 -
0.8	1.0 Points removed: the	2.5	2.5 Full credit:	4.8	5.0 Section 2 -
1.0	1.0 Full credit: the report	2.5	2.5 Full credit:	5.0	5.0 None.
0.8	1.0 Points removed: the	1.5	2.5 Points rerr	3.8	5.0 Section 2 -
1.0	1.0 Full credit: the report	2.3	2.5 Strong im	4.8	5.0 Section 3 -
1.0	1.0 Full credit: average po	2.3	2.5 Strong im	4.6	5.0 Section 1:
1.0	1.0 Full credit: the report	1.0	2.5 Only prop	3.3	5.0 Section 1:
1.0	1.0 Full credit: average po	2.5	2.5 Full credit:	4.8	5.0 Section 1:
1.0	1.0 Full credit: per-state p	2.5	2.5 Full credit:	5.0	5.0 No deduct
0.0	1.0 Not demonstrated: n	0.0	2.5 Not demoi	1.0	5.0 Section 1:
1.0	1.0 Full credit: the report	2.0	2.5 Good impr	4.3	5.0 Section 1:
1.0	1.0 Full credit: per-state p	2.5	2.5 Full credit:	5.0	5.0 No deduct
1.0	1.0 Full credit: state pow	2.0	2.5 Relevant ir	4.3	5.0 Section 1:
0.0	1.0 Not demonstrated: tl	0.8	2.5 Only brief	2.3	5.0 Section 2:
1.0	1.0 Full credit: per-state p	2.0	2.5 Good impr	4.3	5.0 Section 1:
1.0	1.0 Full credit: per-state p	2.3	2.5 Strong im	4.8	5.0 Section 3 -
1.0	1.0 Full credit: the report	2.5	2.5 Full credit:	5.0	5.0 No deduct
1.0	1.0 Full credit: CSV-based	1.8	2.5 Improvem	3.9	5.0 Section 1 -
1.0	1.0 Full credit: the report	1.5	2.5 Improvem	3.8	5.0 Section 1 -
1.0	1.0 Full credit: the report	2.3	2.5 Strong im	4.8	5.0 Section 3 -
1.0	1.0 Full credit: the report	2.5	2.5 Full credit:	5.0	5.0 No deduct
1.0	1.0 Full credit: the report	1.8	2.5 The report	4.3	5.0 Section 3 -
1.0	1.0 Full credit: the report	1.8	2.5 The report	4.3	5.0 Section 3 -
1.0	1.0 Full credit: the report	1.8	2.5 The report	4.3	5.0 Section 3 -
1.0	1.0 Full credit: the report	2.5	2.5 Full credit:	4.8	5.0 Section 1 -
1.0	1.0 Full credit: the report	2.5	2.5 Very stron	4.8	5.0 Section 1 -
1.0	1.0 Full credit: the energy	2.3	2.5 Good impr	4.8	5.0 Section 3 -
0.8	1.0 The energy section is	2.5	2.5 Strong im	4.1	5.0 Section 1 -
1.0	1.0 Full credit: the report	2.5	2.5 Full credit:	5.0	5.0 None
1.0	1.0 Full credit: the energy	1.5	2.5 Good disc	4.0	5.0 Section 3 -
0.8	1.0 Average power, cycle	0.0	2.5 not demor	2.1	5.0 Section 1 -
1.0	1.0 Full credit: the report	2.5	2.5 Full credit:	5.0	5.0 None

Baseline and optimized code are both present and readable. The optimization is solid, but it does not reach the target. This is one of the strongest submissions in the batch: baseline, optimized code, and comparative energy data. The submission includes a strong baseline and a meaningful event-driven optimization. No major grading issues. Baseline functionality is present, but the Q2 battery-energy calculation is wrong and the optimization is large. The submission is complete; the only anomaly is that the bundled README files are template leftovers. No major anomalies. Baseline and improved materials are both present, but the first sketch still violates the sensor-before-Wi-Fi rule. The report is complete, but the first sketch still starts the radio before sensing.

No major anomalies.

No major anomalies.

The improvement discussion is coherent, but it remains report-heavy rather than code-complete.

No major anomalies.

No major anomalies.

The analysis is detailed, but the optimized implementation is not packaged as a standalone improved code.

No major anomalies.

Theoretical section 4 is ignored for grading. Section 3 is the main gap: the report recomputes energy, but the code does not.

No major numeric anomaly stands out. The main limitation is that Section 3 remains a design discussion instead of an implementation.

No major anomaly. The baseline and optimized sketches are well aligned with the rubric, but Q2 should be more detailed.

No major anomaly. The submission is split across multiple archives, but the baseline and improvement are both present.

No major anomaly. The report is substantial, but the strongest credit comes from the optimized code and the energy analysis.

The deep-sleep power reported in Q2 is the main anomaly: it is far above the shared sleep band, so the energy calculation is off.

No major anomaly. The optimization set is strong and well documented, with both code and comparison data.

No major anomaly. The theoretical WSN section is ignored for grading, and the rest of the submission is complete.

The main limitation is document accessibility: the DOCX exists, but the report is much harder to verify than the PDF.

No major anomaly. The baseline and optimized sketches are both delivered, and the report clearly separates them.

Baseline and energy sections are solid. The improvement section is credible, but it stays report-level rather than code-level.

The baseline report is usable, but section 3 needs an implemented optimization pass with recomputed energy.

The optimization work is strong and quantified. The main issue is the exact message string spelling in the code.

Complete and well-structured submission with baseline and two optimized code paths.

The code archive shows a workable baseline and improved sketches, but without the report the energy analysis is missing.

The report is mostly complete and the improvement direction is strong. The main gap is the missing standard deviation.

Complete submission with baseline and two optimized code paths.

The report is usable for grading, but the improvement section needs a clearer runnable optimized version.

The baseline implementation is acceptable, but the submission is incomplete for grading because the energy analysis is missing.

The report is coherent and the optimization ideas are relevant, but the improvement section would benefit from more data.

Calibrated against the shared CSVs; no major anomaly beyond the lack of payload compression and motion detection.

No major issues in the evaluated code/report combination.

Important mismatch: the folder contains multiple code versions, but the first evaluated sketch is V2, so later versions are not checked.

The main code is functional, but the light measurement is only a raw ADC proxy and the optimization section is missing.

No major anomaly; the remaining gap is that the improved payload is still verbose even though the rest of the code is optimized.

No major issues in the evaluated optimized sketch or report.

Important grading note: later optimized artifacts exist in the folder, but the first code file is baseline-only, so the improvement is not evaluated.

Important grading note: the folder also contains an improved sketch, but the first code file is baseline-only, so the improvement is not evaluated.

Important grading note: later optimized artifacts exist, but the first code file is baseline-only, so I graded against the baseline.

Main anomaly is the payload formatting: the code/report insert spaces around the required string tokens.

Baseline and improved sketches are both delivered. Main misalignment is that sensor acquisition happens before the radio is initialized.

Good package overall: source, Wokwi files, timing exports, and the baseline/improved pair are all included.

The main misalignment is the hardware choice: the implementation is functionally close, but the sensor power consumption is high.

No major anomalies. The baseline and optimized versions, the recomputed energy, and the event-driven optimization are all present.

The report is strong on analysis, but the improvement story is mostly report-side. The instructor should verify the code.

The submission is usable for grading, but it does not satisfy the energy-CSV cross-check requirement and the energy analysis is missing.

No major anomalies. Baseline and optimized estimates are both explicit, and the comparison across versions is clear.

h the stronger event-driven design described in the rubric.
discussion are all present and aligned with the rubric.
anomaly stands out.
gely speculative rather than implemented.
related to the challenge.
i order.

e submission.

he actual improved code is only hinted via commented switches.
instead of a delivered optimized implementation.
e backed by a visible shared-CSV artifact.
ifacts are both present.
the recomputed energy figures.
ergy total should be treated cautiously.
material present.
omplete.
n the PDF-based submissions.
tes the improvement idea from the baseline.
r than a separately deliverable optimized sketch.
ergy to reach the rubric target.
aseline description, which should be checked against the source code.

id improvement rubric cannot be verified.
alone optimized code package.

and a tighter baseline-vs-improved comparison.
gy section is missing.
it from a clearer runnable implementation and explicit baseline-vs-improved evidence.
on-interrupt wake-up in the improved version.

ter V3 improvements were not counted.
on is not implemented in the evaluated sketch.
f the energy-saving path is implemented.

so I did not count those later improvements.
/, so it was not counted.
gainst that version.
, so the exact challenge format is not preserved.
s after Wi-Fi is enabled in the baseline path, but the improved energy comparison is otherwise well suppo
. The improvement is credible, but it is still a timer-driven node rather than an interrupt-driven one.
air is not the required PIR plus photoresistor configuration.
wake-up path are all present and consistent.
erify the absence of a separate improved sketch before giving full credit for section 3.
he improvement section is limited to baseline sleep tuning.
ons is coherent.

rted.

10847211 Simone Teggi	1.5	1.5 Full credit: the baselir
10816921 Federico Trubian	1.5	1.5 Full credit: the baselir
11156535 Torodd Urdshals	0.8	1.5 Partial credit only: th
11152089 Christian Vaktel	1.3	1.5 Mostly satisfies the n
11115207 Juan David Vargas Galindo - Cristian Davic	1.30	1.50 Mostly satisfied, but
11046642 Shaghayeh Vesali	1.50	1.50 Satisfied: the baseline
10781960 Andrea Vavassori - Marco Gad	1.30	1.50 Points removed: the
10787431 Davide Vendramin - Matteo Taschini	1.50	1.50 Satisfied: the report,
10712273 Matteo Vicentini	1.5	1.5 Full credit: Wokwi ba
10676235 Marco Vitali	1.50	1.50 Full credit: the baselir
11118429 Guangyi Wang	1.3	1.5 Points removed: sens
11119541 Yizhe Wang	1.50	1.50 all required motion-s
11119918 Yunxuan Wang	0.00	1.50 not demonstrated
11154359 TianQi Wu	1.50	1.50 Satisfied: baseline Wc
10863400 Yuan Wang	1.5	1.5 Full credit: baseline V
11052807 Erdal Yalcin	1.5	1.5 Full credit: baseline V
11015522 Yawen Yan	1.5	1.5 Full credit: baseline V
11057250 Arzu Yildiz	0.7	1.5 Partially satisfied: the
11027977 Yu Xincheng	0.5	1.5 Not demonstrated: tl
11061927 Haoyu Zhang	1.5	1.5 Full credit: baseline V
11033400 Hezhi Zhang	1.3	1.5 Points removed beca
11044575 Jie Zhang	1.3	1.5 Points removed beca
11107611 Ning Zhang	1.5	1.5 Full credit: baseline V
11048104 Ting Zhang	1.5	1.5 Full credit: baseline V
11171264 ZHANG WENZHU	1.5	1.5 Satisfied: the report s
11121670 Xinyue Zhang	1.5	1.5 Full credit: the Wokw
11120269 Zhao Zhiyi	1.3	1.5 Points removed: the
11083784 Zihao Zheng	1.3	1.5 Mostly satisfied, but
11090815 Zhu Wen	1.3	1.5 Points removed: the
11044962 Mohammad Javad Zandiyeh	1.5	1.5 Full credit: the report
10798825 Andrea Giovanni Zanelli	1.3	1.5 Points removed beca
11109586 SeyedAman Zargari - Amirhossein Dibaj	1.5	1.5 Full credit: PIR + phot
10793061 Marco Zheng	1.3	1.5 Mostly satisfied, but
10835836 Andrea Ziccardi	1.5	1.5 Satisfied: the submis

1.0	1.0 Full credit: the power	2.3	2.5 Good impr	4.8	5.0 Section 3 -
1.0	1.0 Full credit: the energy	2.3	2.5 Strong im	4.8	5.0 Section 3 -
0.8	1.0 The energy calculatio	0.0	2.5 not demor	1.6	5.0 Section 1 -
0.9	1.0 The energy math and	2.3	2.5 The report	4.5	5.0 Section 1:
0.90	1.00 Energy math and tim	2.00	2.50 Only the d	4.20	5.00 Section 1:
0.90	1.00 Minus 0.1: the power	2.50	2.50 Satisfied: t	4.90	5.00 Section 2 -
1.00	1.00 Full credit: per-state	2.30	2.50 Strong im	4.60	5.00 Section 1 -
0.90	1.00 Strong energy derivat	2.30	2.50 Delta-base	4.70	5.00 Section 2:
1.0	1.0 Full credit: per-state	2.3	2.5 Good impr	4.8	5.0 Section 3 -
1.00	1.00 Full credit: the report	2.50	2.50 Full credit:	5.00	5.00 No deduct
0.8	1.0 Mostly satisfied: the	2.3	2.5 Good impr	4.4	5.0 Section 1 -
0.90	1.00 energy model and life	2.30	2.50 clear recor	4.70	5.00 Section 2 h
0.00	1.00 not demonstrated	0.00	2.50 not demor	0.00	5.00 No deliver
0.90	1.00 Mostly satisfied: the	2.00	2.50 Partial cre	4.40	5.00 0.1 remov
1.0	1.0 Full credit: per-state	2.3	2.5 Strong im	4.8	5.0 Section 3 -
1.0	1.0 Full credit: per-state	1.0	2.5 Only parti	3.5	5.0 Section 3 -
1.0	1.0 Full credit: per-state	2.0	2.5 Good impr	4.5	5.0 Section 3 -
0.6	1.0 Partial: the cycle mod	1.5	2.5 Improvem	2.8	5.0 Section 1 -
0.5	1.0 Partial: the report giv	1.0	2.5 Only parti	2.0	5.0 Section 1 -
1.0	1.0 Full credit: per-state	2.3	2.5 Strong im	4.8	5.0 Section 3 -
0.8	1.0 Points removed: the	2.3	2.5 Good impr	4.4	5.0 Section 1 -
1.0	1.0 Full credit: the timing	1.5	2.5 Only parti	3.8	5.0 Section 1 -
1.0	1.0 Full credit: per-state	2.5	2.5 Full credit:	5.0	5.0 No deduct
1.0	1.0 Full credit: per-state	2.3	2.5 Strong im	4.8	5.0 Section 3 -
0.9	1.0 Mostly satisfied: the	2.0	2.5 Partial cre	4.4	5.0 Section 2 -
0.9	1.0 Mostly complete: the	2.0	2.5 Partial cre	4.4	5.0 Section 2 -
1.0	1.0 Full credit: repeated t	2.0	2.5 Partial cre	4.3	5.0 Section 1 -
1.0	1.0 Full credit: the report	2.0	2.5 Good reco	4.3	5.0 Section 1 -
0.8	1.0 Points removed: the	1.8	2.5 Points rerr	3.9	5.0 Section 1 -
1.0	1.0 Full credit: per-state	2.5	2.5 Full credit:	5.0	5.0 No deduct
0.9	1.0 Points removed beca	2.3	2.5 Points rerr	4.5	5.0 Section 1 -
0.9	1.0 Mostly satisfied: per-	2.5	2.5 Full credit:	4.9	5.0 Section 2 -
0.9	1.0 Mostly complete, but	1.5	2.5 Partial imp	3.7	5.0 Section 1 -
0.8	1.0 Partially satisfied: the	2.0	2.5 Partially sa	4.3	5.0 Section 2 -

The baseline is fine, and the improvement section is well argued. The remaining gap is that the wake-up st
Baseline and improved sketches are both present. The improvement is convincing, but it is change-detecti
Main misalignment is the sensor substitution: the project is structurally close to the assignment, but it do
The submission is substantially complete on the baseline specification and energy estimation, but the grac
The submission includes a Wokwi ZIP, report, and baseline sketch, but `OnDataSent` contains an extra ser
Main anomaly: the baseline sketch enables Wi-Fi before the sensor read, while the improved sketch correc
The submission is strong overall and includes both baseline and optimized code. Major anomaly: the runr
This is a strong submission with both baseline and improved code plus a detailed report. Theoretical sectic
Evaluated using the nested-folder person code 10712273 as instructed; the outer folder suffix 2661978 was
The previous empty-folder evaluation is stale; this submission contains both baseline and optimized Woki
Main anomaly: the sketch enables Wi-Fi before sensor reads, which weakens the section 1 ordering requir
This submission is evaluable and the nested-folder first student code is 11119541. The main remaining gap
Submission folder is empty; no evaluable report or code artifacts were provided.

Section 1 is solid and the leader-code-dependent timing/energy constants are correct for person code 111
Calibrated with the shared rows/examples; improvement code is present and the recomputed energy is tr
The submission is solid for the baseline challenge, but the improvement section is not really implemented
The baseline is clean and the improvement analysis is credible, but the optimized sketch is not delivered a
Major mismatch: the report is framed as the challenge solution, but the delivered hardware model does n
Major inconsistency: the report describes a full node, but the packaged sketch does not implement the re
The submission is well aligned overall; the improvement is meaningful, but it still stops short of the payloa
Improvement artifacts are split across multiple zip bundles, which makes the optimized workflow harder t
The baseline analysis is usable, but the improvement section is mostly prospective rather than an execut
Strong, well-aligned submission; the improved code and the recomputed analysis are both present and cc
The baseline and optimized modes are both present and readable; the remaining gap is the lack of a true i
The main grading anomaly is that the submission folder contains only the PDF and the Wokwi-link text, so
Major anomaly: the bundle contains only the baseline sketch even though Section 3 describes optimizatio
The actual sketch and Wokwi diagram fit the assignment, but the README inside the project is unrelated I
The baseline implementation is solid and the code archive is present, but the message format should be al
Baseline motion-sensor functionality is present, but the energy section is not backed by the requested sha
The submission is complete and internally consistent. The only minor limitation is that it is report-heavy r
Major inconsistency: the PDF baseline narrative says sensing happens before Wi-Fi init, but the first evalu
Main anomaly: the optimized implementation is present and aligned with the rubric, but the submission c
The submission has the baseline node and code, but the message string and the improvement section sho
Main anomaly: the report's Q2 battery-energy line is internally inconsistent (`5835%5000` vs final 15836 J),

checked against the shared CSV workflow, and the improvement section remains partially a report-level sketch.

The sketch is well aligned with the rubric.

The student wires the PIR to D2, so that mapping should be checked.

The student is in a true wake-up redesign.

The sketch is limited to neighboring entries.

The student uploads file; the improvement path is credible but still timer-based rather than interrupt-driven.

The student is saving optimizations in an active sketch.

The student and the Section 3 optimization remains partial rather than event-driven. Theoretical section 4 was ignored.

for scoring.

The program includes a photoresistor that is not used by the code. Theoretical section 4 was ignored for scoring and CSV baseline, so the instructor should verify the source measurement file used for section 2.

The inconsistency is that the PDF claims an optimized implementation, while the code remains a baseline loop with

and sketches are present and the report is strong, but the first-code rule means the baseline score follows the rubric set.

scored.

etch rather than a fully delivered optimized implementation.

d for scoring.

.

th fixed 2 s deep sleep.

hat earlier version.

student_name	person co	q1	q2	q3		
Yehia Abdallah - Amr Abdelhalim	11137282	1.5	1.00	2.3	4.80	no explicit
Raul Agolli	10783671	1.5	1	2.5	5.00	Not max b
Edoardo Albini	10844134	1.5	1.00	2.3	4.80	does not i
Hadi El Hajj	11140550	1.3	1.00	1.50	3.80	Baseline fu
Simone Amighini	10832147	1.5	1.00	1.5	4.00	The code i
Marco Apollonio - Francesco Azzarà	10764083	1.5	0.90	2.3	4.70	Small pow
Giacomo Aragnetti	10892569	1.5	1.00	2.3	4.80	Ok but mis
Francesco Armuzzi	10922631	1.5	0.80	2.0	4.30	no improv
Andrea Ardu - Laura Maria Bosco	10708185	1.5	1.00	2.5	5.00	
Bruno Arena - Matteo Cavalleri	10806605	1.5	1.00	2.5	5.00	
Zhanat Argimbayeva	11059833	1.5	0.80	2.3	4.60	Missing W
Yasin Asgari - Armin Mardoukhi	11019965	1.5	0.90	2.3	4.70	Missing W
Alperen Aydin - Kayra Uckilinc	11061975	1.5	1.00	2.0	4.50	Missing ak
Riccardo Baiardi	10704990	1.5	0.70	2.3	4.50	Missing pa
Ermir Balliu - Marco Scianna	11140947	1.5	1.00	2.0	4.50	Missing ak
Fernando Javier Barcelata Fuentes	11026049	1.5	0.90	2.5	4.90	
Giulia Barresi + Cesar Adrian Carbajal	11068760	1.5	1.00	2.30	4.80	Missing pa
Gabriele Benedetti + Giuseppe Bucchi	10829488	1.5	1.00	2.30	4.80	Missing pa
Matteo Bergagna	10771308	1.5	1.00	2.40	4.90	Missing de
Rebecca Besana + Noemi Fazzini	10828027	1.5	1.00	2.00	4.50	Thereticall
Simone Bianchessi	10716383	1.3	1.00	2.00	4.30	Integer usi
Bidini Edoardo - Oglier Badessi Mattia	10917333	1.5	1.00	2.50	5.00	
Alberto Biffi	10677341	1.5	1.00	2.50	5.00	
Yiran BI + Shuai ZOU	11030334	1.5	0.90	1.50	3.90	Mismatch
Mattia Brianti - Alex Hathaway	10773859	1.5	1.00	2.30	4.80	ayload size
Mattia Angelo Brunetti	10497245	1.5	0.80	2.00	4.30	Missing idl
CAI CHUANHAO	11166508	1.3	0.80	2.30	4.40	Missing so
Pietro Callandrone - Lorenzo Bianco	10774521	1.5	1.00	2.30	4.80	Missing pa
Andrea Callari - Simone Checchia	10764548	1.5	1.00	2.50	5.00	
Ronald Cammaroto	10768882	1.5	1.00	2.00	4.50	Missing co
Caon Mattia - Bertazzoli Cova Marco	10853745	1.5	1.00	2.40	4.90	The report
Morena Caterina Cardenas Estrada - S	10797131	1.5	0.90	2.50	4.90	Small erroi
Marco Carminati - Om Mohan Deshm	10790205	1.5	0.90	2.00	4.40	window is
Claudia Carraro - Chiara Cavallo	10910881	1.3	1.00	2.00	4.30	Code not g
Stefano Castiglione - Samuele Dario S	10796163	1.5	1.00	2.30	4.80	Missing pa
Keyan Cheng	11154362	1.5	1.00	2.30	4.80	Missing pa
Federico Ciarniello - Andrea Mori	10861756	1.5	1.00	2.30	4.80	Missing pa
Yuri Cinesi Ferrari	10666869	1.5	1.00	2.50	5.00	
Marco Coviello	10857924	1.5	1.00	2.30	4.80	Missing pa
Alessandro Cirillo - Jaime Iriarte	10850482	1.5	1.00	2.50	5.00	
Gabriele Clara Di Gioacchino - Andrea	10717254	1.5	1.00	2.30	4.80	Missing pa
Martin Corchuelo Verjano - Hamidrez	10914222	1.5	1.00	2.30	4.80	Missing pa
Alessandro Andrea Maria Costa	10774792	1.5	1.00	2.50	5.00	
Daniele Crippa	10838600	1.3	1.00	2.50	4.80	Missing pa
Alessandro Cupini	10975089	1.5	0.80	2.00	4.30	Sensing pf
Osman Bugra DAGLI - Erdem Cobano	11006775	1.0	0.70	1.50	3.20	Missing co
Dai Run Jie - Simone	10766478	1.3	0.90	2.00	4.20	Missing pa
Marco Stefano De Bernardi - Federico	10711043	1.5	1.00	2.00	4.50	Missing pa
Riccardo De Giacomo - Fabrizio Grima	11101495	1.5	1.00	2.50	5.00	Missing pa
Zarlene Justrem De Mesa - Guglielmo	10683374	1.5	1.00	2.50	5.00	

: recomputed energy plot is provided/comparison. Payload size reduction not proposed
because students didn't implement the energy improvements on waking up when needed, they apply a logic to
implement motion-interrupt-triggered deep sleep
functionality is mostly present, but `esp32-pro.ino` enables Wi-Fi before sensor reading and the payload size
is not reimplemented in any sort, small comment last point
power consumption error on TX power, no wakeup based on PIR sensor
missing the use of the interrupt based deep sleep, no consideration on the payload size
desired implementation is delivered and the expected TX-power reduction, payload reduction, and PIR-interrupt

wakeup trigger based, payload size as well
wakeup trigger based, payload size as well. No graph shown even if requested
missing a logic to reduce the unnecessary send, no payload size reduction
payload size reduction, missing deep sleep trigger based
missing a logic to reduce the unnecessary send, no payload size reduction

payload size reduction, missing deep sleep trigger based
payload size reduction, missing deep sleep trigger based
deep sleep trigger based
even if the solution has been reported however the code implementation and the energy computation do not
reflect the value of deep sleep rather than appropriate value, so deep sleep is not compliant with the requirements. does

on computation and code execution, simple comment no re-implementation, still week considerations for
the solution is unchanged, wake-up remains timer-based instead of PIR-interrupt-driven, and no recomputed improvement
on the boot period from energy estimations, the code is reimplemented but not attached to the submission on
the power state and mismatch between the code implemented. Missing payload size reduction, missing
payload size reduction, missing deep sleep trigger based

comments on recomputation energy good the implementation on trigger wake up
missing: contained the correct implementation however a small penalty 0.1 for not delivering the codebase
missing on the power measurement, trigger based interrupt stated but not implemented
missing not the full startup cost because timing starts after `delay(80)`, Wi-Fi reset, and `delay(100)` in code, Table
given, missing improvement code, small error as turn on wifi before sensing
payload size reduction, missing deep sleep trigger based
payload size reduction, missing deep sleep trigger based
payload size reduction, missing deep sleep trigger based

payload size reduction, missing deep sleep trigger based

payload size reduction, missing deep sleep trigger based
payload size reduction, missing deep sleep trigger based

payload size reduction, missing deep sleep trigger based, however reported as possible condition
base not equal boot phase, no new implementation generic discussion simple optimization no delta sending
code comment, missing energy computation, missing new energy estimation and discussion, improvement
payload size reduction, missing deep sleep trigger based or anything related to delta sending
payload size reduction, missing deep sleep trigger based or anything related to delta sending consistent with
payload size reduction, missing deep sleep trigger based or anything related to delta sending consistent with

ic on deltas which is not sufficient however i didn't removed full mark as they have briefly stated in the pc

rings use spaces instead of the exact required underscore format. just a single meas is performed no val

ipt wake-up are absent.

involved it, laking of other optimization like any event driven wake up or transmission. Optimization cod
not implement the expected payload reduction or interrupt-driven deep sleep, and the light-sleep sketch

or full mark

red power/energy plot is provided.

nly reported in the pdf hard to read. Imporvements not compete as missing deep sleep and payload size
deep sleep trigger based

3 4 gives `WiFi Off + Idle 2` mean `7093.1 us` despite a stated max of `6197 us` Improvements margibnalyy

l or any optimization

nts are submitted but not enough

h the movement

h the movement

If as possible optimization, they also didn't reduced the payload size

dated approach using multiple measurement, bad result reporting missing some states. The new implem

e even if reported is missing

1 disconnects Wi-Fi without reinitializing ESP-NOW for later cycles.

reduction

efficient only proposed the duty cycle increase

ventilation is not performed just duty cycle changed

Luca De Nicola	10808901	1.5	1.00	2.10	4.60	Missing p
Alessia Di Capua - Francesco Fioretti	10712008	1.5	1.00	2.00	4.50	Missing p
Leonardo Di Giovanni	10888593	1.3	1.00	2.00	4.30	WiFi turne
Zhenze Ding - Zhaojun Liu	11053611	1.5	1.00	2.30	4.80	Missing p
Angelo Di Rosa - Livia Polverari	10702601	1.5	1.00	2.40	4.90	Small pen
Luca Donato	10660050	1.5	1.00	2.30	4.80	Missing p
Arturo Amoretti - Kristi Duro	10799588	1.5	1.00	2.30	4.80	Missing p
Mohamed Youssef Mohamed Youssef	11101890	1.0	0.20	0.50	1.70	Wrong X, \
Amr Elmasry - Amjed Elsadig	11036293	1.5	1.00	2.00	4.50	Boot + Wif
Tommaso Ernst - Alessandro Paolo G	11164465	1.5	1.00	2.30	4.80	Missing p
Fang Junhua - Chai Ke	11078923	1.5	1.00	2.30	4.80	Missing p
Yuhang Fang	11117655	1.5	1.00	2.30	4.80	Missing p
Federico Fasola - Marco Fontana	10841085	1.5	1.00	2.30	4.80	
Alessandro Fedele - Dario Crosa	10767814	1.5	1.00	2.50	5.00	
Feijoo Sandoya Jennifer Valeria	10810258	1.3	1.00	2.00	4.30	Partially in
Lucia Fernandez Giner	11123116	1.3	1.00	2.30	4.60	Partially in
Giovanni Ferola - Lorenzo Della M	10770813	1.5	1.00	2.50	5.00	
Ferrando Alessandro	10938858	1.5	1.00	2.50	5.00	
Ferrari Matteo	10903183	1.5	1.00	2.30	4.80	Missing p
Edoardo Figini - Greta Fornaciari	10771695	1.5	0.90	2.00	4.40	Missing p
Foini Lorenzo - Gotta Filippo	10828129	1.5	1.00	2.50	5.00	
Matteo Franken - Lorenzo Chiroli	10831046	1.5	1.00	2.30	4.80	Missing p
Galbiati Samuele	10773862	0.0	0.50	1.50	2.00	Missing ev
Dinar Gareev - Yiming Hou	11063702	1.5	1.00	2.50	5.00	
Giacomo Genco - Valeria Iagulli	10805688	1.5	1.00	2.00	4.50	Missing p
Zihao Geng - Marica Balestra	11109083	1.3	1.00	2.00	4.30	Missing p
Roberta Gennari	10797129	1.5	1.00	2.30	4.80	Missing p
Ghiaroni Matteo	11102420	1.5	1.00	2.30	4.80	Missing p
Eugenia Ghiglione - Andrea Cucchetti	10711947	1.5	1.00	2.30	4.80	Missing p
Davide Ghisolfi	10839162	1.5	1.00	2.00	4.50	Missing p
Carmen Giaccotto	10709837	1.5	1.00	2.30	4.80	Missing p
Andrea Gianfelice	10846339	1.5	1.00	2.00	4.50	Missing p
Simone Giuliano	10811640	1.5	1.00	2.30	4.80	Missing p
Davide Greco	10732218	1.5	1.00	2.30	4.80	Missing p
Enrico Grillo	10778074	1.5	1.00	2.30	4.80	Missing p
Samuele Grisoni	10810440	1.5	1.00	2.50	5.00	
Luca Guerra	10720381	1.5	1.00	2.50	5.00	
Mengzhen Han & Shiming Gao	11087290	1.3	1.00	2.00	4.30	Missing p
HAN ZHAOHUI	11167997	1.5	1.00	2.50	5.00	
Jinxu Hong	11110511	1.3	1.00	2.30	4.60	Missing p
Rong Huang	10948935	1.5	1.00	2.30	4.80	Missing p
Jialin Hu - Ziyin You	11063424	1.5	1.00	2.00	4.50	Missing p
Hu Kaipeng	11036887	1.5	1.00	2.00	4.50	Missing p
Peng Hu	11032747	1.0	1.00	2.00	4.00	Penalizatic
Manoueil Michael Halim Riad Hanna	10808527	1.5	1.00	2.50	5.00	Small pen
Alex IORTOMAN	11147720	1.5	1.00	2.50	5.00	
Alireza Jahandoost	11051292	1.5	1.00	2.30	4.80	Missing p
JIANG ZHENGLIN	11030873	1.3	1.00	2.00	4.30	WiFi enabl
JIA_TONGXIN	11019981	1.3	1.00	2.30	4.60	Missing p
Yihang Ji	11162444	1.3	1.00	2.30	4.60	Missing p
Razia Jafari	11121758	1.5	0.80	2.00	4.30	Wrong Psl

payload size reduction, missing deep sleep trigger based, no new code delivered just in pdf no fully energy
payload size reduction, missing deep sleep trigger based or anything related to delta sending consistent with
done before the actual logic, waste of energy no code delivery. Missing payload size reduction, missing de
payload size reduction, missing deep sleep trigger based
realization due to incomplete report of Q3
payload size reduction, missing deep sleep trigger based
payload size reduction, missing deep sleep trigger based
wifi on before sensing, value in voltage no lux, no energy consumption plot and incomplete 3 line for Q3 no
`ON` merged in power estimation (still fine), missing energy recomputation, Missing payload size reducti
payload size reduction, missing deep sleep trigger based
payload size reduction, missing deep sleep trigger based
payload size reduction, missing deep sleep trigger based

implemented improvement, no code no new energy estimation on top of enlarging the duty cycle
implemented improvement, no code since was 1 line change however sending if motion

payload size reduction, missing deep sleep trigger based
payload size reduction, missing deep sleep trigger based or anything related to delta sending consistent with

payload size reduction, missing deep sleep trigger based
anything

payload size reduction, missing deep sleep trigger based or anything related to delta sending
payload size reduction, missing deep sleep trigger based or anything related to delta sending
payload size reduction, missing deep sleep trigger based
payload size reduction, missing deep sleep trigger based
payload size reduction, missing deep sleep trigger based
payload size reduction, missing deep sleep trigger based, missing implementation
payload size reduction, missing deep sleep trigger based
payload size reduction, missing deep sleep trigger based, missing code
payload size reduction, missing deep sleep trigger based
payload size reduction, missing deep sleep trigger based
payload size reduction, missing deep sleep trigger based

payload size reduction, missing deep sleep trigger based or anything related to delta sending consistent with

payload size reduction, missing deep sleep trigger based. WiFi on before sensing opposite of Q2
payload size reduction, missing deep sleep trigger based.
payload size reduction, missing deep sleep trigger based or anything related to delta sending consistent with
payload size reduction, missing deep sleep trigger based, missing code improvements
on because code is not provided, is provided just as wokwi link
realization because code is no

payload size reduction, missing deep sleep trigger based.
done before sensing lack of optimized code, no PIR based wakeup or turnoff of the sensor.
payload size reduction, missing deep sleep trigger based. WiFi on before sensing
payload size reduction, missing deep sleep trigger based. WiFi on before sensing
sleep, no energy recomputation

recomputation.

h the movement

ep sleep trigger based, no new code deleivered just in pdf no fully energy recomputation.

code no computation

on, missing deep sleep trigger based, no original code deleivered just in pdf no fully energy recomputatic

h the movement

h the movement

h the movement

in.

Sevval Karadeniz - Haga Zenab Moha	11058806	1.5	1.00	2.50	5.00	
Ozgur Efehan KATI - Firat Adem BILGE	11035971	1.5	1.00	2.30	4.80	Missing pæ
Mobin Khatib - Ali Edareh Heidarabad	11114435	1.5	1.00	2.50	5.00	
Fatemeh Kianiarchegani	11111583	1.5	1.00	2.30	4.80	Missing pæ
Saltuk Bugra Kirazci	11143015	1.3	0.30	1.50	3.10	Missing co
Nicolas Koron	11187268	1.5	1.00	2.50	5.00	
Lorena Lam Díez - Sofía Merino López	11145475	1.5	1.00	2.00	4.50	Missing pæ
Francesco Lazzaro - Andrés Felipe For	11095668	1.5	1.00	2.30	4.80	Missing pæ
Oscar Lindqvist	11191868	1.5	1.00	1.50	4.00	Code desc
LI MING - LIU XIANGYU	11033376	1.3	0.90	2.00	4.20	WiFi on be
Yuanyuan Li	11110488	1.3	0.90	1.80	4.00	missing co
Eric Lombardo Facciale - Filippo Palaz	11161809	1.5	1.00	2.30	4.80	Missing pæ
Esteban Lopez Barreto - Ma Chong	11154567	1.5	1.00	2.30	4.80	Missing pæ
Ingrid Lyngstad	11157354	1.3	1.00	1.80	4.10	WiFi on be
Gabriel Leonardi	10813646	1.5	1.00	2.50	5.00	
Alessio Lika	10849638	1.4	1.00	2.40	4.80	Missing de
Elena Lin - Jiayi Zheng	10802468	1.5	1.00	2.00	4.50	Missing pæ
Ivana Lin	10607397	0.0	1.00	0.50	1.50	Code miss
Susanna Lo Cascio	10861416	1.5	1.00	2.50	5.00	
Lo Monaco Francesco Pio	10852883	1.5	1.00	2.30	4.80	Missing pæ
Samuele Lo Piccolo	10796320	1.5	1.00	2.30	4.80	
Stefano Longhi - Beatrice Spazzadesco	10782022	1.5	1.00	2.50	5.00	
Lorenzi Davide	10502570	1.3	1.00	2.30	4.60	Void setup
Davide Lupo	10738139	1.5	1.00	2.50	5.00	
Edoardo Menegotto	10730038	1.5	1.00	2.50	5.00	
Mobina Mobaraki	11034150	1.5	1.00	2.30	4.80	Missing pæ
Felicia Murkes	11189222	1.5	1.00	2.50	5.00	
Enrico Mantegazza	10862592	1.5	1.00	2.30	4.80	Missing pæ
Francesca Maragoni	10911594	1.3	1.00	2.00	4.30	
Francesco Marchi	10773779	1.5	1.00	2.50	5.00	
Giovanni Miccio	10819854	1.5	1.00	2.30	4.80	Missing pæ
Gabriele Michieletto	10806788	1.5	1.00	2.00	4.50	Missing pæ
Andrea Midulla	10813424	1.5	1.00	2.50	5.00	
Andrea Luca Miele	10727881	1.5	1.00	2.00	4.50	Missing pæ
Lorenzo Francesco Luigi Minorini	10710384	1.5	1.00	2.50	5.00	
Monzani Mario	10845860	1.5	1.00	2.30	4.80	
Motti Filippo	10816840	1.5	1.00	2.00	4.50	Missing pæ
Muyololo Emmanuel	11009540	1.1	1.00	2.00	4.10	WiFi on be
NGUYEN HAO JOHNY VINH	11152090	1.1	1.00	1.50	3.60	WiFi on be
NGUYEN MINH MAN	11026772	1.5	1.00	2.30	4.80	Missing pæ
NUYENS GOTELLI IGNACIA	11173956	1.5	1.00	2.00	4.50	Missing pæ
Nardi Andrea	10777415	1.5	1.00	2.50	5.00	
Neri Tommaso	10817232	1.5	1.00	2.30	4.80	Missing pæ
Nipoti Alessandro	10861979	1.5	1.00	2.30	4.80	Missing pæ
ORTIZ BALLESTEROS ANGELICA	11191225	1.5	1.00	2.50	5.00	
Giovanni Orciuolo	10994077	1.5	1.00	2.30	4.80	Missing pæ
Xiaohan Pan - Mengshuang Tang	11032539	1.3	1.00	2.30	4.60	Raw Lux væ
Zahra Sadat Parviss	11115876	1.5	1.00	2.00	4.50	Missing pæ
Anchen Peng - Qianyi Wang	11031890	1.5	1.00	2.30	4.80	Missing pæ
Luca Pezzoli - Mattia Rainieri	11135422	1.5	1.00	2.50	5.00	
Mariyam Rashid Pitafi	11119281	1.5	1.00	2.30	4.80	

payload size reduction, missing deep sleep trigger based.

payload size reduction, missing deep sleep trigger based.

code comment in report, missing energy consumption computation, missing recomputation of energy -0.5 r

payload size reduction, missing deep sleep trigger based and missing code

payload size reduction, missing deep sleep trigger based

described briefly and missing code implementation and energy computation

before sensing, Idle / boot like wifi on state, incomplete proposed limited

code implementation and solution not complete

payload size reduction, missing deep sleep trigger based

payload size reduction, missing deep sleep trigger based

before sensingSolution to point 3 super limited briefly discussed good energy recomputation still not enough

deliverables penalty on wrong submission

payload size reduction, missing deep sleep trigger based, no event based trigger approach used

working completely, optimization briefly discuss no code provided no energy estimation

payload size reduction, missing deep sleep trigger based

Missing payload size reduction, missing deep sleep trigger based

payload size reduction, missing deep sleep trigger based

payload size reduction, missing deep sleep trigger based

payload size reduction, missing deep sleep trigger based

payload size reduction, missing deep sleep trigger based, no code

payload size reduction, missing deep sleep trigger based, no code

payload size reduction, missing deep sleep trigger based, no event based trigger approach used

before sensor reading and lux as raw adc value Missing payload size reduction, missing deep sleep trigger based

before sensor reading and lux as raw adc value Missing payload size reduction, missing deep sleep trigger based

payload size reduction, missing deep sleep trigger based

payload size reduction, missing deep sleep trigger based, missing conditional sending

payload size reduction, missing deep sleep trigger based

payload size reduction, missing deep sleep trigger based

payload size reduction, missing deep sleep trigger based

al, Missing payload size reduction, missing deep sleep trigger based

payload size reduction, missing deep sleep trigger based, no event based trigger approach used

payload size reduction, missing deep sleep trigger based

missing comment of new implementation -0.5 missing

zh

used, no event based trigger approach used, no code delivered

used, no event based trigger approach used, no code delivered, no energy recomputation

Mattia Panzeri - Diego Butti	10777877	1.5	1.00	2.30	4.80	
Eliseo Paoletti - Gabriele Nicolo	10778484	1.5	1.00	2.50	5.00	
Salvatore Passaro	10697294	1.5	1.00	2.50	5.00	
Giacomo Pasturensi	10742323	1.5	1.00	2.00	4.50	Missing pā
Lorenzo Pedranzini	10778819	1.5	1.00	2.50	5.00	
Jacopo Pelagalli	10828995	1.5	1.00	2.50	5.00	
Simone Perugini	10773372	1.5	1.00	2.30	4.80	Missing pā
Mario Pesca	10777399	1.5	1.00	2.30	4.80	Missing pā
Alessandro Pessina	10619593	1.5	1.00	2.50	5.00	
Davide Petruzzi	10900762	1.5	1.00	2.30	4.80	Missing pā
Alberto Pini	10823873	1.5	1.00	2.50	5.00	
Pietro Pizzoccheri	10797420	1.5	1.00	2.50	5.00	
Alessandro Posani	10705152	1.5	1.00	2.50	5.00	
Marco Puddu	10855966	1.5	1.00	2.50	5.00	
JinWen Qi	11048190	1.5	1.00	2.00	4.50	Missing pā
Yongqi Qiu	10962875	1.5	1.00	2.30	4.80	Missing co
Peng Rao	11022931	1.5	1.00	2.50	5.00	
Amir Hossein Rashidy	11046392	1.5	1.00	2.50	5.00	
Rashwan Mohannad Mohamed Fawz	11128210	1.5	1.00	2.30	4.80	Missing pā
Muhammad Rian	11111393	1.5	0.90	2.20	4.60	Deepsleep
Alessia Radicula	10770385	1.5	1.00	2.50	5.00	
Alessia Ragosta	10837597	1.5	1.00	2.50	5.00	
Gabriele Riboldi - Luca Villa	10768645	1.5	1.00	2.00	4.50	Considerat
Flavio Riccobono	10793255	1.5	1.00	2.50	5.00	
Rodari Simone	10863960	1.5	1.00	2.50	5.00	
Alessandra Romano	10724267	1.5	1.00	2.30	4.80	Missing pā
Alessio Vittorio Rosano	10789693	1.5	1.00	2.50	5.00	
Tommaso Rossetti	10851929	1.5	1.00	2.50	5.00	
Andrea Rossi	10799170	1.5	1.00	2.50	5.00	
Matteo Rossi	10798975	1.5	1.00	2.50	5.00	
Andrea Rubagotti	10780379	1.5	1.00	2.50	5.00	
Davide Rutigliano	10772022	1.5	1.00	2.50	5.00	
Maciej Krzysztof Rzewuski	10847803	1.5	0.00	0.50	2.00	Missinge e
Amin Sakhaei	11119582	1.5	1.00	2.50	5.00	
Anasuya Satapathy	11175090	1.5	1.00	2.30	4.80	Missing pā
Ghazal Sepehrirad	11010656	1.5	1.00	2.50	5.00	
Sajjad Shaffaf	11063739	1.1	1.00	2.50	4.60	
Dalila Samr	10495016	1.3	1.00	2.00	2.00	sending ra
Andrea Santaniello	10830365	1.5	1.00	2.50	5.00	
Edoardo Sartore	10825131	1.5	1.00	2.50	5.00	
Andrea Sau	10812603	1.5	1.00	2.50	5.00	
Giacomo Scampini	10764570	1.5	1.00	2.30	4.80	Missing pā
Andrea Gaetano Senatore	10698107	1.5	1.00	2.50	5.00	
Gabriele Bogdan Sepe	10744406	1.5	1.00	2.50	5.00	
Golsa Shokri	10821199	1.5	1.00	2.50	5.00	
Gabriele Lorenzo Singe	10766937	1.5	1.00	2.50	5.00	
Francesco Street	10769571	1.5	1.00	2.50	5.00	
Cristian Summa	10830061	1.5	1.00	2.50	5.00	
Kelly Tran	11151327	1.5	1.00	2.50	5.00	
Orhun Tunc	11120812	1.3	1.00	2.00	4.30	WiFi on be
Michele Tavecchio	10832525	1.5	1.00	2.50	5.00	

payload size reduction, missing deep sleep trigger based, missing conditional sending

payload size reduction, missing deep sleep trigger based

payload size reduction, missing deep sleep trigger based

payload size reduction, missing deep sleep trigger based

payload size reduction, missing deep sleep trigger based, missing conditional sending, missing full code delivery reimplementation

payload size reduction, missing deep sleep trigger based

outlier, improvements ok in logic limited in implementation discussion and energy computation

discussion on energy consumption missing, no code reimplementation

payload size reduction, missing deep sleep trigger based

energy consumption, missing q3 only theoretical

payload size reduction, missing deep sleep trigger based

with lux, missing code implementation and energy re-estimation is incomplete

payload size reduction, missing deep sleep trigger based

before sensing, partially complete no reimplementation just increases sleep time

vered

Simone Teggi	10847211	1.5	1.00	2.30	4.80	Missing pa
Federico Trubian	10816921	1.5	1.00	2.30	4.80	Missing pa
Torodd Urdshals	11156535	0.8	0.80	1.50	3.10	Wrong de
Christian Vaktel	11152089	1.4	1.00	2.20	4.60	Penalty on
Juan David Vargas Galindo - Cristian D	11115207	1.3	1.00	2.00	4.30	Problem w
Shaghayeh Vesali	11046642	1.5	1.00	2.50	5.00	
Andrea Vavassori - Marco Gad	10781960	1.3	1.00	2.30	4.60	raw lux val
Davide Vendramin - Matteo Taschini	10787431	1.5	1.00	2.30	4.80	Missing pa
Matteo Vicentini	10712273	1.5	1.00	2.30	4.80	Missing pa
Marco Vitali	10676235	1.5	1.00	2.50	5.00	
Guangyi Wang	11118429	1.4	0.90	2.30	4.60	Idle/boot i
Yizhe Wang	11119541	1.5	1.00	2.30	4.80	
Yunxuan Wang	11119918	1.3	0.90	2.00	4.20	Wifi on be
TianQi Wu	11154359	1.5	0.90	2.00	4.40	lowers TX
Yuan Wang	10863400	1.5	1.00	2.30	4.80	no pir wak
Erdal Yalcin	11052807	1.5	1.00	0.00	2.50	
Yawen Yan	11015522	1.5	1.00	1.80	4.30	Limited op
Arzu Yildiz	11057250	0.8	0.50	1.80	3.10	Code wror
Yu Xincheng	11027977	1.3	0.50	0.80	2.60	Limites/sr
Haoyu Zhang	11061927	1.5	1.00	2.50	5.00	
Hezhi Zhang	11033400	1.3	0.90	2.30	4.50	Bot/idle no
Jie Zhang	11044575	1.4	1.00	2.30	4.70	
Ning Zhang	11107611	1.5	1.00	2.50	5.00	
Ting Zhang	11048104	1.3	1.00	2.20	4.50	Wrong lux
ZHANG WENZHU	11171264	1.4	1.00	2.20	4.60	Missing de
Xinyue Zhang	11121670	1.5	1.00	2.00	4.50	Missing de
Zhao Zhiyi	11120269	1.3	1.00	1.80	4.10	Correct, bu
Zihao Zheng	11083784	1.5	1.00	2.00	4.50	Penalty or
Zhu Wen	11090815	1.3	0.60	1.00	2.90	Incomplet
Mohammad Javad Zandiyeh	11044962	1.5	1.00	2.50	5.00	
Andrea Giovanni Zanelli	10798825	1.5	1.00	2.30	4.80	
SeyedAman Zargari - Amirhossein Dib	11109586	1.5	1.00	2.50	5.00	
Marco Zheng	10793061	1.3	1.00	2.50	4.80	
Andrea Ziccardi	10835836	1.5	1.00	2.30	4.80	Missing ex
Cezar Gjinaj	10683981	1.5	1.00	2.50		
Simone Azzoni	10831959	1.5	1.00	2.50		
	11021337	1.5	1.00	2.00		Missing co
	10764433	1.5	1.00	2.30		no rip wak
	10839190	1.3	1.00	2.00		
	11062471	1.5	1.00	2.00		
Zhaochen Qiao	11021721	1.5	1.00	1.20		Missing pc
Alice Berta	10889590	1.5	1.00	2.30		

payload size reduction, missing deep sleep trigger based
payload size reduction, missing deep sleep trigger based
sign of project, wrong power estimation, the system design is incomplete no new code no clear how timing
missing reimplementation code and main code submission
with nested send in on data sent callback, no conditional sending

code sent, no pir wakeup
payload size reduction, missing deep sleep trigger based
payload size reduction, missing deep sleep trigger based

missing, Q1 code poorly commented

before sensing, power a bit off, Missing payload size reduction, missing deep sleep trigger based
power and extends deep sleep
wakeup

optimization, no implementation
no sensor, partial energy consumption estimation
no improvement, partial energy consumption study

not present, no PIR

no sensor, missing full submission
no deliver no rip wake
no alive, missing new energy computation
not missing actual improvement
no code not delivered, no conditional send
no energy, wifi on before sensing, weak improvements

explicit energy computation

conditional sending
wakeup

point 3 only theoretical discussion proposed

g are com

Person Code 1	Person Code 2	Q1	Q2	Q3	Q4	Tot
10780379	10799787	1.5	1	2.5	2	7
10975089		1.5	0.8	2	0.75	5.05
11119541	11115748	1.5	1	2.3	2	6.8
11030873		1.3	1	2	2	6.3
10666869		1.5	1	2.5	2	7
11061975	11049291	1.5	1	2	2	6.5
10716383		1.3	1	2	2	6.3
10768645	10768851	1.5	1	2	2	6.5
10770813	10799739	1.5	1	2.5	2	7
10962875		1.5	1	2.3	0.75	5.55
11015522		1.5	1	1.8	2	6.3
10845860		1.5	1	2.3	0.75	5.55
11027977		1.3	0.5	0.8	1.25	3.85
10764083	10796209	1.5	0.9	2.3	2	6.7
11101495	11101728	1.5	1	2.5	2	7
10994077		1.5	1	2.3	2	6.8
10771695	10862436	1.5	0.9	2	2	6.4
10796320		1.5	1	2.3	1.75	6.55
10683374	10764592	1.5	1	2.5	2	7
11030334	11024119	1.5	0.9	1.5	1.5	5.4
10787431	10785383	1.5	1	2.3	2	6.8
10816840	10797758	1.5	1	2	2	6.5
10796163	10772778	1.5	1	2.3	2	6.8
10676235		1.5	1	2.5	2	7
10819854	10846682	1.5	1	2.3	2	6.8
10720381		1.5	1	2.5	2	7
10838600		1.3	1	2.5	2	6.8
10764570		1.5	1	2.3	2	6.8
10851929	10860913	1.5	1	2.5	2	7
10702601	10793007	1.5	1	2.4	2	6.9
10660050		1.5	1	2.3	1.75	6.55
10798825	10742249	1.5	1	2.3	2	6.8
11171264	11168874	1.4	1	2.2	2	6.6
11095668	11068591	1.5	1	2.3	2	6.8
10828027	11178565	1.5	1	2	2	6.5
10774521	11024306	1.5	1	2.3	2	6.8
10802468	10811585	1.5	1	2	2	6.5
10683981		1.5	1	2.5	2	7
10773862		0	0.5	1.5	1.5	3.5
10857924		1.5	1	2.3	2	6.8
10790205	11055131	1.5	0.9	2	1.25	5.65
10789693	10762257	1.5	1	2.5	2	7
11083784		1.5	1	2	2	6.5
10831959	10829712	1.5	1	2.5	2	7
10837597		1.5	1	2.5	1.5	6.5
11128210		1.5	1	2.3	1.25	6.05
11173956	11188762	1.5	1	2	1.25	5.75
10770385		1.5	1	2.5	2	7
10797420	10831941	1.5	1	2.5	2	7
10825131		1.5	1	2.5	2	7

10853745	10929780	1.5	1	2.4	2	6.9
11120812		1.3	1	2	2	6.3
10619593		1.5	1	2.5	2	7
10828995		1.5	1	2.5	2	7
10730038		1.5	1	2.5	2	7
11166508		1.3	0.8	2.3	2	6.4
11022931		1.5	1	2.5	2	7
11167997	10935200	1.5	1	2.5	2	7
11111393	11113677	1.5	0.9	2.2	1.25	5.85
11063702	11097125	1.5	1	2.5	2	7
10781960	10794601	1.3	1	2.3	2	6.6
10767814	10796197	1.5	1	2.5	2	7
10808901		1.5	1	2.1	2	6.6
10798975	10765707	1.5	1	2.5	2	7
11031890	11031608	1.5	1	2.3	1.25	6.05
11051292	11041035	1.5	1	2.3	2	6.8
11048104		1.3	1	2.2	1.5	6
11090815		1.3	0.6	1	0	2.9
10698107	10728121	1.5	1	2.5	2	7
11053611	11030423	1.5	1	2.3	2	6.8
10900762	11164535	1.5	1	2.3	2	6.8
10850482	11154921	1.5	1	2.5	2	7
11019965	11060212	1.5	0.9	2.3	0.75	5.45
11033376	11062619	1.3	0.9	2	2	6.2
10922631		1.5	0.8	2	2	6.3
10797131	11111318	1.5	0.9	2.5	2	6.9
10783671		1.5	1	2.5	2	7
11006775	11045157	1	0.7	1.5	2	5.2
11044962		1.5	1	2.5	2	7
10813424	10842807	1.5	1	2.5	2	7
11034150	10822446	1.5	1	2.3	2	6.8
11044575		1.4	1	2.3	2	6.7
11157354		1.3	1	1.8	1.25	5.35
10732218	10806471	1.5	1	2.3	2	6.8
11154567	11162609	1.5	1	2.3	1.25	6.05
11101890		1	0.2	0.5	1.5	3.2
10806605	10768765	1.5	1	2.5	1.5	6.5
10817232		1.5	1	2.3	2	6.8
11021337	11175090	1.5	1	2	2	6.5
11109083	10832004	1.3	1	2	2	6.3
11143015		1.3	0.3	1.5	1.25	4.35
10830061	10796466	1.5	1	2.5	2	7
11110511	11102304	1.3	1	2.3	2	6.6
11152090		1.1	1	1.5	#N/A	#N/A
11068760	11001933	1.5	1	2.3	2	6.8
10793061		1.3	1	2.5	2	6.8
10797129	10710507	1.5	1	2.3	2	6.8
11019981	11093193	1.3	1	2.3	1.25	5.85
11189222	11150698	1.5	1	2.5	2	7
10711043	10698676	1.5	1	2	2	6.5
11120269		1.3	1	1.8	2	6.1

10773859	10829946	1.5	1	2.3	2	6.8
11110488	11112969	1.3	0.9	1.8	1.25	5.25
10911594	10901067	1.3	1	2	1.25	5.55
10813646	10768605	1.5	1	2.5	1.75	6.75
11119281		1.5	1	2.3	2	6.8
10677341		1.5	1	2.5	2	7
11046392	11049700	1.5	1	2.5	1.25	6.25
10766937		1.5	1	2.5	2	7
10708185	10793514	1.5	1	2.5	2	7
11058806	11091338	1.5	1	2.5	2	7
11115876	11107660	1.5	1	2	2	6.5
11032747		1	1	2	2	6
11111583	11062873	1.5	1	2.3	1.25	6.05
10948935		1.5	1	2.3	2	6.8
11147720	11118976	1.5	1	2.5	2	7
11152089	11152488	1.4	1	2.2	2	6.6
10777877	10806864	1.5	1	2.3	2	6.8
10811640	10790904	1.5	1	2.3	1.25	6.05
11151327	11157416	1.5	1	2.5	1.25	6.25
10766478	10719178	1.3	0.9	2	1.25	5.45
10764433	10809273	1.5	1	2.3	2	6.8
10839162	10723736	1.5	1	2	2	6.5
10839190	10859045	1.3	1	2	2	6.3
11187268		1.5	1	2.5	2	7
11063424	11022646	1.5	1	2	2	6.5
11118976	11147720	1.5	1	2.5	2	7
10844134		1.5	1	2.3	2	6.8
11137282	11116374	1.5	1	2.3	1.5	6.3
10697294		1.5	1	2.5	2	7
10744406	10709235	1.5	1	2.5	2	7
10849638		1.4	1	2.4	2	6.8
11009540		1.1	1	2	2	6.1
10847211	10768863	1.5	1	2.3	2	6.8
10805688	10765606	1.5	1	2	2	6.5
10769571	10797678	1.5	1	2.5	1.5	6.5
10705152	10740222	1.5	1	2.5	2	7
10829488	10795884	1.5	1	2.3	2	6.8
10502570	10860834	1.3	1	2.3	2	6.6
10711947	10813527	1.5	1	2.3	2	6.8
11121670		1.5	1	2	1.75	6.25
10777415	10767901	1.5	1	2.5	2	7
11078923	11110613	1.5	1	2.3	2	6.8
11109586	11109540	1.5	1	2.5	2	7
11062471	10795184	1.5	1	2	2	6.5
11048190		1.5	1	2	2	6.5
11033400		1.3	0.9	2.3	2	6.5
10910881	10856023	1.3	1	2	1.25	5.55
10831046	10797603	1.5	1	2.3	2	6.8
11026772		1.5	1	2.3	2	6.8
11021721	10972650	1.5	1	1.2	1.25	4.95
11140947	10815900	1.5	1	2	0.75	5.25

10846339		1.5	1	2	2	6.5
10773779	10767244	1.5	1	2.5	1.25	6.25
11135422	11011342	1.5	1	2.5	2	7
10832525	10777259	1.5	1	2.5	2	7
10764548	10764758	1.5	1	2.5	2	7
11140947	10815900	1.5	1	2	0.75	5.25
10903183		1.5	1	2.3	2	6.8
10806788		1.5	1	2	2	6.5
11102420	10846947	1.5	1	2.3	2	6.8
10782022	10766466	1.5	1	2.5	2	7
11161809	11084799	1.5	1	2.3	2	6.8
11164465	10809515	1.5	1	2.3	2	6.8
10889590	10854816	1.5	1	2.3	2	6.8
10773372		1.5	1	2.3	2	6.8
11039171	10799588	1.5	1	2.3	2	6.8
11154359	11151255	1.5	0.9	2	2	6.4
10861416		1.5	1	2.5	2	7
11032539	11035477	1.3	1	2.3	2	6.6
10727881		1.5	1	2	2	6.5
10808527	10796043	1.5	1	2.5	1.75	6.75
10724267		1.5	1	2.3	2	6.8
10888593		1.3	1	2	1.25	5.55
10810440	10795876	1.5	1	2.5	2	7
10704990		1.5	0.7	2.3	2	6.5
10917333	10724359	1.5	1	2.5	2	7
10841085	10767413	1.5	1	2.3	2	6.8
11175090	11021337	1.5	1	2.3	2	6.8
10778484	10848911	1.5	1	2.5	2	7
10821199		1.5	1	2.5	1.25	6.25
11117655		1.5	1	2.3	2	6.8
11046642	11130192	1.5	1	2.5	2	7
11026049		1.5	0.9	2.5	1.5	6.4
10710384	10721613	1.5	1	2.5	2	7
11057250	11057249	0.8	0.5	1.8	0.75	3.85
10861979	11108477	1.5	1	2.3	2	6.8
10778074	10772713	1.5	1	2.3	2	6.8
10778074	10772713	1.5	1	2.3	2	6.8
10938858		1.5	1	2.5	2	7
10768882		1.5	1	2	2	6.5
10812603	10859228	1.5	1	2.5	2	7
10816921		1.5	1	2.3	2	6.8
11123116		1.3	1	2.3	2	6.6
10712273		1.5	1	2.3	2	6.8
10774792		1.5	1	2.5	2	7
10823873	10822654	1.5	1	2.5	2	7
10772022	11095908	1.5	1	2.5	1.5	6.5
10709837	10773289	1.5	1	2.3	2	6.8
11063739	11060363	1.1	1	2.5	2	6.6
10712008	10766440	1.5	1	2	1.25	5.75
10855966		1.5	1	2.5	2	7
11162444		1.3	1	2.3	2	6.6

11156535		0.8	0.8	1.5	1	4.1
10777399		1.5	1	2.3	2	6.8
10862592	10772098	1.5	1	2.3	2	6.8
10863400		1.5	1	2.3	1.25	6.05
10742323		1.5	1	2	1.25	5.75
10793255		1.5	1	2.5	2	7
10861756	11212726	1.5	1	2.3	2	6.8
10832147		1.5	1	1.5	2	6
10717254	10716103	1.5	1	2.3	2	6.8
11191225	11180610	1.5	1	2.5	2	7
11087290	11111794	1.3	1	2	1.25	5.55
10914222	11057471	1.5	1	2.3	2	6.8
10738139	10769569	1.5	1	2.5	2	7
11059833		1.5	0.8	2.3	1.5	6.1
11119582	11121331	1.5	1	2.5	2	7
11107611		1.5	1	2.5	2	7
11035971	11035164	1.5	1	2.3	2	6.8
10852883		1.5	1	2.3	2	6.8
10828129	10774461	1.5	1	2.5	1.25	6.25
10863960	10766370	1.5	1	2.5	2	7
11036293	11061031	1.5	1	2	1.25	5.75
11140550		1.3	1	1.5	2	5.8
10799170	10797304	1.5	1	2.5	2	7
10778819		1.5	1	2.5	2	7
11119918		1.3	0.9	2	2	6.2
11010656	11044024	1.5	1	2.5	2	7
10607397		0	1	0.5	0.75	2.25
11145475	11156291	1.5	1	2	2	6.5
11154362		1.5	1	2.3	2	6.8
10497245		1.5	0.8	2	1.5	5.8
10771308		1.5	1	2.4	2	6.9
10830365		1.5	1	2.5	2	7
11061927	11028952	1.5	1	2.5	2	7
10892569		1.5	1	2.3	1.75	6.55
10810258		1.3	1	2	1.25	5.55
11114435	11036028	1.5	1	2.5	2	7
11036887		1.5	1	2	1.25	5.75
11121758		1.5	0.8	2	1.25	5.55
10847803		1.5	0	0.5	1.75	3.75
11115207	11097500	1.3	1	2	0.75	5.05
11052807		1.5	1	0	1.25	3.75
11118429		1.4	0.9	2.3	2	6.6
11191868		1.5	1	1.5	0	4
11159736		0	0	0	2	2
10495016		1.3	1	2	2	6.3
10835836		1.5	1	2.3	2	6.8

ID	Ora di inizio	Ora di completar	Posta elettronica	Nome	Ora ultima	Name and	Name and
1	3/18/26 13:11:41	3/18/26 13:58:36	10764083@polim	Marco Apollonio	Marco Ap	Francesco	
2	3/18/26 18:58:34	3/18/26 19:22:49	10838600@polim	Daniele Crippa	Daniele Crippa		
3	3/19/26 11:43:42	3/19/26 12:09:02	10768645@polim	Gabriele Riboldi	Gabriele R	Luca Villa	
4	3/20/26 22:35:22	3/20/26 22:44:49	10828027@polim	Rebecca Besana	Rebecca B	Noemi Faz	
5	3/22/26 0:05:39	3/22/26 1:16:11	10810258@polim	Jennifer Valeria Feijo	Jennifer Valeria Feijo		
6	3/22/26 15:50:06	3/22/26 16:24:30	11036887@polim	Hu Kaipeng	Kaipeng Hu		
7	3/21/26 15:18:31	3/22/26 17:51:18	10839162@polim	Davide Ghisolfi	Davide Gh	Tony Davi	
8	3/23/26 10:31:10	3/23/26 11:05:33	10780379@polim	Andrea Rubagotti	Andrea Ru	Maria Cor	
9	3/23/26 11:19:15	3/23/26 11:30:02	10975089@polim	Alessandro Cupini	Alessandro Cupini		
10	3/23/26 14:11:10	3/23/26 14:27:41	11119541@polim	Wang Yizhe	Yizhe Wan	Jiashen Zh	
11	3/23/26 17:30:57	3/23/26 17:33:26	10666869@polim	Yuri Cinesi Ferrari	Yuri Cinesi Ferrari		
12	3/23/26 17:29:53	3/23/26 17:40:23	11030873@polim	Jiang Zhenglin	JIANG ZHEN	None	
13	3/23/26 17:26:57	3/23/26 18:02:02	11044575@polim	Jie Zhang	JIE ZHANG		
14	3/23/26 21:43:15	3/23/26 22:02:24	11061975@polim	Alperen Aydin	Alperen A	Kayra ÜÇK	
15	3/24/26 12:15:42	3/24/26 12:29:33	10716383@polim	Simone Bianchessi	Simone Bianchessi		
16	3/24/26 12:55:40	3/24/26 13:45:32	11015522@polim	Yawen Yan	Yawen Yan		
17	3/24/26 13:43:17	3/24/26 14:05:20	10863960@polim	Simone Rodari	Simone R	Gabriele P	
18	3/24/26 15:47:06	3/24/26 16:02:40	11152089@polim	Christian Vaktel	Christian V	Magnus L	
19	3/24/26 20:00:32	3/24/26 20:09:11	10962875@polim	Qiu Yongqi	Yongqi Qiu		
20	3/24/26 23:01:20	3/24/26 23:30:09	11110511@polim	Hong Jinxu	Jinxu Hon	Tianyu Ga	
21	3/25/26 10:09:23	3/25/26 10:12:17	11027977@polim	Yu Xincheng	XINCHEN YU		
22	3/25/26 9:29:01	3/25/26 10:52:29	10802468@polim	Lin Elena	Elena Lin	Jiayi Zhen	
23	3/25/26 10:54:32	3/25/26 11:16:55	10917333@polim	Edoardo Bidini	Edoardo B	Mattia Og	
24	3/25/26 11:21:30	3/25/26 11:33:49	10711043@polim	Marco Stefano De B	Marco Ste	Federico M	
25	3/25/26 13:52:07	3/25/26 14:43:14	10845860@polim	Mario Monzani	Mario Monzani		
26	3/25/26 19:28:09	3/25/26 19:44:25	11101495@polim	Riccardo De Giacomo	Riccardo D	Fabrizio G	
27	3/25/26 19:28:40	3/25/26 19:45:26	11068760@polim	Giulia Barresi	Giulia Barr	Cesar Adri	
28	3/26/26 4:34:28	3/26/26 5:19:10	10994077@polim	Giovanni Orciuolo	Giovanni Orciuolo		
29	3/26/26 10:36:30	3/26/26 10:51:14	10799739@polim	Lorenzo Della Mater	Giovanni F	Lorenzo D	
30	3/26/26 17:35:45	3/26/26 17:51:19	10771695@polim	Edoardo Figini	Edoardo F	Greata Fo	
31	3/26/26 17:59:10	3/26/26 18:30:06	10698107@polim	Andrea Gaetano Sen	Andrea Ga	Matteo Ca	
32	3/26/26 18:24:44	3/26/26 18:50:40	10796320@polim	Samuele Lo Piccolo	Samuele Lo Piccolo		
33	3/26/26 18:58:57	3/26/26 19:14:27	11187268@polim	Nicolás Koron	Nicolas Koron		
34	3/26/26 19:56:58	3/26/26 20:30:27	11039171@polim	Kristi Duro	Kristi Durc	Arturo Am	
35	3/26/26 21:04:37	3/26/26 21:11:12	11030334@polim	Bi Yiran	Yiran BI	Shuai Zou	
36	3/26/26 20:58:13	3/26/26 21:21:59	10683374@polim	Zarlene Justrem De M	Zarlene Ju	Guglielmo	
37	3/27/26 10:09:57	3/27/26 10:22:29	10787431@polim	Davide Vendramin	Davide Ve	Matteo Ta	
38	3/27/26 10:44:00	3/27/26 10:50:31	10816840@polim	Filippo Motti	Filippo M	Matteo P	
39	3/27/26 10:48:39	3/27/26 10:56:18	10796163@polim	Stefano Castiglione	Stefano C	Samuele L	
40	3/27/26 10:57:02	3/27/26 11:15:12	10676235@polim	Marco Vitali	Marco Vitali		
41	3/27/26 11:17:08	3/27/26 11:23:34	10819854@polim	Giovanni Miccio	Giovanni M	Luca Pera	
42	3/27/26 11:20:49	3/27/26 11:46:11	10773859@polim	Mattia Brianti	Mattia Bri	Alex Hatha	
43	3/27/26 11:16:30	3/27/26 11:46:15	10720381@polim	Luca Guerra	Luca Guerra		
44	3/27/26 10:29:46	3/27/26 14:34:27	11150698@polim	Lars Aron Vilhelm Lui	Felicia Mu	Aron Lunc	
45	3/27/26 14:34:51	3/27/26 14:43:41	11188762@polim	Fernanda Vicente Go	Fernanda	Ignacia Nu	
46	3/27/26 14:35:06	3/27/26 14:57:24	10764570@polim	Giacomo Scampini	Giacomo Scampini		
47	3/27/26 18:12:27	3/27/26 18:28:33	10851929@polim	Tommaso Rossetti	Tommaso	Giulia Put	
48	3/27/26 19:17:20	3/27/26 19:35:44	10702601@polim	Angelo Di Rosa	Angelo Di	Livia Polve	
49	3/27/26 20:33:45	3/27/26 20:44:29	10660050@polim	Luca Donato	Luca Donato		
50	3/27/26 21:02:44	3/27/26 21:41:31	11097125@polim	Yiming Hou	Dinar Gare	Yiming Ho	

Person Code	Person Code 2	tear	Direct	trar	Multi-hop	Which str:	Which performance metric are you o
10764083	10796209		4400	5600		Multi-hop	Network Lifetime
10838600			4400	5600		Multi-hop	Minimize maximum per-node energy
10768645	10768851		4400	5600		Multi-hop	explained in next question
10828027	11178565		4400	5600		Multi-hop	Minimize maximum per-node energy
10810258			4400	5600		Direct	Minimize total communication energy
11036887			4400	5600		Direct	Minimize total communication energy
10839162	10723736		4400	5600		Multi-hop	Minimize maximum per-node energy
10780379	10799787		4400	5600		Multi-hop	network lifetime
10975089			3800	5000		Direct	Minimize total communication energy
11119541	11115748		4400	5600		Multi-hop	Minimize maximum per-node energy
10666869			4400	5600		Multi-hop	Minimize maximum per-node energy
11030873	None		4400	5600		Multi-hop	Minimize maximum per-node energy
11044575			4400	5600		Multi-hop	Maximize the battery lifetime of the e
11061975	11049291		4400	5600		Multi-hop	Minimize maximum per-node energy
10716383			4400	5600		Multi-hop	Minimize maximum per-node energy
11015522			4400	5600		Multi-hop	Minimize maximum per-node energy
10863960	10766370		4400	5600		Multi-hop	Minimize maximum per-node energy
11152089	11152488		4400	5600		Multi-hop	Minimize maximum per-node energy
10962875			3201	2602		Multi-hop	Minimize total communication energ
11110511	11102304		4400	5600		Multi-hop	Minimize maximum per-node energy
11027977			4400	5600		Direct	Minimize total communication energy
10802468	10811585		4400	5600		Multi-hop	Minimize the energy of the battery-p
10917333	10724359		4400	5600		Multi-hop	Maximize the network lifetime
10711043	10698676		4400	5600		Multi-hop	Minimize maximum per-node energy
10845860			5600	10400		Direct	Minimize total communication energ
11101495	11101728		4400	5600		Multi-hop	The motion sensor lifetime
11068760	11001933		4400	5600		Multi-hop	Minimize maximum per-node energy
10994077			4400	5600		Multi-hop	Minimize maximum per-node energy
10770813	10799739		4400	5600		Multi-hop	Minimize energy consumed by the se
10771695	10862436		4400	5600		Multi-hop	Minimize maximum per-node energy
10698107	10728121		4400	5600		Multi-hop	Minimize maximum per-node energy
10796320			3800	5600		Multi-hop	Maximize System Lifetime
11187268			4400	5600		Multi-hop	Minimize sensor energy
11039171	10799588		4400	5600		Multi-hop	Minimize maximum per-node energy
11030334	11024119		4400000	5600000		Multi-hop	Minimize maximum per-node energy
10683374	10764592		4400	5600		Multi-hop	Minimize maximum per-node energy
10787431	10785383		4400	5600		Multi-hop	Minimize maximum per-node energy
10816840	10797758		4400	5600		Multi-hop	Maximize the timelife of the motion s
10796163	10772778		4400	5600		Multi-hop	Minimize maximum per-node energy
10676235			4400	5600		Multi-hop	Maximize network lifetime
10819854	10846682		4400	5600		Multi-hop	Minimize maximum per-node energy
10773859	10829946		4400	5600		Multi-hop	Minimize maximum per-node energy
10720381			4400	5600		Multi-hop	Minimize maximum per-node energy
11189222	11150698		4400	5600		Multi-hop	Minimize the maximum sensor energ
11188762	11173956		4400	5600		Direct	Minimize total communication energ
10764570			4400	5600		Multi-hop	Individual node lifetime (specifically, t
10851929	10860913		4400	5600		Multi-hop	Maximize presence sensor (battery) li
10702601	10793007		4400	5600		Multi-hop	Minimize maximum per-node energy
10660050			1400	5600		Multi-hop	Minimize maximum per-node energy
11063702	11097125		4400	5600		Multi-hop	Minimize maximum per-node energy

If you selected 'Other', please specify

Maximize the battery lifetime of the energy-cor

.

when i choose the strategy for long-term opera

-

-

Minimize total communication energy

We are optimizing network lifetime, especially t

Minimize total communication energy

NA

N/A.

/

Maximize the battery lifetime of the energy-cor

Only battery energy consumption is coming fro

I selected the second option. No need to specif

not selected

Didn't select other

nothing

/

Not applicable (minimizing total communication

Minimize the energy of the battery-powered nc

Maximize the network lifetime, that is the time

No

None

The motion sensor lifetime in respect to total n

.

Not applicable

Another option (better than the others) could b

-

For long term operation of the system the limit

Maximize System Lifetime to provide the longe

I prioritize minimizing the sensor energy becaus

"Minimize maximum per-node energy" is the p

N/A

Minimize maximum per-node energy

Maximize the timelife of the motion sensor, sin

We selected "minimize maximum per-node ene

Maximize network lifetime, by maximizing the l

.

-

-

What we want to do is minimize the maximum

No

Individual node lifetime (specifically, the batter

Maximize presence sensor (battery) lifetime

-

Minimize maximum per-node energy, since the

I did not select 'Other2 but I can specify the rea

Provide a brief justification. Your explanation must clearly link the selection to the question.	Direct TX	Multi-Hop	Question
Multi-hop takes 5.6mJ total vs direct's 4.4mJ, but cuts the battery-power	0.25	0.25	2
The sensor is the energy critical node, because bulbs are directly connect	0.25	0.25	2
I chose multi-hop because it greatly reduces the energy used by the batte	0.25	0.25	2
Even if the total energy is higher, multi-hop is better because distributes t	0.25	0.25	2
Multi-hop uses more energy. we have 4x0.8mJ TX and 4x0.6mJ RX becaus	0.25	0.25	1
Direct transmission (4400 µJ) is preferred because it consumes less total e	0.25	0.25	1
The hop by hop strategy minimizes the energy spent by each node to ser	0.25	0.25	2
We optimized the motion sensor, the only energy-constrained node, con	0.25	0.25	2
Direct is more efficient at 40m. Shorter hops reduce amplifier energy, but	0	0	1
Multi-hop minimizes max per-node energy, 3800uJ (sensor, direct) vs 140	0.25	0.25	2
The sensor is likely the only component on a battery, meaning that lessen	0.25	0.25	2
Bulbs use wall power; the sensor relies on a battery. Multi-hop drops its c	0.25	0.25	2
Multi-hop cuts the sensor energy from 3.8 mJ to 0.8 mJ by shifting relayin	0.25	0.25	2
Only battery energy consumption is from the sensor itself. The bulbs are	0.25	0.25	2
Even if the total energy consumed by the network is higher, the energy "	0.25	0.25	2
Multi-hop reduces sensor TX energy from 3800 µJ to 800 µJ (4.75x), minin	0.25	0.25	2
The motion sensor is the only battery powered device. For more battery	0.25	0.25	2
Multi-hop method since it minimizes the maximum per-node energy by c	0.25	0.25	2
Multi-hop is selected because it gives lower total communication energy	0	0	1
Multi-hop drops the battery sensor's load from 3.8mJ to 0.8mJ. Shifting t	0.25	0.25	2
Direct transmission is preferred since it consumes less total energy than r	0.25	0.25	1
Multi hop reduces sensor's transmission distance from 40m->10m and er	0.25	0.25	2
Multi-hop transmission reduces the energy consumption at the motion s	0.25	0.25	2
Multi-hop is preferred since it reduces per-node energy (0.8 vs 3.8 mJ), ex	0.25	0.25	2
Since E_c and E_rx must be paid at every hop, multi-hop is disadvantaged	0	0	1
sensor E_sensor=kEC+KepsD^2, grows quadratically with distance. E_sen	0.25	0.25	2
Multi-hop is preferable for long-term operation because it spreads energy	0.25	0.25	2
Hue bulbs have effectively infinite power from sockets. Multi-hop minimi	0.25	0.25	2
Motion sensor is powered by a battery and the bulbs are grid connected,	0.25	0.25	2
We chose multi-hop strategy since the only battery powered node is the	0.25	0.25	2
With the first strategy the motion sensor consumes a total of 3.8mJ, whil	0.25	0.25	2
With multi-hop the motion sensor is consuming 4.75 times less so the sys	0	0.25	2
the total energy expenditure is higher, but the sensor consumes less ene	0.25	0.25	2
Direct TX places a 3.8mJ peak energy burden on the battery sensor. Mult	0.25	0.25	2
Multi-hop reduces motions sensor energy from 3.8x10^6 nJ to 0.8x10^6 nJ (	0	0	2
The sensor transmits to bulb 1 consuming 0,8 mJ which significantly allow	0.25	0.25	2
The sensor is the only component powered by battery. In direct transmis	0.25	0.25	2
Using hop-by-hop the sensor will consume 0,8 mJ instead of 3,8 mJ, this v	0.25	0.25	2
Multi-hop uses more total energy, but distributes it across nodes. The se	0.25	0.25	2
Aqara sensor is battery operated, so we maximize its lifetime. Direct: 3.8 r	0.25	0.25	2
In general, multi-hop extends system life by spreading the load despite h	0.25	0.25	2
Sensor POV: direct communication -> 3.8mJ, Hop-by-Hop -> 0.8mJ. Minin	0.25	0.25	2
Multi-hop drops the sensor's energy from 3800 to 800 microjoules. Since	0.25	0.25	2
Direct uses least power, multi-hop saves sensor battery. Bulbs aren't batt	0.25	0.25	2
Direct is preferred since it minimizes total energy per message (4400 vs 56	0.25	0.25	1
Multi-hop is better because it saves the sensor's battery. The smart bulbs	0.25	0.25	2
Multi-hop cuts sensor TX energy from 3.8 to 0.8 mJ (-79%). Since it is the	0.25	0.25	2
the amount of energy consumed is split almost equally to each node inst	0.25	0.25	2
I would choose multi-hop to extend sensor lifetime. It minimizes per-nod	0	0.25	2
The sensor's battery can drain and become a bottleneck, to maximize ne	0.25	0.25	2

[illegible]

51	3/27/26 22:08:52	3/27/26 22:33:23	11171264@polim Zhang Wenzhu	WENZHU ZIXIAOWEN
52	3/27/26 22:20:58	3/27/26 22:41:37	11095668@polim Francesco Lazzaro	Francesco Andres Fe
53	3/28/26 9:52:28	3/28/26 9:54:36	10774521@polim Pietro Callandrone	Pietro Call Lorenzo B
54	3/28/26 10:30:41	3/28/26 10:37:44	10683981@polim Cezar Gjinaj	Cezar Gjinaj
55	3/28/26 10:38:40	3/28/26 10:41:06	10773862@polim Samuele Galbiati	Samuele Galbiati
56	3/28/26 10:34:07	3/28/26 10:50:00	10857924@polim Marco Coviello	Marco Coviello
57	3/28/26 12:24:22	3/28/26 14:17:05	10790205@polim Marco Carminati	10790205 Marco Car
58	3/28/26 14:40:04	3/28/26 14:53:36	10789693@polim Alessio Vittorio Rosa	Alessio Vit Andrea Co
59	3/28/26 14:42:20	3/28/26 15:06:15	10831959@polim Simone Azzoni	Simone AzMartino A
60	3/28/26 14:59:19	3/28/26 15:14:24	11083784@polim Zheng Zihao	Zihao Zheng
61	3/28/26 15:07:41	3/28/26 15:18:15	11120812@polim Orhun Tunc	Orhun Tunc
62	3/28/26 15:30:44	3/28/26 15:35:55	11128210@polim Mohannad Mohame	Mohannad Mohame
63	3/28/26 15:09:08	3/28/26 15:45:03	10837597@polim Alessia Ragosta	Alessia Ragosta
64	3/28/26 15:59:30	3/28/26 16:36:43	10853745@polim Mattia Caon	Mattia CarMarco Ber
65	3/28/26 16:43:05	3/28/26 16:48:12	10797420@polim Pietro Pizzoccheri	Pietro PizzLorenzo B
66	3/28/26 16:45:32	3/28/26 16:53:44	10825131@polim Edoardo Sartore	Edoardo Sartore
67	3/28/26 16:28:47	3/28/26 16:55:23	10770385@polim Alessia Radicula	Alessia Radicula
68	3/28/26 17:04:48	3/28/26 17:15:57	10817232@polim Tommaso Neri	Tommaso Neri
69	3/28/26 17:28:00	3/28/26 17:35:32	10619593@polim Alessandro Pessina	Alessandro Pessina
70	3/28/26 17:43:25	3/28/26 17:50:27	10828995@polim Jacopo Pelagalli	Jacopo Pelagalli
71	3/28/26 18:54:12	3/28/26 19:14:38	10900762@polim Davide Petruzzi	Davide Pe'Matteo Gi
72	3/28/26 19:12:14	3/28/26 19:19:44	10730038@polim Edoardo Menegotto	Edoardo Menegotto
73	3/28/26 19:57:28	3/28/26 20:31:36	10772022@polim Davide Rutigliano	Davide Ru Emiliano F
74	3/28/26 21:17:43	3/28/26 21:32:29	10697294@polim Salvatore Passaro	Salvatore Passaro
75	3/28/26 21:24:32	3/28/26 21:37:32	11022931@polim Peng Rao	Peng Rao
76	3/28/26 18:36:44	3/28/26 21:51:35	11166508@polim Cai Chuanhao	CAI CHUANHAO
77	3/29/26 0:26:28	3/29/26 0:48:59	11111393@polim Muhammad Rian	Muhamm Aakash Bh
78	3/29/26 9:53:24	3/29/26 10:06:22	10742249@polim Francesco Giuseppe	Andrea GiFrancesco
79	3/29/26 11:22:56	3/29/26 11:45:25	10767814@polim Alessandro Fedele	AlessandrDario Cros
80	3/29/26 11:47:41	3/29/26 12:01:28	10808901@polim Luca De Nicola	Luca De Nicola
81	3/29/26 12:01:47	3/29/26 12:04:33	10798975@polim Matteo Rossi	Matteo REdoardo P
82	3/29/26 12:05:45	3/29/26 12:06:04	10781960@polim Andrea Vavassori	Andrea VaMarco Ga
83	3/29/26 11:36:20	3/29/26 12:07:48	11167997@polim Han Zhaohui	ZHAOHUI ZICHEN LI
84	3/29/26 12:06:39	3/29/26 12:25:42	11031608@polim Peng Anchen	Qianyi WaAnchen Pe
85	3/29/26 12:41:25	3/29/26 12:53:57	11051292@polim Alireza Jahandoost	Alireza JahAli Nabipo
86	3/29/26 12:53:36	3/29/26 12:58:04	11048104@polim Ting Zhang	Zhang Ting
87	3/29/26 9:54:15	3/29/26 13:24:38	11090815@polim Zhu Wen	Wen Zhu
88	3/29/26 14:35:51	3/29/26 14:51:40	11030423@polim Liu Zhaojun	Liu ZhaojuDing Zhen
89	3/29/26 14:58:34	3/29/26 15:48:56	10850482@polim Alessandro Cirillo	AlessandrJaime Iriar
90	3/29/26 14:38:00	3/29/26 15:53:19	11157416@polim Jenny Hoa Thi Hong	Jenny HoaKelly Tran
91	3/29/26 15:59:52	3/29/26 16:10:43	11019965@polim Yasin Asgari	Yasin AsgaArmin Ma
92	3/29/26 16:25:58	3/29/26 16:27:23	10832525@polim Michele Tavecchio	Michele TAlessandr
93	3/29/26 16:12:41	3/29/26 16:37:18	11033376@polim Li Ming	MING LI XIANGYU I
94	3/29/26 11:18:37	3/29/26 16:50:56	11109083@polim Geng Zihao	Zihao Gen Marica Ba
95	3/29/26 16:07:48	3/29/26 16:57:18	10797131@polim Morena Caterina Car	Morena CShashi Bh
96	3/29/26 16:35:49	3/29/26 17:01:33	10783671@polim Raul Agolli	Raul Agolli
97	3/29/26 16:27:04	3/29/26 17:06:45	10922631@polim Francesco Armuzzi	Francesco Armuzzi
98	3/29/26 17:03:56	3/29/26 17:10:19	11006775@polim Osman Bugra Dagli	OSMAN BIERDEM CC
99	3/29/26 16:31:31	3/29/26 17:21:09	10808527@polim Manoueil Michael H	Manoueil Federico F
100	3/28/26 13:20:28	3/29/26 17:22:02	11034150@polim Mobina Mobaraki	Mobina MFederico C
101	3/29/26 17:17:24	3/29/26 17:31:16	10793061@polim Marco Zheng	Marco Zheng

11171264	11168874	4400	5600	Multi-hop Minimize total communication energy
11095668	11068591	4400	5600	Multi-hop Overall Cost Minimization
10774521	11024306	4400	5600	Multi-hop Minimize maximum per-node energy
10683981		4400	5600	Multi-hop Minimize sensor energy
10773862		4400000	5600000	Multi-hop Minimize maximum per-node energy
10857924		4400	5600	Multi-hop Energy needed to send a packet from
11055131	Om Mohan Deshm	4400	5600	Direct Minimize total communication energy
10789693	10766257	4400	5600	Multi-hop Minimize maximum per-node energy
10831959	10829712	4400	5600	Multi-hop Minimize motion sensor energy
11083784		4400	5600	Multi-hop Minimize maximum per-node energy
11120812		4400	5600	Multi-hop Minimize maximum per-node energy
11128210		4400	5600	Direct Minimize total communication energy
10837597		3800	5000	Multi-hop Maximize the network lifetime
10853745	10929780	4400	5600	Multi-hop Network lifetime
10797420	10831941	4400	5600	Multi-hop sensor node battery life
10825131		4400	5600	Multi-hop Minimize maximum per-node energy
10770385		4400	5600	Multi-hop Maximize network lifetime
10817232		4400	5600	Multi-hop Minimize maximum per-node energy
10619593		4400	5600	Multi-hop Minimize maximum per-node energy
10828995		4400	5600	Multi-hop Minimize maximum per-node energy
10900762	11164535	4400	5600	Multi-hop Minimize maximum per-node energy
10730038		4400	5600	Multi-hop Minimize the communication energy
10772022	11095908	3800	5000	Multi-hop Maximize the Lifetime of the Network
10697294		4400	5600	Multi-hop Minimize maximum per-node energy
11022931		4400	5600	Multi-hop Minimize maximum per-node energy
11166508		4400	5600	Multi-hop Minimize maximum per-node energy
11111393	11113677	4400	5600	Direct Minimize total communication energy
10798825	10742249	4400	5600	Multi-hop Minimize maximum per-node energy
10767814	10796197	4400	5600	Multi-hop Minimize maximum per-node energy
10808901		4400	5600	Multi-hop Minimize maximum per-node energy
10798975	10765707	4400	5600	Multi-hop Maximizing Network Lifetime
10781960	10794601	4400	5600	Multi-hop Minimize maximum per-node energy
11167997	10935200	4400	5600	Multi-hop Minimize maximum per-node energy
11031890	11031608	4400	5600	Direct Minimize total communication energy
11051292	11041035	4400	5600	Multi-hop Minimize maximum per-node energy
11048104		3800	5000	Multi-hop Minimize maximum per-node energy
11090815		3800	5000	Direct Minimize maximum per-node energy
11030423	11053611	4400	5600	Multi-hop Minimize total communication energy
10850482	11154921	4400	5600	Multi-hop Minimize the energy consumed by th
11157416	11151327	4400	5600	Direct Minimize total communication energy
11019965	11060212	3800	5000	Direct Minimize total communication energy
10832525	10777259	4400	5600	Multi-hop Network lifetime
11033376	11062619	4400	5600	Multi-hop Minimize maximum per-node energy
11109083	10832004	4400	5600	Multi-hop Minimize maximum per-node energy
10797131	11111318	4400	5600	Multi-hop minimize maximum per-node energy
10783671		4400	5600	Multi-hop other
10922631		4400	5600	Multi-hop Maximize system lifetime
11006775	11045157	4400	5600	Multi-hop sensor node lifetime
10808527	10796043	3800	5000	Multi-hop Minimize maximum per-node energy
11034150	10822446	4400	5600	Multi-hop We are optimizing the battery lifetime
10793061		4400	5600	Multi-hop Minimize maximum per-node energy

Total monetary cost (batteries vs. grid)

Didn't select other

Minimize sensor energy

The performance metric being optimize (minimize)

/

-

Minimize motion sensor energy

N/A

no

Maximize the network lifetime considering in the network

We maximize network lifetime to preserve the network

the performance metric we are optimizing is the network lifetime

Minimize maximum per-node energy

Maximize network lifetime means minimize the energy consumption

I want to minimize the tx power on the only battery

-

We have selected minimize maximum per-node energy

Minimize the communication energy only for the network

Maximize the Lifetime of the Network

-

Select: Minimize maximum per-node energy to maximize network lifetime

null

N/A

Didn't choose 'Other'

...

-

Maximizing Network Lifetime (Motion Sensor Based)

NA

The optimized performance metric is minimizing energy consumption

We choose Minimize total communication energy

N/A

Minimize maximum per-node energy

Minimize maximum per-node energy (to maximize network lifetime)

The decision should not be based only on total energy consumption

The strategy is to minimize the energy consumption

I did not choose 'other'

I choose the first one: Minimize total communication energy

Network lifetime, the time at which the first node dies

no

.

The optimized performance metric focuses on maximizing network lifetime

Network Lifetime

Maximize the system lifetime by minimizing the energy consumption

Sensor node lifetime

-

The sensor is the system's "bottle neck" as the network lifetime is limited by the sensor

The question is obligatory to answer: so I'll answer

Multi-hop balances energy among nodes. Although it uses more total en	0.25	0.25	2
We compared average battery prices to grid electricity. The battery cost f	0.25	0.25	2
Multi-hop reduces the maximum per-node energy (sensor: 800 μ J vs 3800	0.25	0.25	2
Bulbs and sink are NOT on battery, we want to maximize sensor life only	0.25	0.25	2
I chose multi-hop strategy because the sensor is the only device power b	0	0	2
In multi-hop, sensors send over shorter distances, reducing energy use a	0.25	0.25	2
We opt for direct transmission strategy that will increase the total lifespa	0.25	0.25	1
Multi-hop reduces sensor energy (3.8 $\rightarrow$ 0.8 mJ) despite higher total cost, v	0.25	0.25	2
The only device that has a battery is the motion sensor, all other devices	0.25	0.25	2
The smart bulb is wired connected device, which is not sensitive to the po	0.25	0.25	2
Multi-hop minimizes max per-node energy. Sensor uses battery, bulbs us	0.25	0.25	2
direct transmission is more energy efficient because the cost of extra rece	0.25	0.25	1
Sensor lasts 3,524,210 transmissions (direct) and 16,740,000 (multi-hop).	0	0	2
Multi-hop cuts sensor energy from 3.8 to 0.8mJ/msg, maximizing network	0.25	0.25	2
choosing multi-hop transmission, we shift the energy consumption away	0.25	0.25	2
Direct tx uses less energy overall. However since bulbs are plugged in and	0.25	0.25	2
Multi-hop reduces sensor TX from 3800 μ J to 800 μ J due to quadratic pat	0.25	0.25	2
I chose multi-hop because it cuts the sensor's Tx energy from 3800 μ J to 8	0.25	0.25	2
Multi-hop cuts the required energy for sensor from 3.80 to 0.80 mJ (E_{tx}	0.25	0.25	2
It's better for the motion sensor, which could be powered by a battery, to	0.25	0.25	2
Multi-hop minimizes the maximum per-node energy. Although total ener	0.25	0.25	2
Only the senso node has power constraint due to the battery. So if we re	0.25	0.25	2
The main reason for this choice is related to the battery of the motion se	0	0	2
bulbs are mains-powered, sensor is battery-powered (from specs)->multi	0.25	0.25	2
Minimize maximum per-node energy. Multi-hop shifts the 3.8mJ load fro	0.25	0.25	2
Multi-hop reduces sensor energy. Despite higher total energy, minimizing	0.25	0.25	2
For long-term operation, the optimal strategy minimizes energy required	0.25	0.25	1
Multi-hop reduces the sensor's energy from 3.8mJ to 0.8mJ. Minimizing t	0.25	0.25	2
The Hop-by-hop strategy is less taxing on the individual nodes. This is pre	0.25	0.25	2
In direct approach most of the energy is consumed by the sensor node, c	0.25	0.25	2
The Hue bulbs are wired. Multi-hop spends more total energy (5.6mJ vs 4	0.25	0.25	2
Multi-hop has higher total energy but spreads the load: max per-node en	0.25	0.25	2
multi-hop distributes the energy consumption among multiple nodes, pr	0.25	0.25	2
Direct transmission is better because it uses less total communication en	0.25	0.25	1
Bulbs are plugged in so their energy don't matter. Multi-hop reduces sen	0.25	0.25	2
Multi-hop reduces per-node energy at the sensor by shortening transmis	0	0	2
Duty cycling via Deep Sleep to minimize maximum per-node energy. Resu	0	0	0
if we consider the ideal communication model, direct transmission is opti	0.25	0.25	0
To demonstrate the advantages of this approach, we compare the energ	0.25	0.25	2
Direct transmission consumes less total energy (4400 μ J) than multi-hop (	0.25	0.25	1
Multi-hop reduces hop distance but needs extra forwarding, adding trans	0	0	1
Direct (3800mJ) drains the 12960J (2xCR2450) of the motion sensor, the o	0.25	0.25	2
For long-term operation, network lifetime matters most. Since the motio	0.25	0.25	2
System lifetime depends on the motion sensor, the critical constrained n	0.25	0.25	2
Multi-hop cuts sensor energy from 3.8M to 800k nJ (-79%). Shifting the lo	0.25	0.25	2
Multi-hop reduces the battery sensor's energy from 3.8mJ to 0.8mJ per n	0.25	0.25	2
Multi-hop minimizes max per-node energy:sensor TX energy drops from	0.25	0.25	2
The sensor uses almost 4.75 times less energy with multi-hop. The physic	0.25	0.25	2
Q1:We ignored sink energyQ2:Direct transmission puts the entire energy	0	0.25	2
Multi-hop cuts sensor distance from 40m to 10m, reducing energy/packe	0.25	0.25	2
The multihop method extends device lifetime by distributing the energy l	0.25	0.25	2

1.5	2
1.5	2
1.5	2
1.5	2
1.5	1.5
1.5	2
0.75	1.25
1.5	2
1.5	2
1.5	2
1.5	2
0.75	1.25
1.5	1.5
1.5	2
1.5	2
1.5	2
1.5	2
1.5	2
1.5	2
1.5	2
1.5	2
1.5	2
1.5	1.5
1.5	2
1.5	2
1.5	2
0.75	1.25
1.5	2
1.5	2
1.5	2
1.5	2
1.5	2
1.5	2
0.75	1.25
1.5	2
1.5	1.5
0	0
1.5	2
1.5	2
0.75	1.25
0.75	0.75
1.5	2
1.5	2
1.5	2
1.5	2
1.5	2
1.5	2
1.5	2
1.5	2
1.5	1.75
1.5	2
1.5	2

102	3/29/26 17:24:23	3/29/26 17:41:23	11120269@polim Zhao Zhiyi	Zhao Zhiyi
103	3/29/26 17:34:30	3/29/26 17:47:17	10813424@polim Andrea Midulla	Andrea MiLorenzo C
104	3/29/26 17:35:09	3/29/26 17:47:55	10806605@polim Bruno Arena	Bruno Are Matteo Ca
105	3/29/26 17:37:12	3/29/26 17:55:06	11044962@polim Mohammad Javad Z	Mohammad Javad Z
106	3/29/26 17:45:53	3/29/26 17:58:47	11157354@polim Ingrid Lyngstad	Ingrid Lyngstad
107	3/29/26 17:13:56	3/29/26 18:12:47	11087290@polim Han Mengzhen	MengzherShiming G
108	3/29/26 17:51:56	3/29/26 18:40:10	10732218@polim Davide Greco	Greco DavBellosi Jac
109	3/29/26 16:11:08	3/29/26 18:40:15	11154567@polim Esteban Lopez Barre	Esteban LcMa Chong
110	3/29/26 18:32:28	3/29/26 19:06:32	11101890@polim Mohamed Youssef M	Mohamed Youssef M
111	3/29/26 17:19:03	3/29/26 19:44:01	10830061@polim Cristian Summa	Cristian SuMathias R
112	3/29/26 19:29:56	3/29/26 19:46:54	11143015@polim Saltuk Bugra Kirazci	Saltuk Bug-
113	3/27/26 11:53:50	3/29/26 19:52:51	11152090@polim Hao Johny Vinh Ngu	Hao Johny Vinh Ngu
114	3/29/26 19:51:58	3/29/26 20:16:48	10816921@polim Federico Trubian	Federico Trubian
115	3/29/26 20:05:51	3/29/26 20:23:04	10861416@polim Susanna Lo Cascio	Susanna Lo Cascio
116	3/29/26 20:42:50	3/29/26 20:51:23	11078923@polim Junhua Fang	FANG JUN/CHAI KE
117	3/29/26 17:44:21	3/29/26 20:54:02	11044024@polim Hedieh Raeisi	Ghazal SerHedieh Ra
118	3/29/26 20:02:23	3/29/26 21:01:51	11058806@polim Sevval Karadeniz	Sevval KarHaga Zena
119	3/29/26 20:37:58	3/29/26 21:07:24	10797129@polim Roberta Gennari	Roberta G Sara Naim
121	3/26/26 15:53:52	3/29/26 21:17:26	11019981@polim Jia Tongxin	Tongxin JiYuedan Ca
122	3/29/26 22:07:21	3/29/26 22:07:25	10708185@polim Andrea Ardu	Andrea ArLaura Mar
123	3/29/26 22:06:43	3/29/26 22:09:39	10813646@polim Gabriel Leonardi	Gabriel LerFederico C
124	3/29/26 21:14:37	3/29/26 22:13:46	11112969@polim Li Xin	Yuanyuan Xin Li
125	3/29/26 22:19:32	3/29/26 22:22:40	11119281@polim Mariyam Rashid Pita	Mariyam Rashid Pita
126	3/29/26 22:16:05	3/29/26 22:31:23	10677341@polim Alberto Biffi	Alberto Biffi
127	3/29/26 22:22:43	3/29/26 22:31:58	11046392@polim Amir Hossein Rashid	Amir Hoss ERFAN MC
128	3/29/26 22:17:11	3/29/26 22:36:34	10911594@polim Francesca Maragoni	Francesca Anna Met
129	3/29/26 22:43:37	3/29/26 22:47:25	10766937@polim Gabriele Lorenzo Sin	Gabriele Lorenzo Sin
130	3/29/26 22:16:38	3/29/26 23:53:09	10821199@polim Golsa Shokri	Golsa Shokri
131	3/29/26 19:48:43	3/30/26 0:53:33	11121331@polim Amirhossein Sharifis	Amin Sakt Amirhosse
132	3/30/26 1:21:41	3/30/26 1:44:55	11032747@polim Hu Peng	Peng Hu
133	3/30/26 0:16:05	3/30/26 1:54:22	11115876@polim Zahra Sadat Parviss	Zahra SadAli Isapou
134	3/30/26 2:16:49	3/30/26 2:56:42	11111583@polim Fatemeh Kianiharche	Fatemeh K Sajad gha
135	3/30/26 2:57:37	3/30/26 3:02:19	10948935@polim Rong Huang	Rong HUANG
136	3/30/26 3:21:22	3/30/26 3:59:39	11117655@polim Fang Yuhang	yuhang fang
137	3/30/26 4:41:35	3/30/26 5:04:46	10709837@polim Carmen Giaccotto	Carmen GiAlessia Fra
138	3/30/26 10:06:07	3/30/26 10:12:46	11159736@polim Muhammad Hassaa	Muhammad Hassaa
139	3/30/26 10:26:50	3/30/26 10:36:23	10832147@polim Simone Amighini	Simone Amighini
140	3/30/26 10:07:27	3/30/26 10:38:26	10777877@polim Mattia Panzeri	Mattia PaiDiego But
141	3/30/26 10:57:49	3/30/26 11:04:26	10811640@polim Simone Giuliano	Simone GiSimone Lu
142	3/30/26 11:08:17	3/30/26 11:20:10	10766478@polim Dai Run Jie Simone	run jie simjiaxin yang
143	3/30/26 11:17:27	3/30/26 11:20:20	10839190@polim Elena Bestetti	Elena BestJacopo Ale
144	3/30/26 11:08:49	3/30/26 11:20:33	10764433@polim Jonny Arbex Marque	Jonny Arb Leonardo
145	3/30/26 11:19:59	3/30/26 11:27:40	11063424@polim Hu Jialin	JIALIN HU ZIYIN YOU
146	3/30/26 11:28:43	3/30/26 11:41:21	10844134@polim Edoardo Francesco A	Edoardo Albini
147	3/30/26 11:21:01	3/30/26 11:53:07	10705152@polim Alessandro Posani	AlessandroRiccardo C
148	3/30/26 11:26:11	3/30/26 12:06:49	11118976@polim Dosilhenbi Yumnam	Dosilhenb pre
149	3/30/26 12:05:48	3/30/26 12:27:46	10847211@polim Simone Teggi	Simone TeMarco Scr
150	3/30/26 8:27:58	3/30/26 12:35:34	10860834@polim Giacomo Milianti	Davide LoiGiacomo I
151	3/30/26 12:25:12	3/30/26 12:40:44	11137282@polim Yehia Mohamed Abc	Yehia AbdAmr Abde
152	3/30/26 12:28:12	3/30/26 12:43:34	10769571@polim Francesco Street	Francesco Andrea Ve
153	3/30/26 12:42:15	3/30/26 12:55:57	10744406@polim Gabriele Bogdan Sep	Gabriele B Matilde B

11120269		4400	5600	Multi-hop	Minimize maximum per-node energy
10813424	10842807	4400	5600	Multi-hop	Minimize maximum per-node energy
10806605	10768765	3800	5000	Multi-hop	Minimize maximum per-node energy
11044962		4400	5600	Multi-hop	Minimize maximum per-node energy
11157354		4400	5600	Direct	Minimize total communication energy
11087290	11111794	4400	5600	Direct	Minimize total communication energy
10732218	10806471	4400	5600	Multi-hop	Minimize maximum per-node energy
11154567	11162609	4400	5600	Direct	Minimize total communication energy
11101890		3800	5000	Multi-hop	Minimize total communication energy
10830061	10796466	4400	5600	Multi-hop	Minimize energy consumption on the
11143015	-	4400	5600	Direct	Minimize total communication energy
296209		4400	5600	Multi-hop	Minimize maximum per-node energy
10816921		4400	5600	Multi-hop	maximize network lifetime
10861416		4400	5600	Multi-hop	Minimize maximum per-node energy
11078923	11110613	4400	5600	Multi-hop	Minimize maximum per-node energy
11044024	11010656	4400	5600	Multi-hop	Minimize maximum per-node energy
11058806	11091338	4400	5600	Multi-hop	Minimize maximum per-node energy
10797129	10710507	4400	5600	Multi-hop	Minimize motion sensor's energy
11019981	11093193	4400	5600	Direct	Minimize total communication energy
10708185	10793514	4400	5600	Multi-hop	Energy consumption of the Motion S
10813646	10768605	4400	10400	Multi-hop	Maximize operating lifetime of the se
11110488	11112969	4400	5600	Direct	Minimize total communication energy
11119281		4400	5600	Multi-hop	Minimize maximum per-node energy
10677341		4400	5600	Multi-hop	Lifetime of the system
11046392	11049700	4400	5600	Direct	Minimize total communication energy
10911594	10901067	4400	5600	Direct	Minimize total communication energy
10766937		4400	5600	Multi-hop	Minimize maximum per-node energy
10821199		4400	5600	Direct	Minimize total communication energy
11119582	11121331	4400	5600	Multi-hop	Minimize maximum per-node energy
11032747		4400	5600	Multi-hop	Minimize maximum per-node energy
11115876	11107660	4400	5600	Multi-hop	Minimize maximum per-node energy
11111583	11062873	4400	5600	Direct	Minimize total communication energy
10948935		4400	5600	Multi-hop	Minimize maximum per-node energy
11117655		4400	5600	Multi-hop	Minimize maximum per-node energy
10709837	10773289	4400	5600	Multi-hop	Minimize maximum per-node energy
11159736		4400	5600	Multi-hop	Minimize maximum per-node energy
10832147		4400	5600	Multi-hop	Minimize maximum per-node energy
10777877	10806864	4400	5600	Multi-hop	Optimize lifetime of the network
10811640	10790904	4400	5600	Direct	Minimize total communication energy
10766478	10719178	4400	5600	Direct	Minimize total communication energy
10839190	10859045	4400	5600	Multi-hop	Minimize maximum per-node energy
10764433	10809273	4400	5600	Multi-hop	Lifetime of the battery - powered sen
11063424	11022646	4400	5600	Multi-hop	Sensor lifetime (battery-powered mo
10844134		4400	5600	Multi-hop	sensor's battery life
10705152	10740222	4400	5600	Multi-hop	Minimize maximum per-node energy
11118976	11147720	4400	5600	Multi-hop	Minimize maximum per-node energy
10847211	10768863	4400	5600	Multi-hop	Minimize sensor node energy consum
10502570	10860834	4400	5600	Multi-hop	Minimize maximum per-node energy
11137282	11116374	4	6	Multi-hop	Minimize maximum per-node energy
10769571	10797678	3800	5000	Multi-hop	Minimize maximum per-node energy
10744406	10709235	4400	5600	Multi-hop	Minimize maximum per-node energy

For long-term operation, the lifetime of the ent

.

-

.

I did not select other.

no

N/A

N/A

Minimize energy consumption on the Motion S

-

.

Maximize the lifetime of the node that runs out

Minimize maximum per-node energy

N/A

-

We should minimize the energy for those node

If retransmission and idle energy were taken int

Although the total network energy is higher in r

Maximizing operating lifetime of the sensor

Minimize total communication energy

N/A

I choose to maximize the lifetime of the system

i chose Minimize total communication energy

Selected minimize maximum per-node energy

.

-

Minimize maximum per-node energy

.

If the optimization metric is the total network e

none

Minimize maximum per-node energy is better.

N/A

none

/

In direct case, we force the motion sensor to us

we did not select other, please refer to the repc

i choose minimize total communication energy

I did not select other

We are optimizing the lifetime of the battery - p

The lifetime of the battery-powered motion ser

The metric I am optimizing is the sensor's batte

The selected performance metric is the minimiz

nil

Since the sensor node is the only battery opera

(cannot deliver form with this block being empt

want to avoid draining the most stressed node

.

Did not select Other, question was flagged as a

Although direct transmission consumes less total energy, energy is taken	0.25	0.25	2
Multi-hop increases total network energy 5.6 vs 4.4 mJ, but the only cons	0.25	0.25	2
We minimize the maximum per-node energy: multi-hop lowers the senso	0	0	2
With multi-hop, the sensor uses 800 μ J per message instead of 3800 μ J, e	0.25	0.25	2
Multi-hop adds extra receive and forwarding energy at each node, makin	0.25	0.25	1
Direct transmission is preferred as it uses less total energy (4400 μ J vs 560	0.25	0.25	1
By using multi-hop, we take advantage of the fact that the intermediate r	0.25	0.25	2
Direct transmission is chosen since it consumes 4.4 mJ vs. 5.6 mJ for mult	0.25	0.25	1
Multi-hop reduces the sensor's energy from 3800 μ J to 800 μ J, balancing th	0	0	2
Transmission energy depends on the distance: direct transmission over 4	0.25	0.25	2
Direct is more efficient (4400 vs 5600 μ J). High circuit energy (E_c) makes m	0.25	0.25	1
Multi-hop reduces the energy used by the battery-powered sensor (0.8 m	0.25	0.25	2
Motion sensor is the only battery powered device in the network, so it's t	0.25	0.25	2
long direct transmission is heavy on the battery of the sensor; using mult	0.25	0.25	2
Lifetime is limited by the first node to deplete its battery. For direct, the p	0.25	0.25	2
The sensor is the only node that runs on battery. Multi-hop minimizes its	0.25	0.25	2
Multihop reduces sensor energy per message from 3.8 mJ to 0.8 mJ (79%	0.25	0.25	2
DIRECTLY: energy provided by the sensor is 3,8 mJ and equals 86% of tota	0.25	0.25	2
Although multi-hop transmission reduces the distance of each individual	0.25	0.25	1
Using the multi-hop strategy cuts sensor energy from 3,800 to 800 μ J by s	0.25	0.25	2
In the direct case, the energy required by the system is lower than the ho	0.25	0	2
Direct transmission is chosen for lower energy use (4.4 mJ vs 5.6 mJ multi	0.25	0.25	1
Sensor uses 800 μ J (multi-hop) vs 3800 μ J (direct). Bulbs are mains-power	0.25	0.25	2
For a longterm operation I choose B even if the total consumption is high	0.25	0.25	2
Direct is kinda better here because it uses lesser total energy than multi-h	0.25	0.25	1
Direct is better: 4.4mJ beats 5.6mJ. Multi-hop fails because the energy sa	0.25	0.25	1
Multi-hop lowers the sensor's energy per message (0.8 mJ vs 3.8 mJ), so i	0.25	0.25	2
I choose direct tx, minimizing tot communication energy, because it costs	0.25	0.25	1
Multi-hop spreads load across nodes, cutting peak energy per node. Net	0.25	0.25	2
Although Multi-hop consumes more total energy, the energy is distribut	0.25	0.25	2
We choose multi-hop because it lowers the maximum per-node energy (1	0.25	0.25	2
Direct minimizes total network energy 4.4<5.6 mJ. Multi maximizes senso	0.25	0.25	1
Choose multi-hop to minimize maximum per-node energy (bottleneck). I	0.25	0.25	2
Direct transmission uses less total energy, but it drains one node much fa	0.25	0.25	2
Multi-hop minimizes per-node energy, cutting sensor use from 3800 to 80	0.25	0.25	2
Multi-hop is better long-term: node sends 10m not 40m, cutting transmi	0.25	0.25	2
Bulbs are not battery powered so they have no energy capacity constrain	0.25	0.25	2
Direct: motion sensor 3.8mJ, sink node 0.6mJ others components zerosM	0.25	0.25	2
assuming we are still the ideal transmission case, direct transmission is th	0.25	0.25	1
Direct transmission consumes 4400 μ J vs 5600 μ J for multi-hop.Sensor TX	0.25	0.25	1
The sensor consumes less in multi-hop (0.8 mJ) than direct (3.8 mJ), so w	0.25	0.25	2
The multi-hop strategy ensures long-term operation by extending batter	0.25	0.25	2
Multi-hop reduces sensor energy from 3.8 mJ to 0.8 mJ. Since the sensor	0.25	0.25	2
Sensor is battery powered, bulbs are powered by the electrical grid. Energ	0.25	0.25	2
Multi-hop is preferred since it minimizes the maximum per-node energy.	0.25	0.25	2
Multi-hop transmission spreads the energy usage across intermediate no	0.25	0.25	2
Direct transmission: $E_{\text{sens}} = k \cdot E_c + k \cdot \epsilon_s \cdot (40)^2 = 3800 \mu\text{J}$ Multi hop tr	0.25	0.25	2
motion sensor is the only one with battery: need minimizing its energy co	0.25	0.25	2
Multi-hop reduces the highest energy spent by one node, so it improves	0	0	2
The battery of the sensor consumes less energy with respect to direct cor	0	0	2
Multi-hop minimizes the motion sensor energy (0.8 mJ vs 3.8 mJ), so it ex	0.25	0.25	2

1.5	2
1.5	2
1.5	1.5
1.5	2
0.75	1.25
0.75	1.25
1.5	2
0.75	1.25
1.5	1.5
1.5	2
0.75	1.25
1.5	2
1.5	2
1.5	2
1.5	2
1.5	2
1.5	2
0.75	1.25
1.5	2
1.5	1.75
0.75	1.25
1.5	2
1.5	2
0.75	1.25
0.75	1.25
1.5	2
0.75	1.25
1.5	2
1.5	2
1.5	2
0.75	1.25
1.5	2
1.5	2
1.5	2
1.5	2
1.5	2
1.5	2
0.75	1.25
0.75	1.25
1.5	2
1.5	2
1.5	2
1.5	2
1.5	2
1.5	2
1.5	2
1.5	2
1.5	2
1.5	1.5
1.5	1.5
1.5	2

154	3/30/26 12:28:31	3/30/26 12:57:45	11009540@polim Emmanuel Muyololc	EMMANUEL MUYOL
155	3/30/26 12:48:26	3/30/26 13:02:33	10849638@polim Alessio Lika	Alessio Lika
156	3/30/26 12:50:58	3/30/26 13:11:57	10805688@polim Giacomo Genco	da dsa
157	3/30/26 11:52:20	3/30/26 13:28:36	10829488@polim Gabriele Benedetti	Gabriele B Giuseppe
158	3/30/26 13:18:00	3/30/26 13:40:28	10795184@polim Silvia Ferrara	Silvia FerrrEren Genis
159	3/30/26 13:40:41	3/30/26 13:59:00	11164465@polim Tommaso Ernst	Tommaso Alessandr
160	3/30/26 14:15:40	3/30/26 14:33:17	10777415@polim Andrea Nardi	Andrea NaFrancesco
161	3/30/26 14:31:32	3/30/26 14:43:45	11109586@polim Seyedaman Zargari	SeyedamaAmirhosse
162	3/30/26 14:05:41	3/30/26 14:55:25	11121670@polim Zhang Xinyue	Zhang Xinyue
163	3/30/26 15:02:36	3/30/26 15:12:25	10846339@polim Andrea Gianfelice	Andrea Gianfelice
164	3/30/26 15:12:25	3/30/26 15:13:46	11033400@polim Zhang Hezhi	ZHANG HEZHI
165	3/30/26 15:15:07	3/30/26 15:16:21	10910881@polim Claudia Carraro	Claudia CaChiara Cav
166	3/30/26 15:06:54	3/30/26 15:23:35	10903183@polim Matteo Ferrari	Matteo Ferrari
167	3/30/26 15:00:27	3/30/26 15:25:14	11048190@polim Qi Jinwen	Jinwen Qi
168	3/30/26 15:26:16	3/30/26 15:33:57	10972650@polim Guan Enci	Zhaochen Enci Guan
169	3/30/26 15:19:04	3/30/26 15:36:21	10831046@polim Matteo Franken	Matteo Fr Lorenzo C
170	3/30/26 15:27:02	3/30/26 15:39:46	10815900@polim Marco Scianna	Ermir BalliMarco Sci
171	3/30/26 15:38:35	3/30/26 15:41:32	11140947@polim Ermir Balliu	Ermir BalliMarco Sci
172	3/30/26 15:32:02	3/30/26 15:44:03	10773779@polim Francesco Marchi	Francesco Juan Marc
173	3/30/26 15:24:28	3/30/26 15:47:24	11026772@polim Minh Man Nguyen	Minh Man Nguyen
174	3/30/26 15:36:05	3/30/26 15:48:07	11135422@polim Luca Pezzoli	Luca PezzrMattia Rai
175	3/30/26 15:37:06	3/30/26 15:49:50	10764548@polim Andrea Callari	Andrea CaSimone Cl
176	3/30/26 15:24:27	3/30/26 15:54:57	10806788@polim Gabriele Maria Michi	Gabriele Michieletto
177	3/30/26 16:01:35	3/30/26 16:02:41	10782022@polim Stefano Longhi	Stefano LcBeatrice S
178	3/30/26 15:50:41	3/30/26 16:03:51	10854816@polim Marco Abbadessa	Marco AblAlice Berta
179	3/30/26 16:06:25	3/30/26 16:17:00	11119918@polim Wang Yunxuan	WANG YUNXUAN
180	3/30/26 16:06:00	3/30/26 16:18:15	10712273@polim Matteo Vicentini	Matteo Vicentini
181	3/30/26 16:05:36	3/30/26 16:18:36	11161809@polim Eric Lombardo Facci	Eric LombFilippo Pal
182	3/30/26 14:31:09	3/30/26 16:21:08	11028952@polim Pang Haonan	Haoyu ZhHaonan P
183	3/30/26 16:22:28	3/30/26 16:32:11	10773372@polim Simone Perugini	Simone Perugini
184	3/30/26 16:29:31	3/30/26 16:40:56	11059833@polim Zhanat Argimbayeva	Zhanat Argimbayeva
185	3/30/26 16:43:52	3/30/26 16:57:31	10704990@polim Riccardo Baiardi	Riccardo Baiardi
186	3/30/26 16:47:49	3/30/26 17:00:50	11114435@polim Mobin Khatib	Mobin KhAli Edareh
187	3/30/26 16:58:26	3/30/26 17:03:57	11102420@polim Matteo Ghiaroni	Matteo GlChiara Far
188	3/30/26 16:55:07	3/30/26 17:04:29	10727881@polim Andrea Luca Miele	Andrea Luca Miele
189	3/30/26 16:45:12	3/30/26 17:05:17	11035477@polim Tang Mengshuang	Xiaohan P Mengshu
190	3/30/26 16:31:41	3/30/26 17:28:44	11154359@polim Wu Tianqi	Tianqi Wu Qinglong I
191	3/30/26 17:32:02	3/30/26 17:38:13	10888593@polim Leonardo Di Giovanr	leonardo di giovanni
192	3/30/26 17:44:54	3/30/26 17:52:11	10823873@polim Alberto Pini	Alberto PiiAlessandr
193	3/30/26 17:57:41	3/30/26 18:01:38	10810440@polim Samuele Grisoni	Samuele CAlessio Gu
194	3/30/26 17:33:20	3/30/26 18:11:39	10841085@polim Federico Fasola	Federico FMarco For
195	3/30/26 17:36:46	3/30/26 18:17:31	10724267@polim Alessandra Romano	Alessandra Romano
196	3/30/26 17:47:13	3/30/26 18:27:56	10717254@polim Gabriele Clara Di Gio	Gabriele C Andrea Pa
197	3/30/26 18:04:08	3/30/26 18:29:32	10835836@polim Andrea Ziccardi	Andrea Ziccardi
198	3/30/26 18:18:04	3/30/26 18:31:16	10710384@polim Lorenzo Francesco Li	Lorenzo FiAndrea Ve
199	3/30/26 18:19:33	3/30/26 18:34:05	10778484@polim Eliseo Paoletti	Eliseo Pao Gabriele N
200	3/30/26 15:50:47	3/30/26 18:39:15	10797304@polim Francesco Sarra	Andrea RoFrancesco
201	3/30/26 18:22:47	3/30/26 18:39:16	10863400@polim Wang Yuan	Yuan Wang
202	3/30/26 18:22:31	3/30/26 18:49:28	11026049@polim Fernando Javier Barc	Fernando Javier Barc
203	3/30/26 18:25:37	3/30/26 18:58:20	11175090@polim Anasuya Satapathy	Anasuya SJiaxun Jiar
204	3/30/26 18:49:54	3/30/26 19:00:55	10771308@polim Matteo Bergagna	Matteo Bergagna

11009540		4400	5600	Multi-hop Minimize maximum per-node energy
10849638		4400	5600	Multi-hop Minimize maximum per-node energy
10805688	10805688	3232	32323	Multi-hop Minimize maximum per-node energy
10829488	10795884	4400	5600	Multi-hop Maximize network lifetime
10795184	11062471	4400	5600	Multi-hop Network lifetime
11164465	10809515	4400	5600	Multi-hop Minimize maximum per-node energy
10777415	10767901	4400	5600	Multi-hop Maximising the operating lifetime of t
11109586	11109540	4400	5600	Multi-hop Node lifetime of the most constrainec
11121670		3800	5600	Multi-hop Minimize maximum per-node energy
10846339		4400	5600	Multi-hop Minimize maximum per-node energy
11033400		4400	5600	Multi-hop Minimize maximum per-node energy
10910881	10856023	4400	5600	Direct Minimize total communication energ
10903183		4400	5600	Multi-hop minimize energy of the battery-powe
11048190		4400	5600	Multi-hop Minimize maximum per-node energy
11021721	10972650	4400	5600	Direct Minimize total communication energ
10831046	10797603	4400	5600	Multi-hop Maximize Aqara Motion Sensor batte
11140947	10815900	3800	5000	Direct Minimize total communication energ
11140947	10815900	3800	5000	Direct Minimize total communication energ
10773779	10767244	4400	5600	Multi-hop Minimize total communication energ
11026772		4400	5600	Multi-hop Minimize total communication energ
11135422	11011342	4400	5600	Multi-hop Maximize network lifetime
10764548	10764758	4400	5600	Multi-hop Network lifetime
10806788		4400	5600	Multi-hop Minimize maximum per-node energy
10782022	10766466	4400	5600	Multi-hop Minimize maximum per-node energy
10854816	10889590	4400	5600	Multi-hop Minimize maximum per-node energy
11119918		4400	5600	Multi-hop Minimize maximum per-node energy
10712273		4400	5600	Multi-hop Minimize maximum per-node energy
11161809	11084799	4400	5600	Multi-hop Minimize maximum per-node energy
11061927	11028952	4400	5600	Multi-hop Minimize maximum per-node energy
10773372		4400	5600	Multi-hop other
11059833		3800	5000	Multi-hop Minimize maximum per-node energy
10704990		4400	5600	Multi-hop Minimizzare il massimo costo per noc
11114435	11036028	4400	5600	Multi-hop Minimize maximum per-node energy
11102420	10846947	4400	5600	Multi-hop Minimize maximum per-node energy
10727881		4400	5600	Multi-hop Minimize maximum per-node energy
11032539	11035477	4400	5600	Multi-hop Minimize maximum per-node energy
11154359	11151255	4400	5600	Multi-hop Minimize maximum per-node energy
10888593		4400	5600	Direct Minimize total communication energ
10823873	10822654	4400	5600	Multi-hop Minimize battery-supplied energy (i.e
10810440	10795876	4400	5600	Multi-hop Minimize maximum per-node energy
10841085	10767413	4400	5600	Multi-hop Optimize sensor battery lifetime
10724267		4400	5600	Multi-hop Minimize maximum per-node energy
10717254	10716103	4400	5600	Multi-hop Network lifetime
10835836		4400	5600	Multi-hop Minimize maximum per-node energy
10710384	10721613	4400	5600	Multi-hop Minimizing the Aqara motion sensor
10778484	10848911	4400	5600	Multi-hop Minimize maximum per-node energy
10799170	10797304	4400	5600	Multi-hop Minimize maximum per-node energy
10863400		4400	5600	Direct Minimize total communication energ
11026049		3800	5000	Multi-hop Minimize the energy of the P1 node
11175090	11021337	4400	5600	Multi-hop Minimize maximum per-node energy
10771308		4400	5600	Multi-hop Minimize the sensor transmission ev

No

I didn't

-

Maximize network lifetime (time to the first "de

For long-term operations we would try to maxi

.

We are maximising the operating lifetime of the

Maximize battery lifespan of the constrained no

no , I choose Minimize maximum per-node en

.

nothing

not selected

minimize energy of the battery-powered node (

null

Minimize total communication energy

Maximize Aqara Motion Sensor battery life by r

\

See above

.

My chosen metric is to maximize the lifetime of

The optimized metric refers to the time before t

Network lifetime (Maximizing sensor battery life

N/A

/

.

While multi-hop uses more total energy, it exte

Not selected.

Minimize maximum per-node energy

/

Maximize the battery lifetime of the motion ser

—

Ipotizzando che i bulb siano alimentati direttan

I didn't choose other but it was mandatoryIn M

I have selected Minimize maximum per-node en

Not applicable

None , didnt select 'other'.

.

Minimize battery-supplied energy (i.e. system l

-

Optimize sensor battery lifetime

The selected performance metric is the minimiz

Network lifetime of the energy-constrained noc

.

Minimizing the Aqara motion sensor p1 energy

na

-

The P1 node is the motion sensor and is the on

N/A

I optimize by minimizing the energy required to

Multi-hop cuts sensor energy from 3.8mJ to 0.8mJ by shortening distance	0.25	0.25	2
Capacity of bulbs are way higher than the motion sensor, the hop-by-hop	0.25	0.25	2
Multi-hop strategy minimizes max per-node energy. It trades higher total	0.25	0.25	2
Direct tx uses less total energy but drains the battery sensor (3.8mJ). Mul	0.25	0.25	2
When the sensor performs a direct communication it consumes more than	0.25	0.25	2
Multi-hop optimizes sensor lifetime: it cuts battery drain (0.8mJ vs 3.8mJ)	0.25	0.25	2
To maximise the lifetime of the sensor's battery (1240mAh): this "costs" in	0.25	0.25	2
Multi-hop cuts the battery-powered sensor's energy cost from 3.8mJ to 0	0.25	0.25	2
By selecting multi-hop transmission, the strategy minimizes maximum per	0	0.25	2
Multi-hop is better; while direct uses less total energy here, multi-hop op	0.25	0.25	2
Multi-hop routing cuts sensor Tx energy from 3800uJ to 800uJ via mains-p	0.25	0.25	2
Direct transmission minimizes total network energy (4400 µJ vs 5600 µJ) s	0.25	0.25	1
bulbs are not battery-powered, the sensor is. multi hop cuts sensor's tx e	0.25	0.25	2
Though total energy is higher (5600uJ vs 4400uJ), Multi-hop drops the ba	0.25	0.25	2
Direct transmission consumes is lower than multi-hop. Considering also t	0.25	0.25	1
Multi-hop reduces the needed TX power since the distance is cut from 40	0.25	0.25	2
Direct is better because it minimizes total communication energy: 3.8 mJ	0	0	1
Direct is better because it minimizes total communication energy: 3.8 mJ	0	0	1
The transmission energy grows quadratically with distance, therefore opt	0.25	0.25	1
The motion sensor is battery-powered while the bulbs are plugged. By ch	0.25	0.25	2
Bulbs/sink node are not battery supplied. The sensor has battery: in the	0.25	0.25	2
Multi-hop optimizes network lifetime by reducing the battery-powered se	0.25	0.25	2
Multi-hop minimizes maximum per-node energy. Routing via mains-pow	0.25	0.25	2
Direct Transmission uses less total energy, but drains the battery-powere	0.25	0.25	2
Multi-hop minimizes max per-node energy. The battery-powered sensor	0.25	0.25	2
While multi-hop uses more total energy, it extends network lifetime by re	0.25	0.25	2
Multi-hop reduces the sensor energy (0.8 mJ vs 3.8 mJ), extending its lifet	0.25	0.25	2
Direct uses less energy (4.4 vs 5.6 mJ) but drains the sensor (3.8 mJ). Mul	0.25	0.25	2
Direct : Energy consumption per message is 3.8 mJ. Multi-hop : Relay nod	0.25	0.25	2
I chose multi-hop because it greatly reduces the energy used by the batte	0.25	0.25	2
Multi-hop reduces sensor TX energy to 800µJ vs 3800µJ (direct), offloadin	0	0	2
Ho scelto di usare la multi-hop transmission perché il motion sensor se ne	0.25	0.25	2
Multi-hop reduces sensor's energy from 3.8mJ to 0.8mJ. This minimizes t	0.25	0.25	2
Direct transmission is globally efficient but burdens single nodes. Multi-h	0.25	0.25	2
Although multi-hop transmission consumes more energy overall, since the	0.25	0.25	2
Under ideal communication, direct transmission drains the sensor batter	0.25	0.25	2
Multi-hop triples node life with only 1.27x total energy. This trade-off is id	0.25	0.25	2
direct transmission is preferred because it minimizes total communicatio	0.25	0.25	1
We minimize power draw at the sensor node (0.8 mJ vs 3.8 mJ), since it is	0.25	0.25	2
Multi hop is preferred because it distributes energy use across node pres	0.25	0.25	2
Since the sensor is the only battery operated component, the multi-hop	0.25	0.25	2
I chose multi-hop mode to optimize network life. Even as energy consum	0.25	0.25	2
Multi-hop minimizes the sensor's per-message energy (3.8→0.8 mJ), exte	0.25	0.25	2
Multi-hop reduces sensor energy by transmitting to B1 over a shorter dis	0.25	0.25	2
In the spec sheet, it is mentioned that the MS uses 2 CR2450 batteries, fo	0.25	0.25	2
The battery-powered sensor uses 3.8mJ for direct transmission. Multi-ho	0.25	0.25	2
"Direct" has lower overall consumption but "multi-hop" reduces the batt	0.25	0.25	2
Direct uses less total energy (4400 µJ vs 5600 µJ), thus minimizing networ	0.25	0.25	1
Since the motion sensor is the only operating on battery, the key matrix i	0	0	2
Multi-hop reduces sensor energy from 3800 to 800 uJ (4.75 times). Relay c	0.25	0.25	2
sensor: battery powered. bulb: electricity grid powered. ignore consumpti	0.25	0.25	2

[illegible]

205	3/30/26 19:02:13	3/30/26 19:09:22	11180610@polim	Daniel Andreas Nage	Angelica ODaniel Näge
206	3/30/26 18:48:27	3/30/26 19:10:51	11046642@polim	Shaghayegh Vesali	shaghayegh akbar
207	3/30/26 19:16:33	3/30/26 19:20:11	10772713@polim	Duilio Diego Giorgett	Enrico Gril Diego Gio
208	3/30/26 19:00:41	3/30/26 19:20:12	10778074@polim	Enrico Grillo	Enrico Gril Diego Gio
209	3/30/26 18:58:13	3/30/26 19:20:48	10774792@polim	Alessandro Andrea M	Alessandro Andrea M
210	3/30/26 19:20:57	3/30/26 19:21:03	10861979@polim	Alessandro Nipoti	Alessandro Xiaoyang
211	3/30/26 19:06:40	3/30/26 19:22:15	11057250@polim	Arzu Yildiz	Arzu YILDI Yaser Enes
212	3/30/26 18:55:20	3/30/26 19:24:04	10778819@polim	Lorenzo Pedranzini	Lorenzo Pedranzini
213	3/30/26 19:22:37	3/30/26 19:40:53	10768882@polim	Ronald Cammaroto	Ronald Cammaroto
214	3/30/26 18:28:44	3/30/26 19:57:11	10711947@polim	Eugenia Ghiglione	Eugenia G Andrea Cu
215	3/30/26 16:59:38	3/30/26 20:06:32	10855966@polim	Marco Puddu	Marco Puddu
216	3/30/26 20:09:15	3/30/26 20:14:50	10812603@polim	Andrea Sau	Andrea Sa Daniele To
217	3/30/26 20:18:56	3/30/26 20:29:10	11123116@polim	Lucía Fernández Gin	Fernandez Giner
218	3/30/26 19:52:24	3/30/26 20:39:28	10861756@polim	Federico Ciarniello	Federico C Andrea M
219	3/30/26 20:55:15	3/30/26 20:58:09	11063739@polim	Sajjad Shaffaf	sajjad sha seyed tah
220	3/30/26 20:48:56	3/30/26 21:01:09	11162444@polim	Yihang Ji	Yihang Ji
221	3/30/26 20:42:50	3/30/26 21:04:41	10712008@polim	Alessia Maria Di Cap	Alessia Di Francesco
222	3/30/26 20:17:34	3/30/26 21:07:00	10862592@polim	Enrico Mantegazza	Enrico Ma Osama Ati
223	3/30/26 21:08:28	3/30/26 21:15:47	11156535@polim	Torodd Urdshals	Torodd Urdshals
224	3/30/26 21:21:16	3/30/26 21:29:29	10777399@polim	Mario Pesca	Mario Pesca
225	3/30/26 21:25:45	3/30/26 21:33:53	10742323@polim	Giacomo Pasturensi	Giacomo Pasturensi
226	3/30/26 20:42:34	3/30/26 21:35:07	11118429@polim	Wang Guangyi	Wang Guangyi
227	3/30/26 21:34:09	3/30/26 21:43:27	10914222@polim	Martín Corchuelo Ve	Martin Co Hamidreza
228	3/30/26 21:53:31	3/30/26 21:55:47	10738139@polim	Davide Lupo	Davide Lu Fabio Agr
229	3/30/26 20:33:34	3/30/26 21:59:16	10793255@polim	Flavio Riccobono	Flavio Riccobono
230	3/30/26 20:47:47	3/30/26 22:08:44	10938858@polim	Alessandro Ferrando	Alessandro Ferrando
231	3/30/26 17:47:23	3/30/26 22:24:09	11035971@polim	Özgür Efehan Kati	Özgür Efehan Firat Aden
232	3/30/26 22:06:05	3/30/26 22:28:04	10495016@polim	Dalila Samr	Dalila Samr
233	3/30/26 22:24:59	3/30/26 22:37:48	10852883@polim	Francesco Pio Lo Mo	Francesco Lo Monaco
234	3/30/26 22:34:09	3/30/26 22:38:21	11107611@polim	Ning Zhang	Ning Zhang
235	3/30/26 22:41:43	3/30/26 22:43:37	10828129@polim	Lorenzo Foini	Lorenzo F Filippo Go
236	3/30/26 22:58:36	3/30/26 23:03:28	11036293@polim	Amr Ahmed Yehia El	Amr Elmas Amjed El
237	3/30/26 22:53:34	3/30/26 23:14:55	10830365@polim	Andrea Santaniello	Andrea Santaniello
238	3/30/26 23:16:06	3/30/26 23:28:26	10607397@polim	Ivana Lin	Ivana Lin
239	3/30/26 23:24:58	3/30/26 23:30:18	11145475@polim	Lorena Lam Diez	Lorena Lar Sofia Meri
240	3/30/26 23:28:09	3/30/26 23:31:22	11154362@polim	Cheng Keyan	Keyan Cheng
241	3/30/26 23:32:34	3/30/26 23:37:05	10497245@polim	Mattia Angelo Brune	Mattia Angelo Brune
242	3/30/26 23:46:03	3/30/26 23:51:03	11121758@polim	Razia Jafari	RAZIA JAFARI
243	3/30/26 23:48:18	3/30/26 23:54:50	10892569@polim	Giacomo Aragnetti	Giacomo /NA
244	3/30/26 23:55:03	3/30/26 23:58:18	11115207@polim	Juan David Vargas G	Juan David Cristian D
245	3/30/26 23:20:25	3/30/26 23:58:57	11140550@polim	Hadi El Hajj	Hadi El Hajj
246	3/30/26 23:54:01	3/30/26 23:59:22	10847803@polim	Maciej Krzysztof Rze	Maciej Rzewuski
247	3/30/26 21:39:54	3/31/26 0:01:18	11052807@polim	Erdal Yalçin	ERDAL YALCIN
247	3/30/26 21:39:54	3/31/26 0:01:18	11052807@polim	Erdal Yalçin	ERDAL YALCIN

11191225	11180610	4400	5600	Multi-hop Battery lifetime of the sensor node.
11046642	1118765	4400	5600	Multi-hop Minimize maximum per-node energy
10778074	10772713	4400	5600	Multi-hop Minimize energy consumed by motion
10778074	10772713	4400	5600	Multi-hop Minimize energy consumed by motion
10774792		4400	5600	Multi-hop Maximize network lifetime
10861979	11108477	4400	5600	Multi-hop Minimize maximum per-node energy
11057250	11057249	3800	5000	Direct Minimize total communication energy
10778819		4400	5600	Multi-hop Minimize communication energy for routing
10768882		4400	5600	Multi-hop minimize communication energy for routing
10711947	10813527	4400	5600	Multi-hop Minimize the energy consumed by the sensor node
10855966		4400	5600	Multi-hop Minimize maximum per-node energy
10812603	10859228	4400	5600	Multi-hop Minimize energy on sensor node
11123116		4400	5600	Multi-hop Minimize maximum per-node energy
10861756	11212726	4400	5600	Multi-hop Minimize maximum per-node energy
11063739	11060363	4400	5600	Multi-hop Minimize maximum per-node energy
11162444		4400	5600	Multi-hop Minimize maximum per-node energy
10712008	10766440	4400	5600	Direct Minimize total communication energy
10862592	10772098	4400	5600	Multi-hop Optimize sensor lifetime
11156535		3800	5600	Direct Minimize total communication energy
10777399		4400	5600	Multi-hop Minimize maximum per-node energy
10742323		4400	5600	Direct Minimize maximum per-node energy
11118429		4400	5600	Multi-hop Minimize maximum per-node energy
10914222	11057471	4400	5600	Multi-hop Battery life of sensor node
10738139	10769569	4400	5600	Multi-hop Minimize maximum per-node energy
10793255		4400	5600	Multi-hop Minimize maximum per-node energy
10938858		4400	5600	Multi-hop Minimize the presence sensor node energy
11035971	11035164	4400	5600	Multi-hop Battery Life
10495016		4400	5600	Multi-hop optimizing Individual Node Lifetime (s)
10852883		4400	5600	Multi-hop Minimize maximum per-node energy
11107611		4400	5600	Multi-hop Minimize maximum per-node energy
10828129	10774461	4400	5600	Direct Minimize total communication energy
11036293	11061031	4400	5600	Direct Minimize total communication energy
10830365		4400	5600	Multi-hop Minimize maximum per-node energy
10607397		3800	5000	Direct Minimize total communication energy
11145475	11156291	4400	5600	Multi-hop Minimize maximum per-node energy
11154362		4400	5600	Multi-hop Minimize maximum per-node energy
10497245		3800	5000	Multi-hop Minimize maximum per-node energy
11121758		4400	5600	Direct Minimize total communication energy
10892569	NA	3800	5000	Multi-hop Minimization of both under hop-skip
11115207	11097500	5000	4400	Multi-hop Minimize total communication energy
11140550		4400	5600	Multi-hop Minimize maximum per-node energy
10847803		3800	5600	Multi-hop Network lifetime
11052807		4400	5600	Direct Minimize maximum per-node energy
11052807		4400	5600	Direct Minimize maximum per-node energy

The battery lifetime is the most important metr

Since the Bulbs and the Sink are wired, the only

Since the Bulbs and the Sink are wired, the only

Maximize network lifetime (intended as the tim

.

No

Aqara is the only node with energy constraints

Minimize communication energy for the Aquar

Minimize the energy consumed by the motion

-

Minimize energy on sensor node. Don't care ab

-

I've not selected 'Other' but this would be our r

Minimize maximum per-node energy (network

For long-term operation, multi-hop transmissio

N/A

While the bulbs are mains powered and have n

I didn't

.

none

The sensor is the only one that uses a battery a

.

Minimize maximum per-node energy

Minimize the presence sensor node energy con

Battery Life

optimizing Individual Node Lifetime (specifically

did not specify other

per-node energy

N/A

-

I have not selected other

--

Minimize maximum per-node energy

no others

In the report is described entirely with data and

Minimize total communication energy

.

We need to check what node will limit lifetime c

no.

no.

Direct transmission uses less total energy but drains the sensor due to lo	0.25	0.25	2
Multi-hop reduces the sensor's energy by using short hops (~4.75 times l	0.25	0.25	2
The strategy that optimizes our decision metric is the second one: transn	0.25	0.25	2
the strategy that optimizes our decision metric is the second one: transn	0.25	0.25	2
Multi-hop maximizes life: the sensor goes from 3.8 to 0.8 mJ/msg, so the	0.25	0.25	2
with direct transmission, the node that consumes the most is the motion	0.25	0.25	2
Direct transmission consumes less energy (3.8 mJ) than multi-hop (5 mJ).	0	0	1
Bulbs are always connected to power grids. Sink node has no energy con	0.25	0.25	2
The multi-hop approach optimize the network lifetime: the sensor node t	0.25	0.25	2
The motion sensor is the only node which has energy budget limited, sinc	0.25	0.25	2
Multi-hop uses more total energy (5600 vs 4400 uJ) but minimizes max pe	0.25	0.25	2
Parameter to minimize is energy spent by sensor node that is battery po	0.25	0.25	2
Multi-hop spreads the energy across different nodes rather than overloa	0.25	0.25	2
Multi-hop increases total energy but slashes sensor TX costs. Since bulbs	0.25	0.25	2
Multi-hop drops the battery sensor's energy from 3800uJ to 800uJ. For th	0.25	0.25	2
In multi-hop transmission, the energy is distributed across the network. I	0.25	0.25	2
Capacity of bulbs are way higher than the motion sensor, the hop-by-hop	0.25	0.25	1
With direct transmission (40m) the sensor consumes 3800 uJ, with multi-	0.25	0.25	2
The best is direct transmission because it minimizes total communication	0	0.25	1
The relay nodes B1 B2, and B3 are mains-powered smart bulbs with no er	0.25	0.25	2
We minimize Max per-node energy for Network Lifetime. Since T_{comm}	0.25	0.25	1
Direct minimizes total energy but drains the sensor first. Multi-hop balan	0.25	0.25	2
The strategy chosen minimized the usage of the sensor which uses a batt	0.25	0.25	2
Multi-hop reduces the sensor energy from 3.8 mJ to 0.8 mJ by shortening	0.25	0.25	2
Multi-hop minimizes max per-node energy: although total energy is high	0.25	0.25	2
Direct TX uses 3.8mJ/msg, but Multi-hop only 0.8mJ. With a 1240mAh ba	0.25	0.25	2
Multi-hop (5600μJ) uses more total energy than direct (4400μJ) but slash	0.25	0.25	2
Multi-hop is better. It optimize individual node lifetime by shifting the 3.8	0.25	0.25	2
Direct tx uses less total energy, but the sensor is the only battery-powere	0.25	0.25	2
Even hop-by-hop consumes more total energy, but it prevents one node	0.25	0.25	2
Direct strategy minimises total communication energy, as shown by resu	0.25	0.25	1
Direct transmission is better since it avoids extra receptions. Multi-hop re	0.25	0.25	1
I choose multi-hop because it reduces the maximum energy used by one	0.25	0.25	2
To minimize total energy, direct transmission is best. Shorter hops save p	0	0	1
Direct transmission uses less total energy, but to extend the battery life o	0.25	0.25	2
Choosing multi-hop, the sensor's transmission distance is reduced from 4	0.25	0.25	2
Multi-hop reduces sensor TX from 3.8 mJ (40m) to 0.8 mJ (10m). Bulbs are	0	0	2
Multi-hop may seem efficient by reducing distance and energy loss, but r	0.25	0.25	1
Again, better described in report. The idea is that the min max target func	0	0.25	2
The hop-by-hop strategy is chosen to optimize Network Lifetime by minir	0	0	1
Multi-hop reduces the maximum energy consumption of any node, there	0.25	0.25	2
Sensor is battery powered, we can assume that network lifetime is depen	0	0.25	2
It is critical for extending the overall network lifetime. In a IoT system, the	0.25	0.25	1
It is critical for extending the overall network lifetime. In a IoT system, the	0.25	0.25	1

1.5	2
1.5	2
1.5	2
1.5	2
1.5	2
1.5	2
0.75	0.75
1.5	2
1.5	2
1.5	2
1.5	2
1.5	2
1.5	2
1.5	2
1.5	2
1.5	2
0.75	1.25
1.5	2
0.75	1
1.5	2
0.75	1.25
1.5	2
1.5	2
1.5	2
1.5	2
1.5	2
1.5	2
1.5	2
1.5	2
1.5	2
0.75	1.25
0.75	1.25
1.5	2
0.75	0.75
1.5	2
1.5	2
1.5	1.5
0.75	1.25
1.5	1.75
0.75	0.75
1.5	2
1.5	1.75
0.75	1.25
0.75	1.25

Person code 1	Pc 2	Total
10764083	10796209	2
10838600		2
10768645	10768851	2
10828027	11178565	2
10810258		1.25
11036887		1.25
10839162	10723736	2
10780379	10799787	2
10975089		0.75
11119541	11115748	2
10666869		2
11030873		2
11044575		2
11061975	11049291	2
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